

ORDER YOUR **KAWHE/COFFEE** IN MĀORI

He mōwai māku I'll have a flat white
He pango poto māku I'll have a short black
He pango roa māku I'll have a long black
He rate pīni māku I'll have a soy latte
He kaputino māku I'll have a cappuccino
He rate māku I'll have a latte
He tiakarete wera māku I'll have a hot chocolate

Rahi Size

Kei te pēhea koe? How's it going?	 (S) Paku	Hei kawē atu To take away
Anei taku kapu mau tonu Here is my reusable cup	 (M) Waenga	Ki konei To have here
	 (L) Nui	

McCafé

1. What's the Māori word for...

(a) "long"?

(b) "hot"?

2. How would you order a large cappuccino?

3. What's the word for chocolate?

AthNLP 2026

Machine Translation and Multilinguality

Antonis Anastasopoulos



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Greg Durrett CS388@UT Austin,
Graham Neubig's Advanced NLP course@CMU
and Philipp Koehn's MT course@JHU***

Machine Translation

Machine Translation

- Intro

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- A historical note

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- Semi-supervised and Unsupervised MT

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 - Both language analysis & generation



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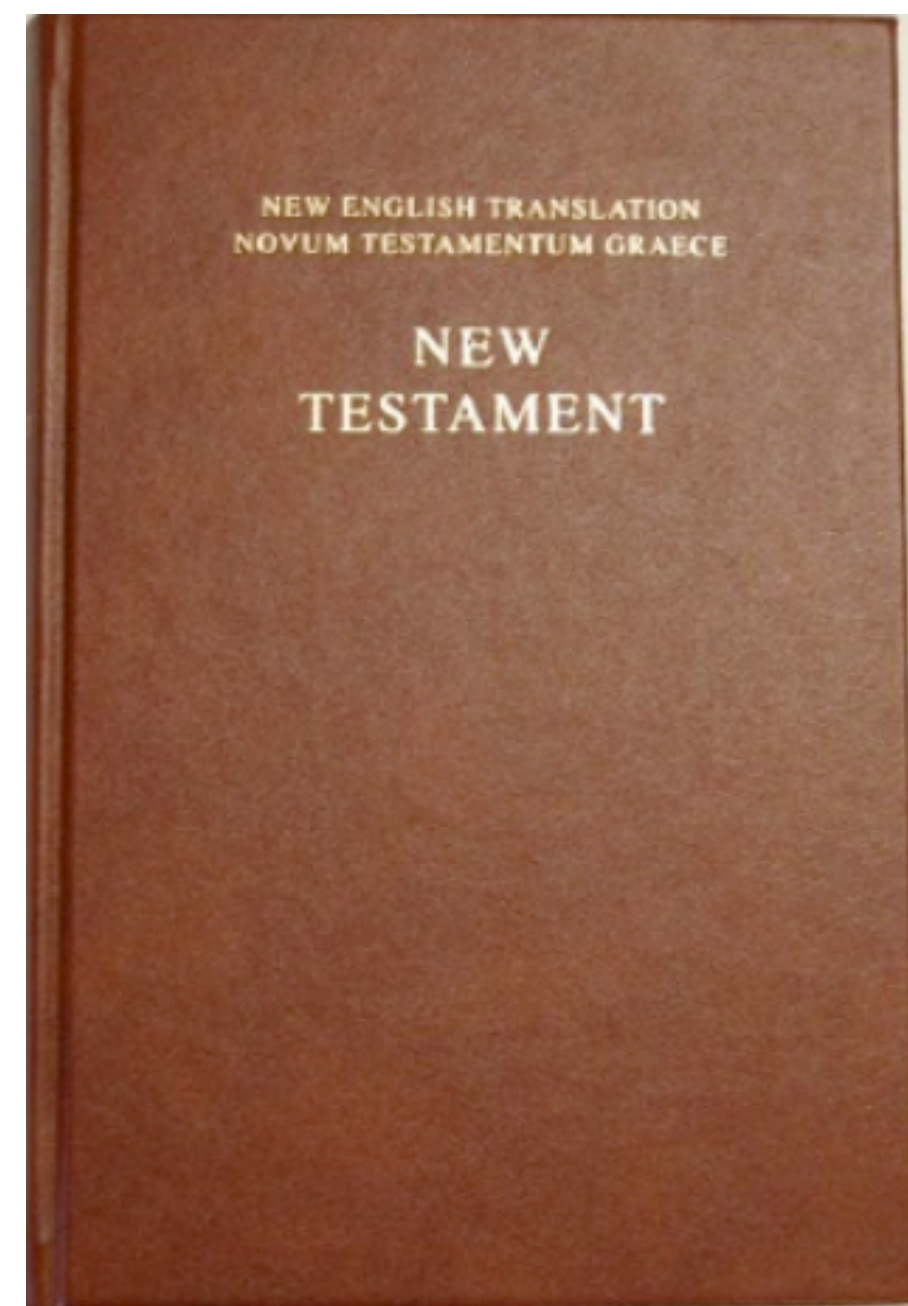


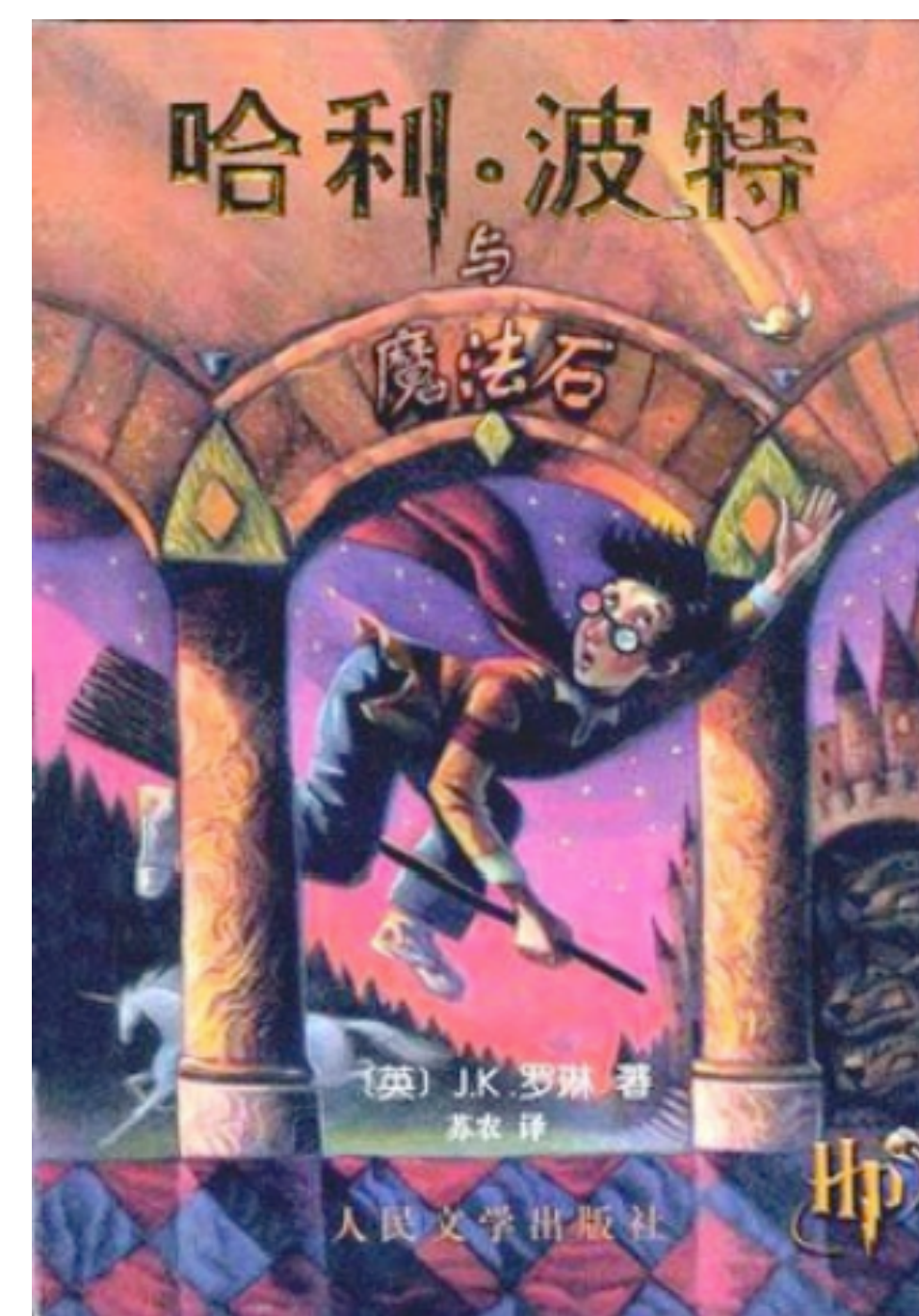
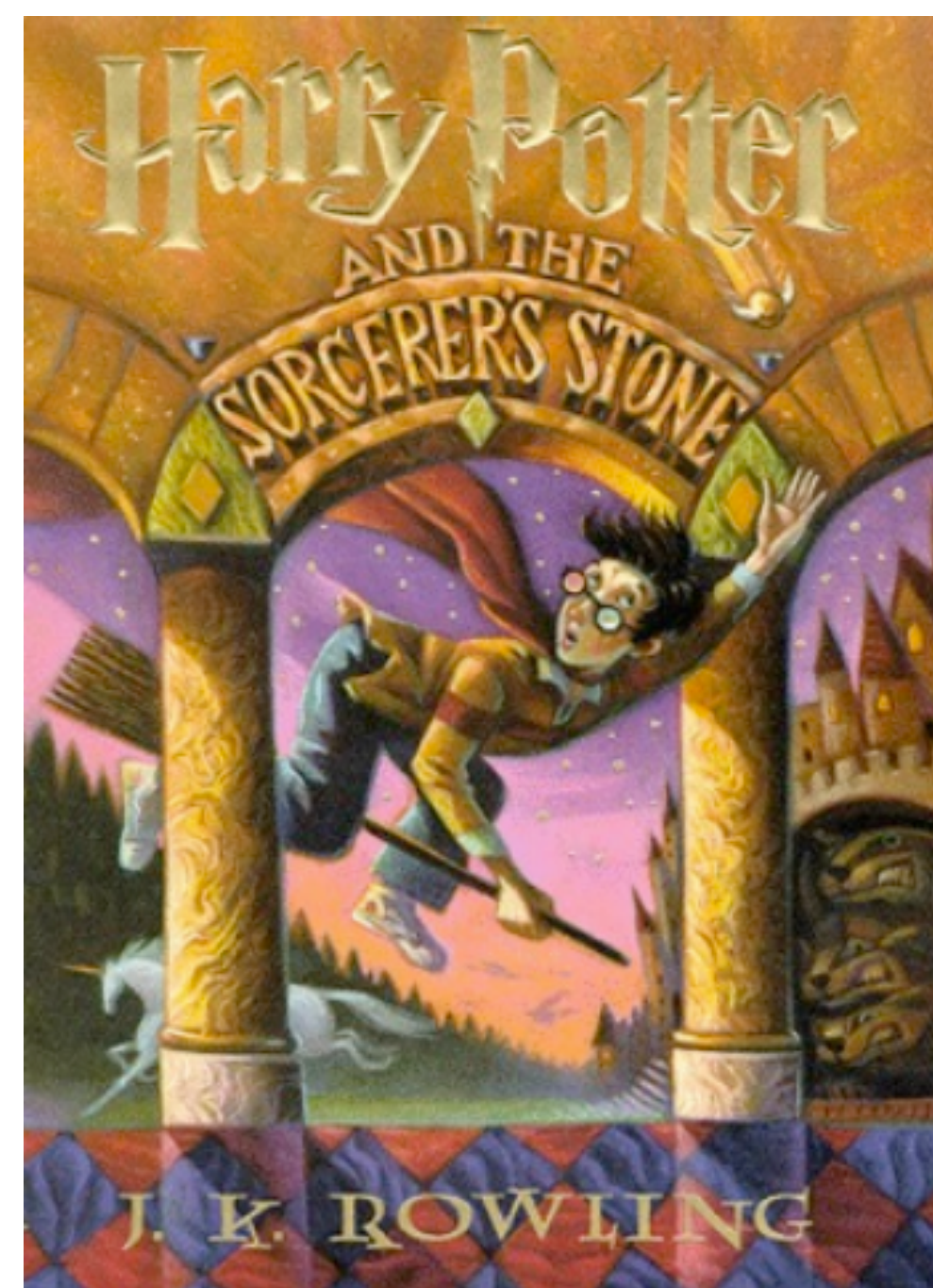
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 - Huge in Europe, growing in Asia
 - Large social/government/military as well as commercial needs









CLASSIC SOUPS

			Sm.	Lg.
清 燉 雞 湯	57.	House Chicken Soup (Chicken, Celery, Potato, Onion, Carrot)	1.50	2.75
雞 飯 湯	58.	Chicken Rice Soup	1.85	3.25
雞 麵 湯	59.	Chicken Noodle Soup	1.85	3.25
廣 東 雲 吞	60.	Cantonese Wonton Soup.....	1.50	2.75
蕃 茄 蛋 湯	61.	Tomato Clear Egg Drop Soup	1.65	2.95
雲 吞 湯	62.	Regular Wonton Soup	1.10	2.10
酸 辣 湯	63. 24	Hot & Sour Soup	1.10	2.10
蛋 花 湯	64.	Egg Drop Soup.....	1.10	2.10
雲 蛋 湯	65.	Egg Drop Wonton Mix.....	1.10	2.10
豆 腐 菜 湯	66.	Tofu Vegetable Soup	NA	3.50
雞 玉 米 湯	67.	Chicken Corn Cream Soup	NA	3.50
蟹 肉 玉 米 湯	68.	Crab Meat Corn Cream Soup.....	NA	3.50
海 鮮 湯	69.	Seafood Soup.....	NA	3.50

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- NMT is the flagship task for NLP Deep Learning
 - RNNs? Encoder-decoder? Attention mechanism?
- NMT research has pioneered many of the recent innovations of NLP Deep Learning

A historical note

1950s: Early Machine Translation

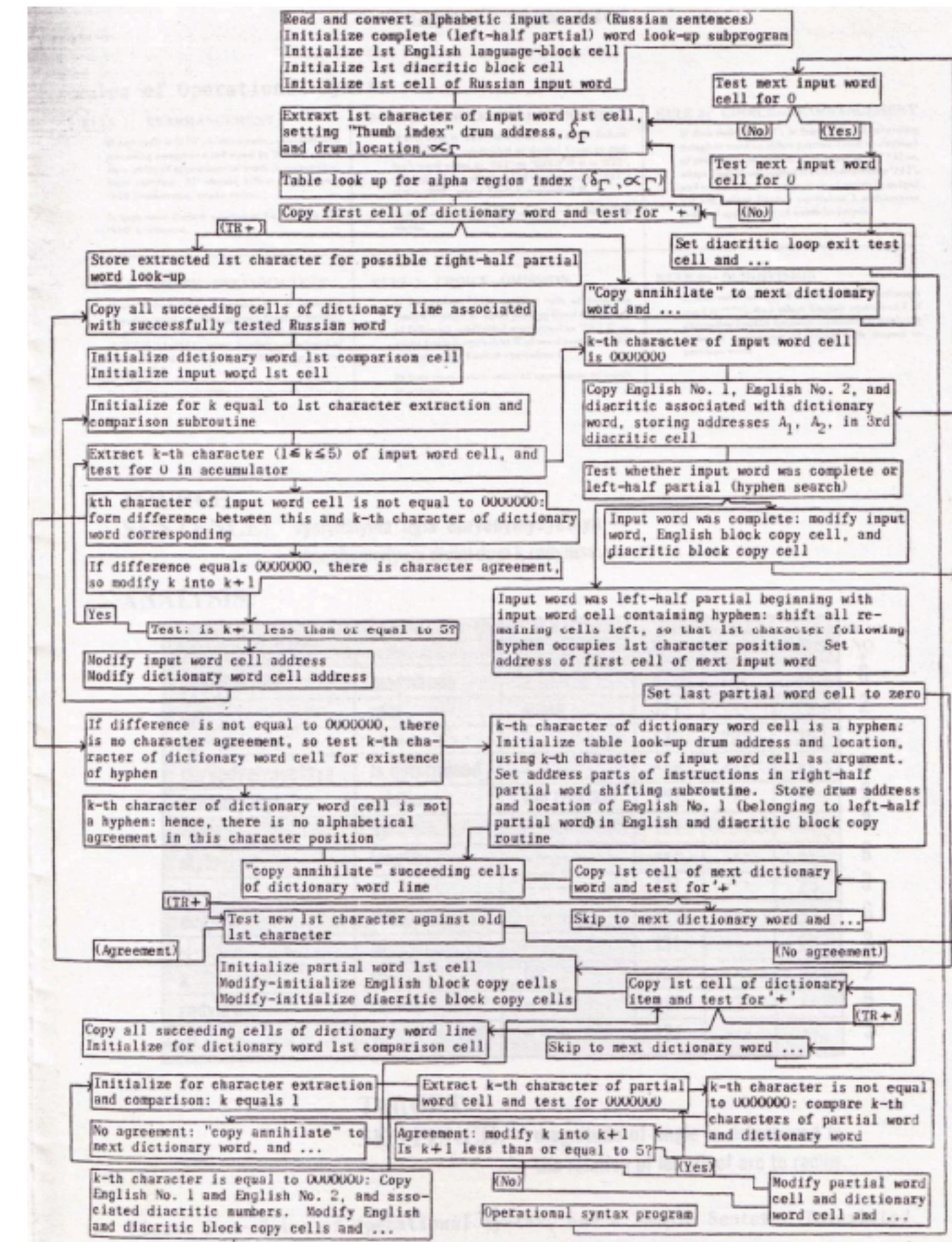


Fig. 7: Flowchart of part of the dictionary lookup procedures (from Sheridan 1955)

Flow chart of the dictionary look-up procedures ([source](#))

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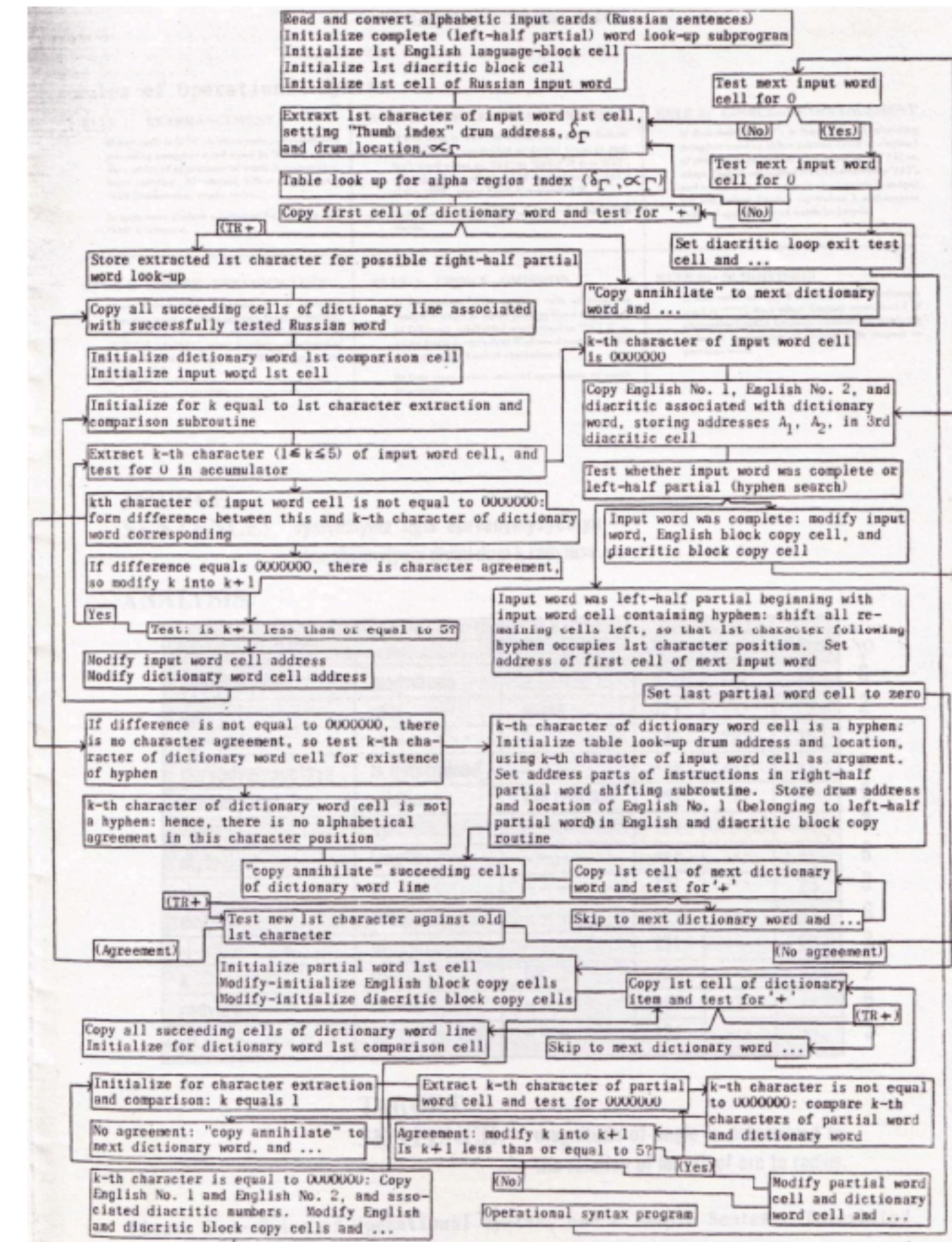


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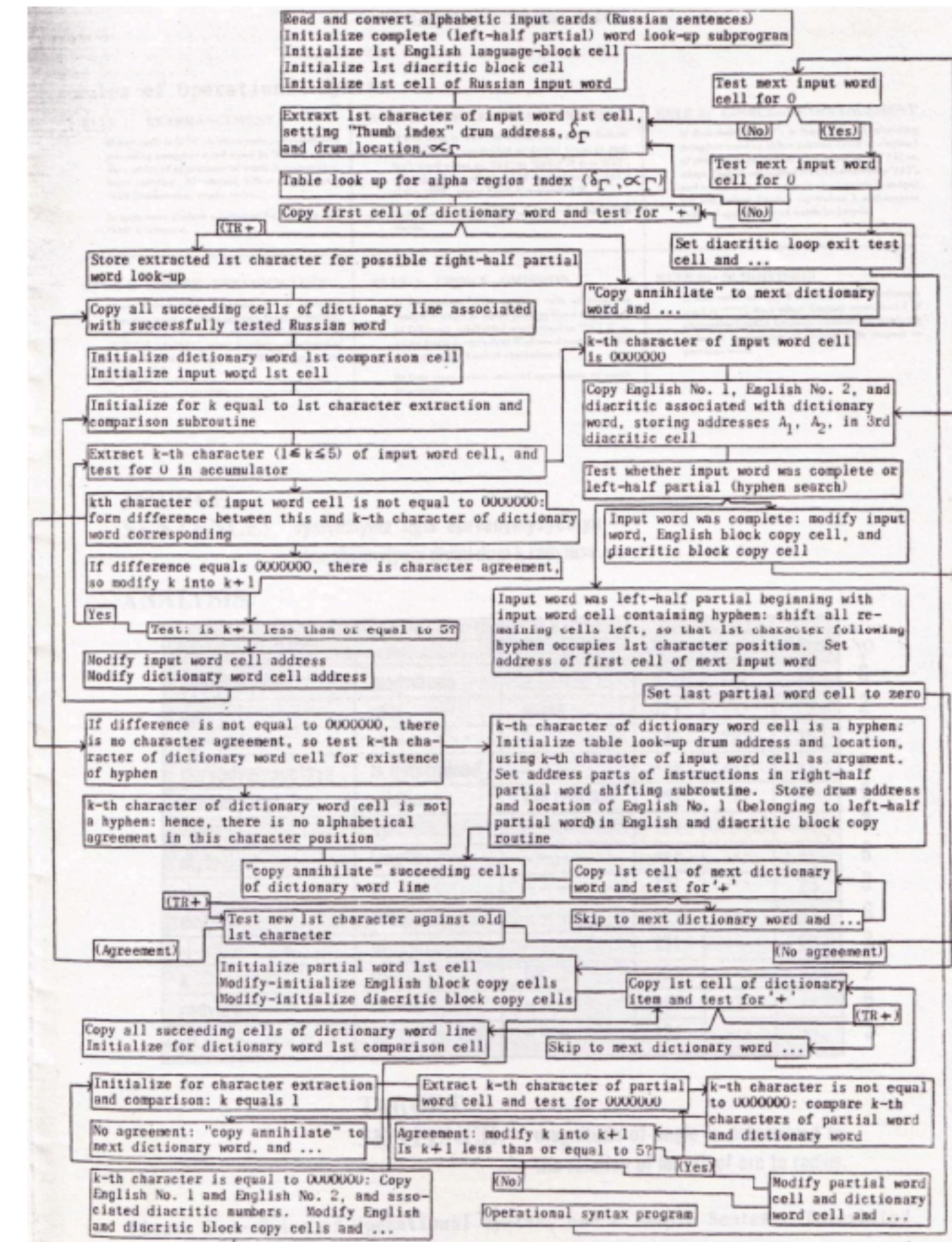


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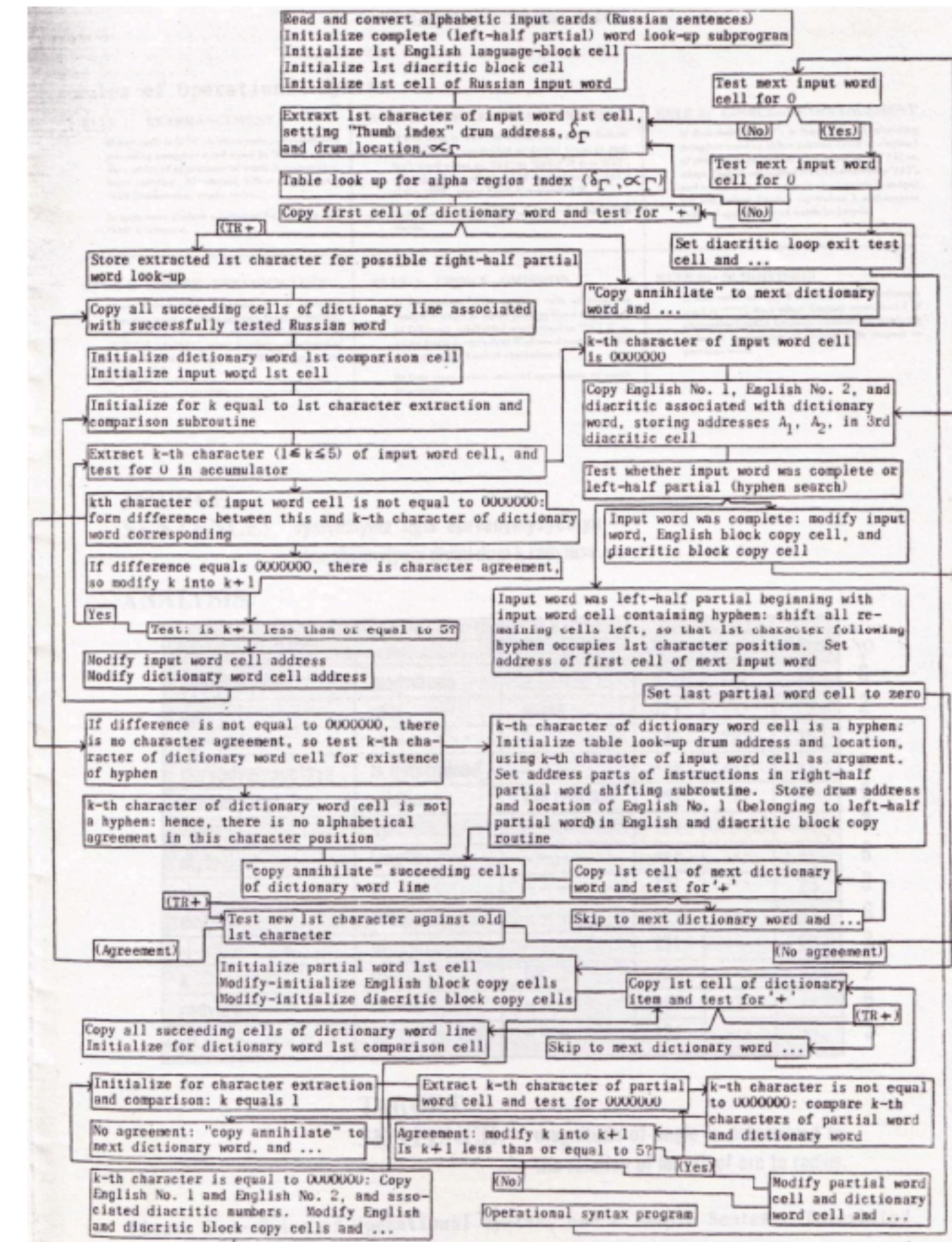


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- Systems were mostly rule-based, using a bilingual dictionary to map Russian words to their English counterparts

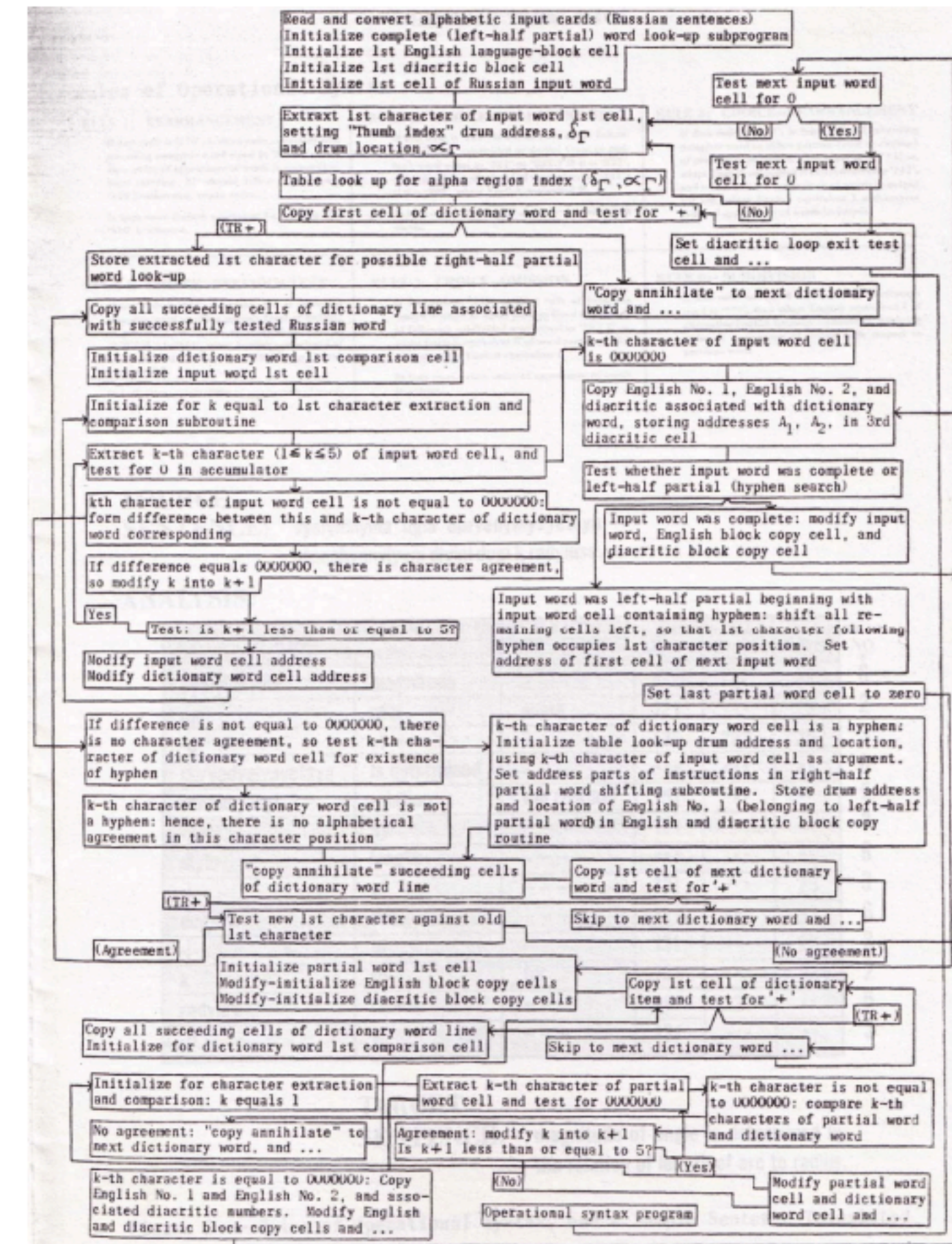


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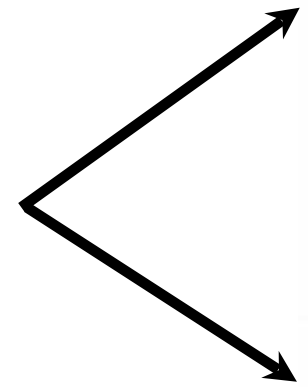
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- Core idea: Learn a probabilistic model from data
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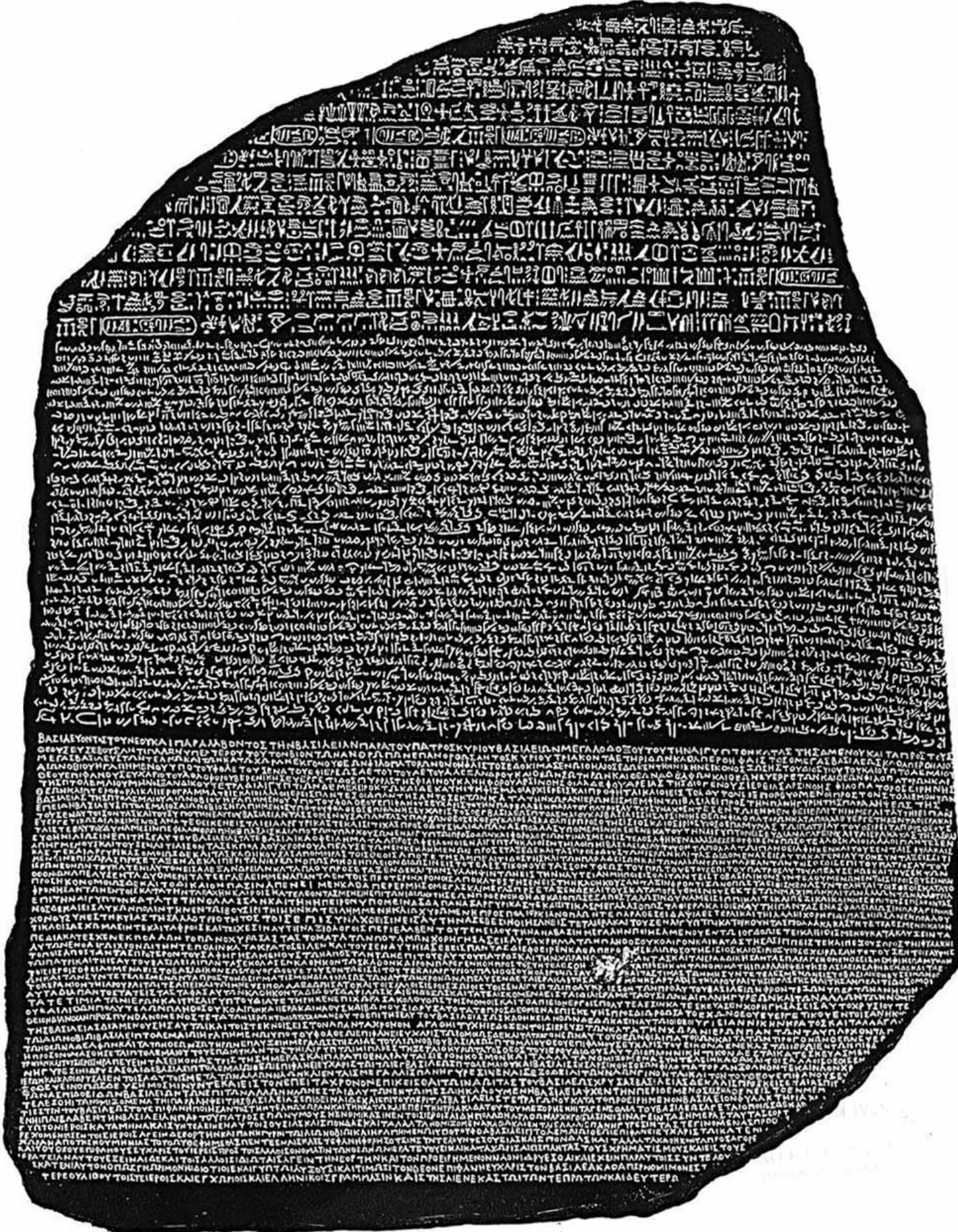
1990s-2010s: Statistical Machine Translation

- Core idea: Learn a probabilistic model from data
- Suppose we're translating French $\rightarrow$ English.
- We want to find best English sentence y , given French sentence x

Egyptian



Greek



One naturally wonders if the problem of translation could conceivably be treated as a problem in cryptography. When I look at an article in Russian, I say: *‘This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode.’*



Warren Weaver to Norbert Wiener, March, 1947

Noisy Channel MT

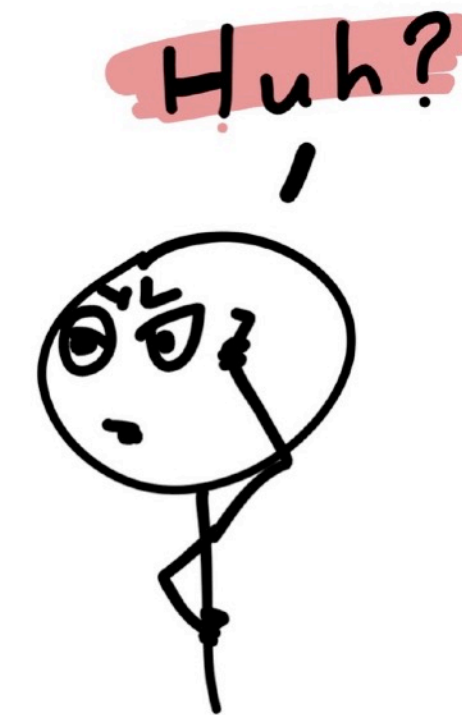
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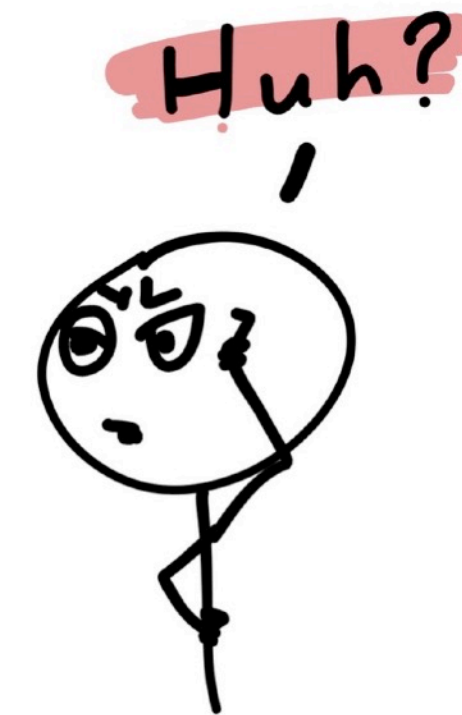


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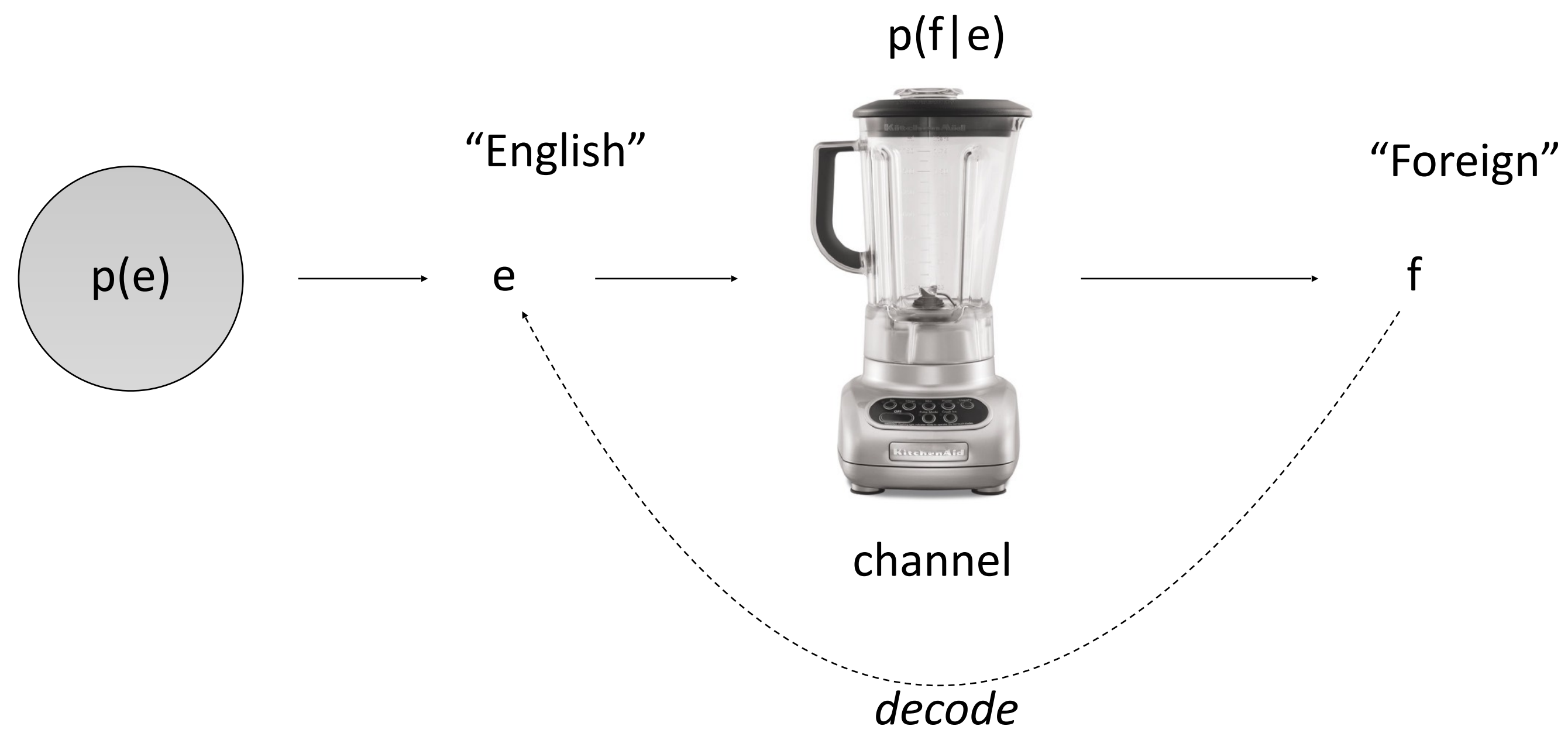
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Possible English translation

Confusing foreign sentence



Noisy Channel MT



Noisy Channel MT

$$\hat{e} = \arg \max_e p(e|f)$$

$$= \arg \max_e \frac{p(e) \times p(f|e)}{p(f)}$$

$$= \arg \max_e \boxed{p(e)} \times \boxed{p(f|e)}$$

“Language Model”

“Translation Model”

Noisy Channel Division of Labor

- Language model – $p(\mathbf{e})$
 - is the translation fluent, grammatical, and idiomatic?
 - use any model of $p(\mathbf{e})$ – typically an n -gram model
- Translation model – $p(\mathbf{f}|\mathbf{e})$
 - translation probability
 - ensures adequacy of translation

Translation Model

- $p(f|e)$ gives the channel probability – the probability of translating an English sentence into a foreign sentence
- f = je voudrais un peu de frommage
- e_1 = I would like some cheese
- e_2 = I would like a little of cheese
- e_3 = There is no train to Barcelona

$p(f|e)$

0.4

0.5

>0.00001

Translation Model

- How do we parameterize $p(\mathbf{f}|\mathbf{e})$?

$$p(f|e) = \frac{\text{count}(f, e)}{\text{count}(e)} \quad ?$$

- There are a lot of sentences: this won't generalize to new inputs

Lexical Translation

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Haus: house, home, shell, household

- Multiple translations
 - Different word senses, different registers, different inflections
 - *house, home* are common
 - *shell* is specialized (the Haus of a snail is its shell)

How common is each?

Translation	Count
house	5000
home	2000
shell	100
household	80

MLE

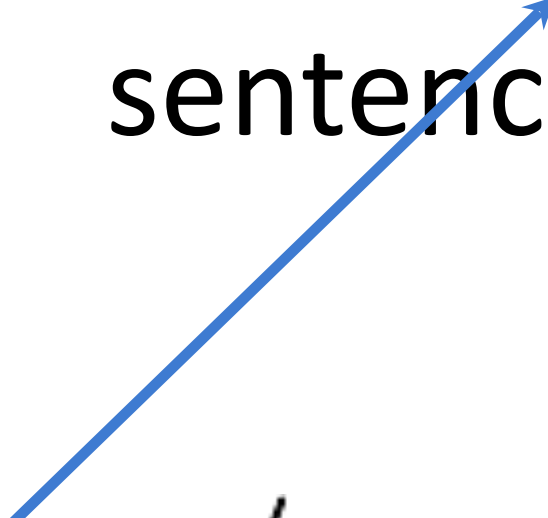
$$\hat{p}_{\text{MLE}}(e \mid \text{Haus}) = \begin{cases} 0.696 & \text{if } e = \text{house} \\ 0.279 & \text{if } e = \text{home} \\ 0.014 & \text{if } e = \text{shell} \\ 0.011 & \text{if } e = \text{household} \\ 0 & \text{otherwise} \end{cases}$$

Lexical Translation

- Goal: a model $p(\mathbf{e}|\mathbf{f},m)$
- where $\mathbf{e}$ and $\mathbf{f}$ are complete English and Foreign sentences

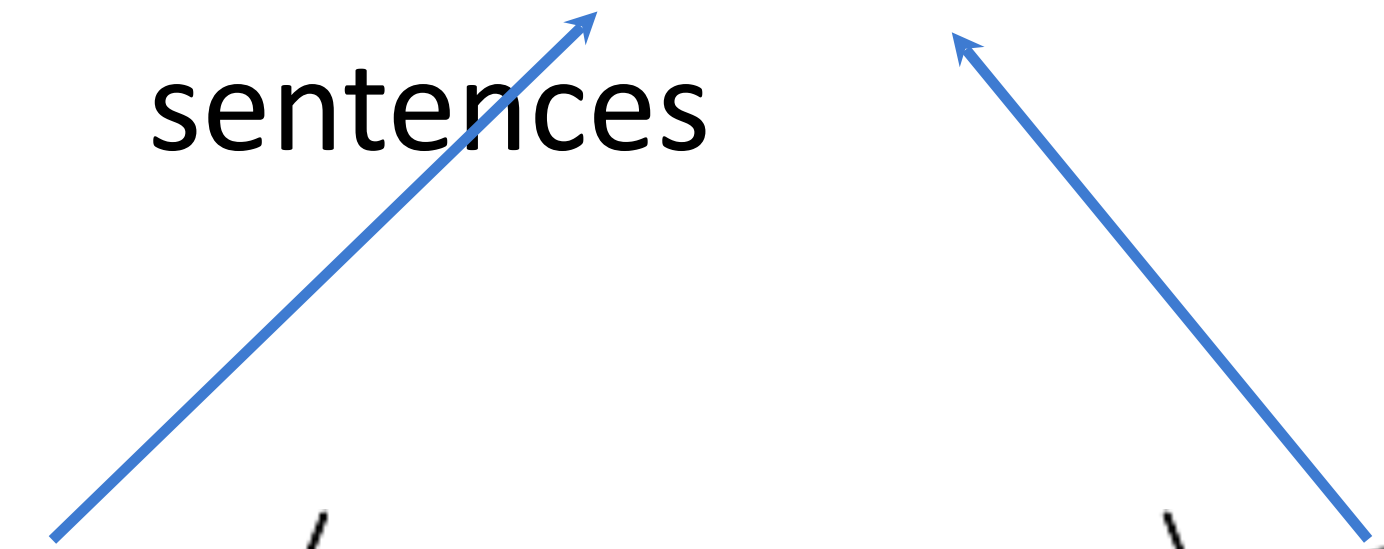
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$$\mathbf{e} = \langle e_1, e_2, \dots, e_m \rangle$$


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 - Thus, we have a latent *alignment* $\mathbf{a}_i$ that indicates which word $\mathbf{e}_i$ “came from.” Specifically it came from $\mathbf{f}_{a_i}$.
 - Given the alignments $\mathbf{a}$, translation decisions are conditionally independent of each other and depend *only* on the aligned source word $\mathbf{f}_{a_i}$.

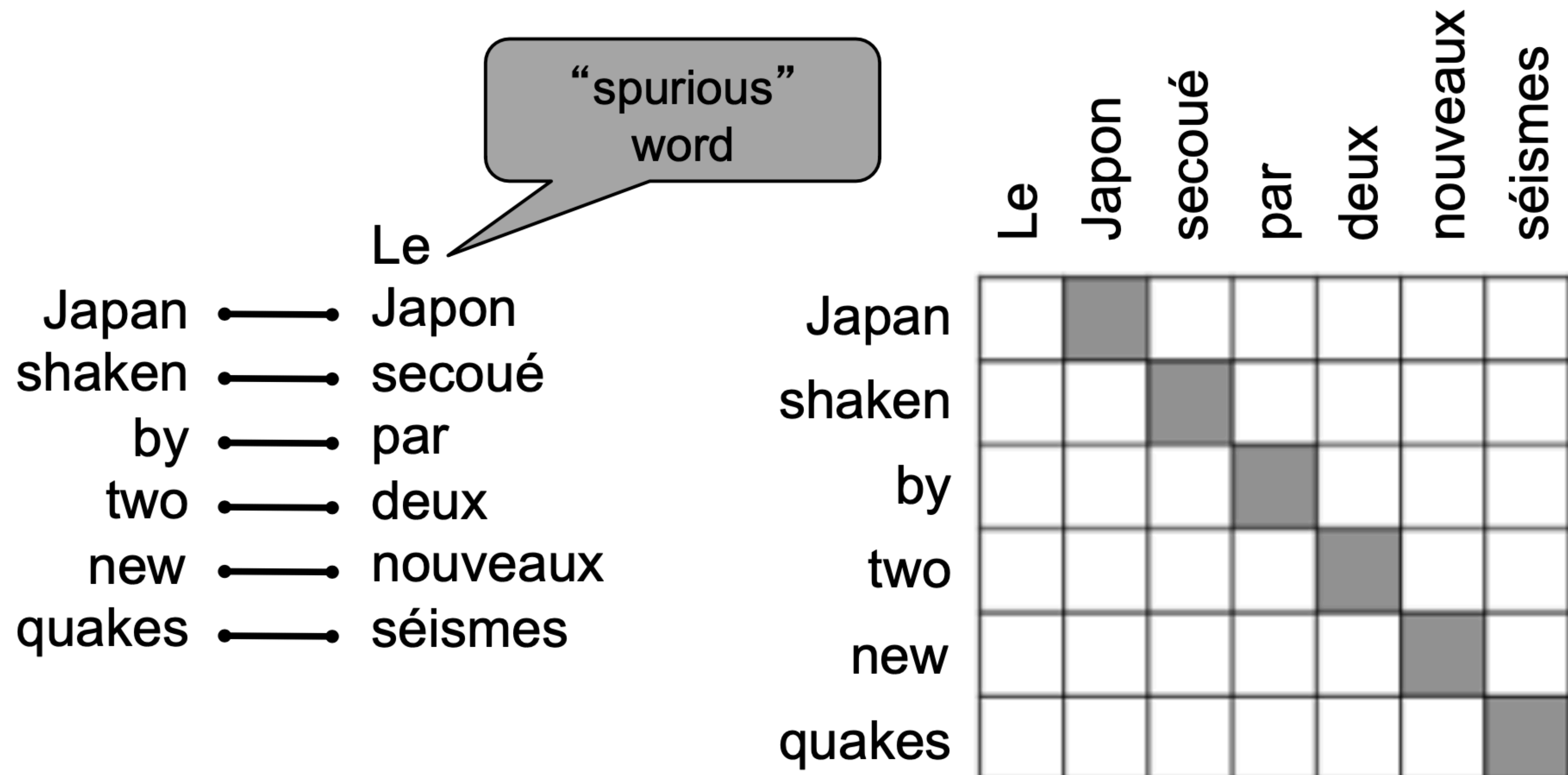
Lexical Translation

- Putting our assumptions together, we have:

$$p(\mathbf{e} \mid \mathbf{f}, m) = \underbrace{\sum_{\mathbf{a} \in [0, n]^m} p(\mathbf{a} \mid \mathbf{f}, m)}_{p(\text{Alignment})} \times \underbrace{\prod_{i=1}^m p(e_i \mid f_{a_i})}_{p(\text{Translation} \mid \text{Alignment})}$$

What is alignment?

- Alignment is the correspondence between particular words in the translated sentence pair.



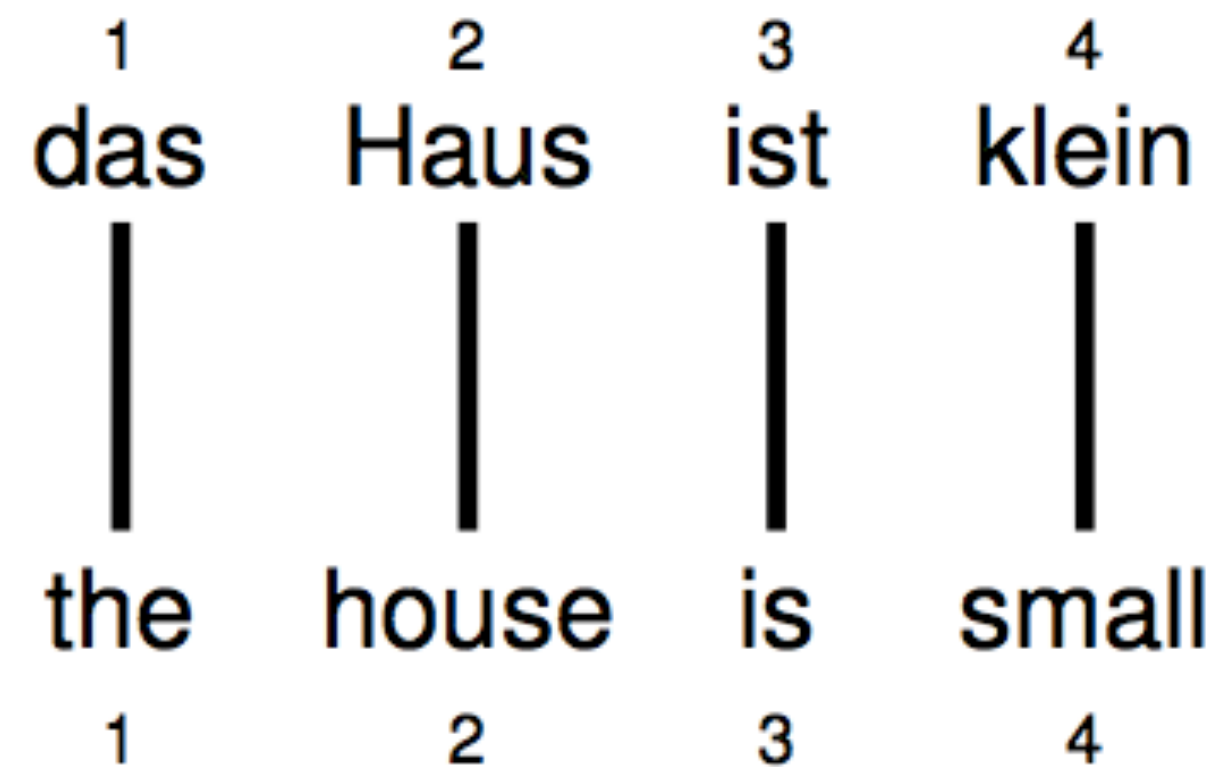
Alignment

$$p(\mathbf{a} \mid \mathbf{f}, m)$$

- Most of the action for the first 10 years of MT was here. Words weren't the problem. Word *order* was hard.

Alignment

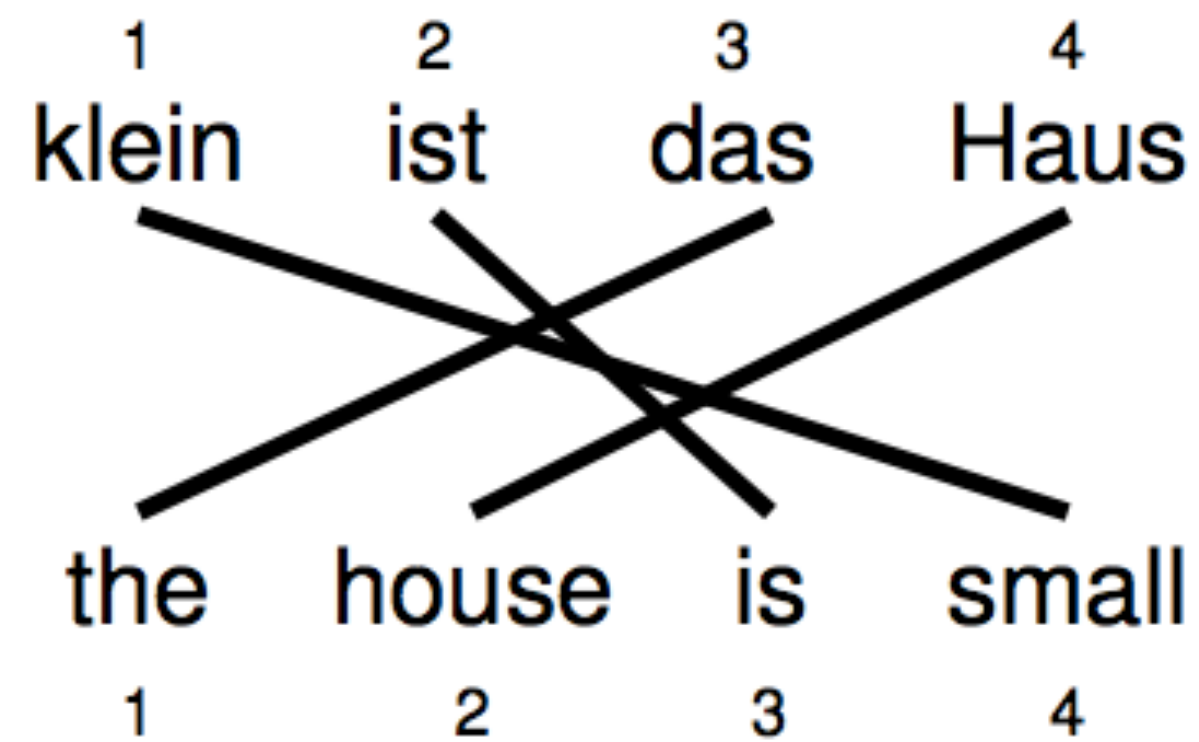
- Alignments can be visualized by drawing links between two sentences, and they are represented as vectors of positions:



$$\mathbf{a} = (1, 2, 3, 4)^{\top}$$

Reordering

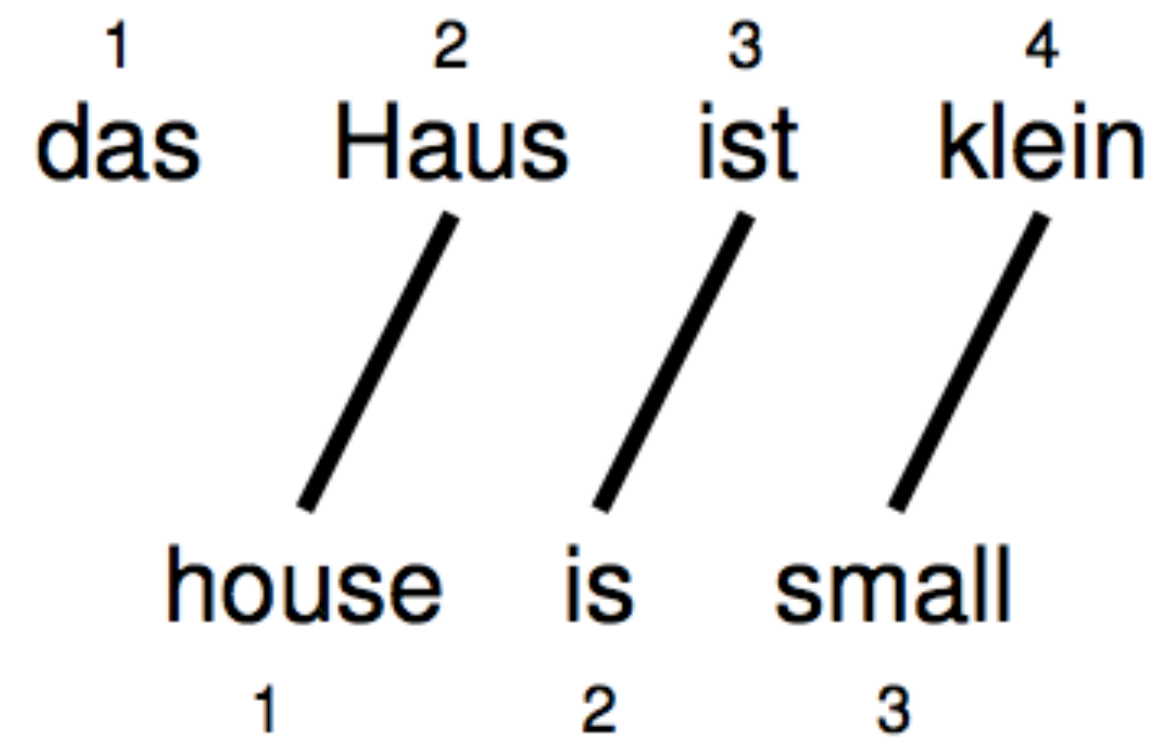
- Words may be reordered during translation



$$\mathbf{a} = (3, 4, 2, 1)^{\top}$$

Word Dropping

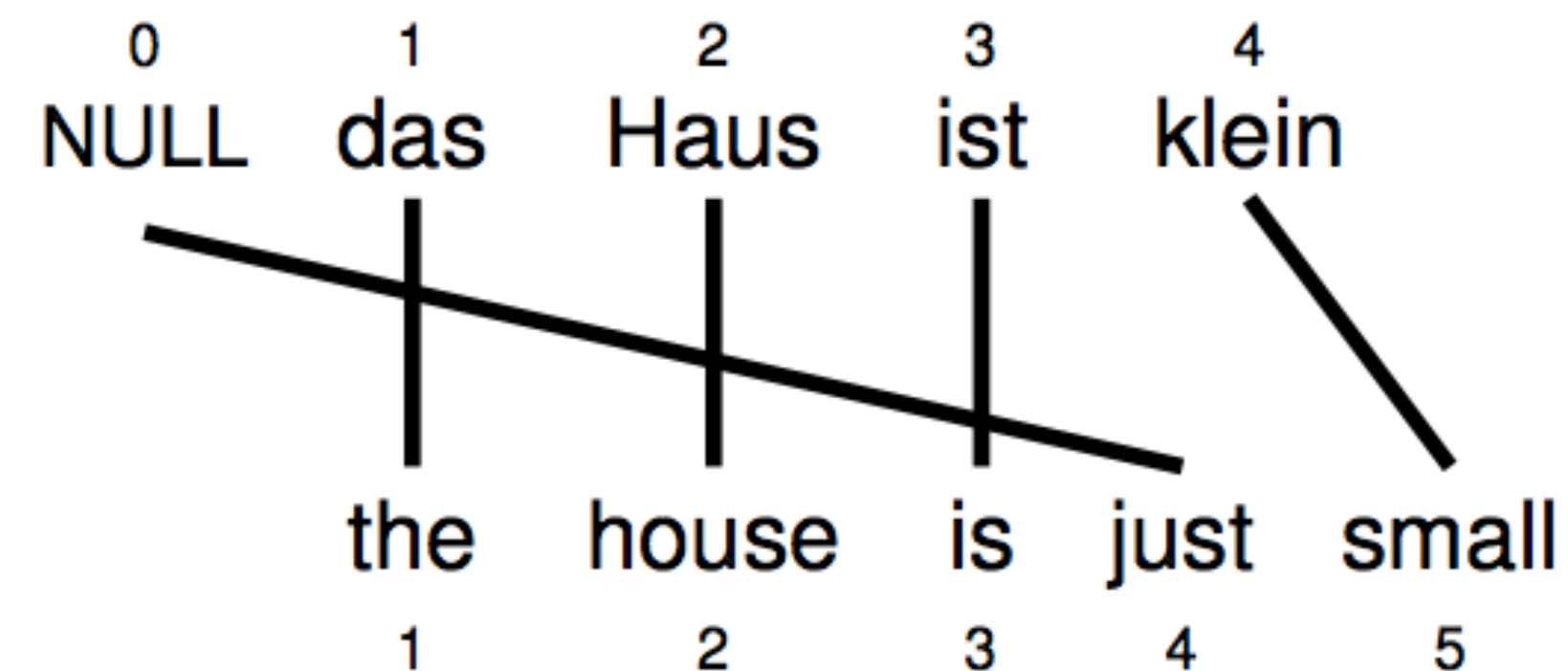
- A source word may not be translated at all



$$\mathbf{a} = (2, 3, 4)^\top$$

Word Insertion

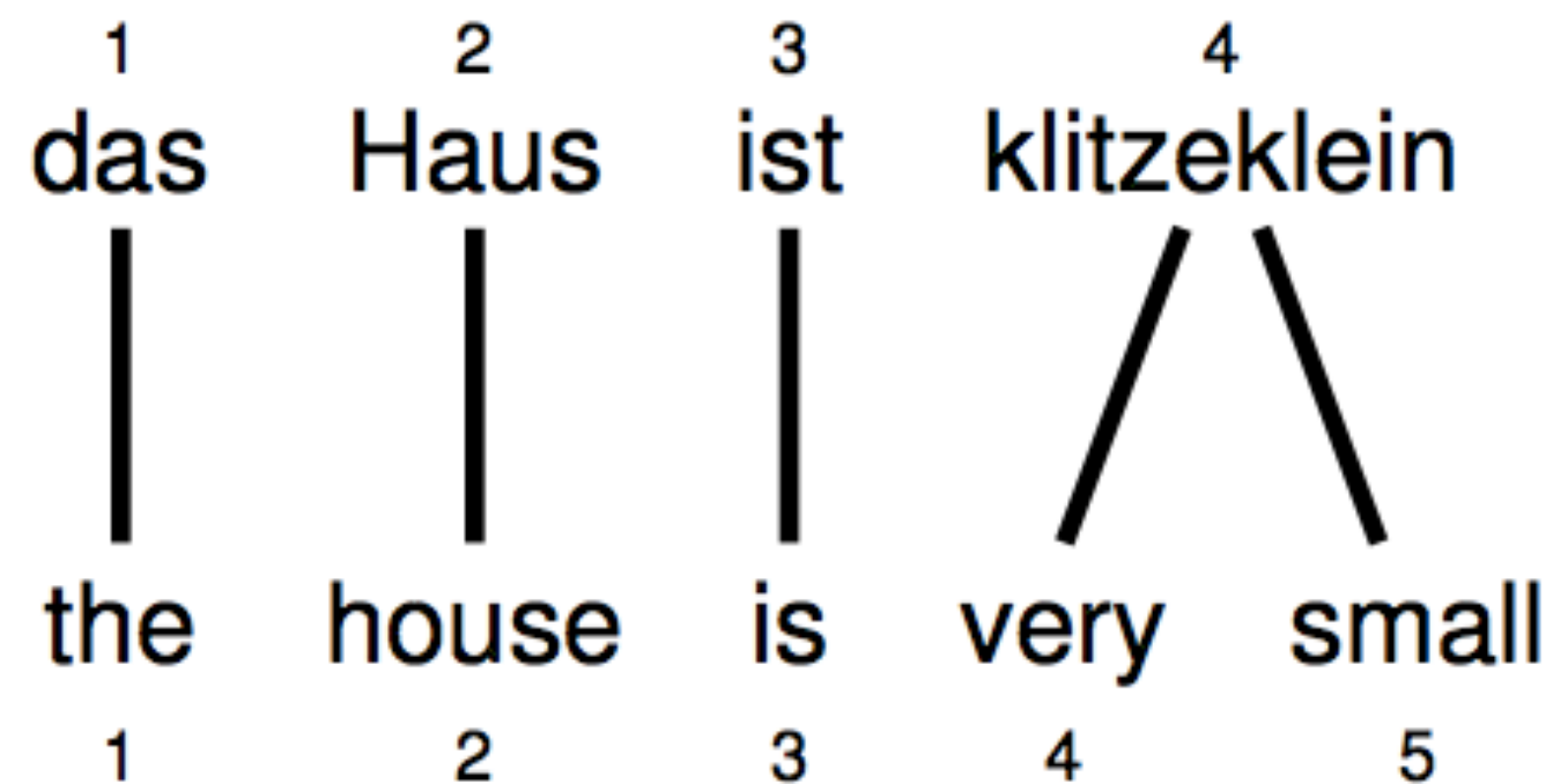
- Words may be inserted during translation
- E.g. English **just** does not have an equivalent
- But these words must be explained – we typically assume every source sentence contains a NULL token



$$\mathbf{a} = (1, 2, 3, 0, 4)^{\top}$$

One-to-many Translation

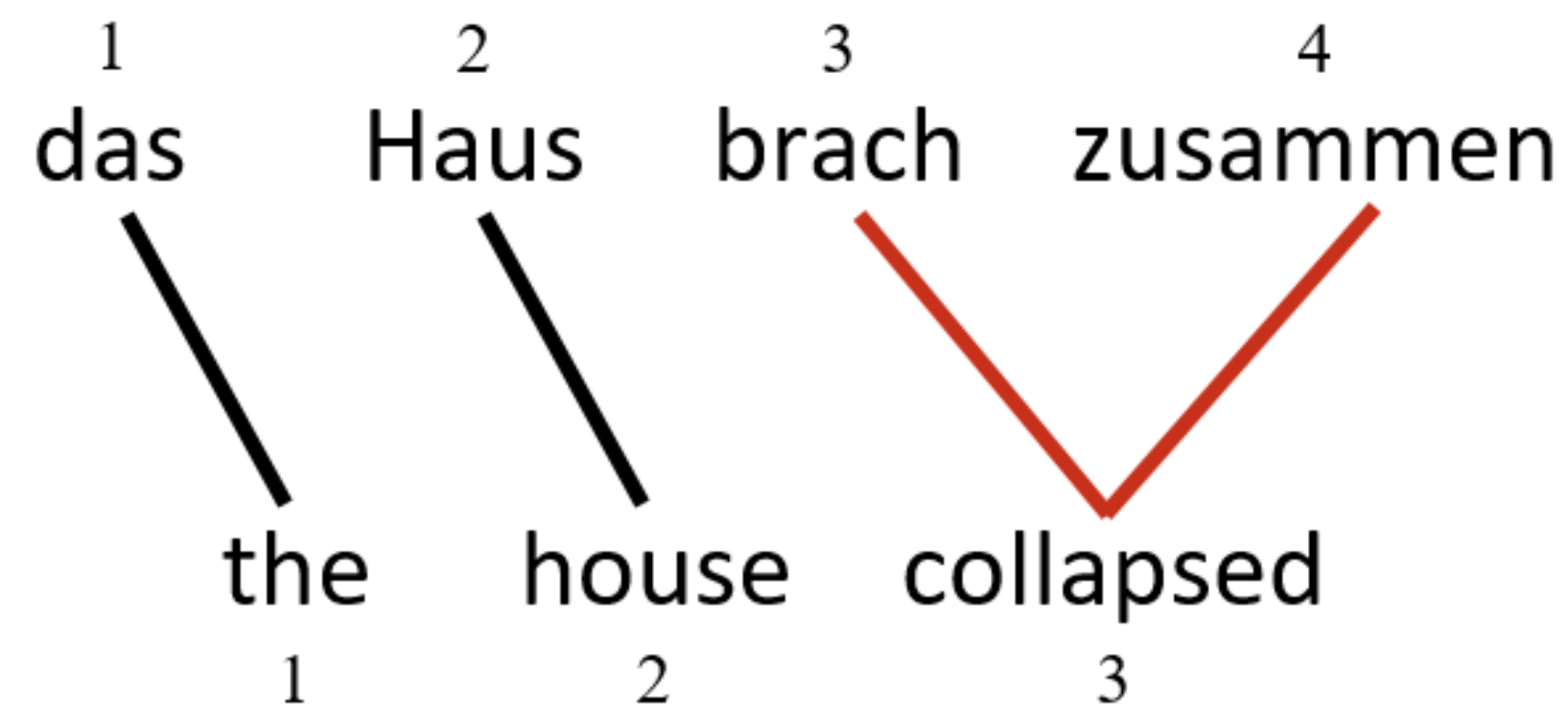
- A source word may translate into **more than one** target word



$$\mathbf{a} = (1, 2, 3, 4, 4)^{\top}$$

Many-to-one Translation

- More than one source word may **not** translate as a unit in lexical translation



$\mathbf{a} = ???$

$\mathbf{a} = (1, 2, (3, 4)^\top)^\top \quad ?$

IBM Model 1

- Simplest possible lexical translation model
- Additional assumptions:
 - The m alignment decisions are independent
 - The alignment distribution for each a_i is uniform over all source words and NULL

for each $i \in [1, 2, \dots, m]$

$$a_i \sim \text{Uniform}(0, 1, 2, \dots, n)$$

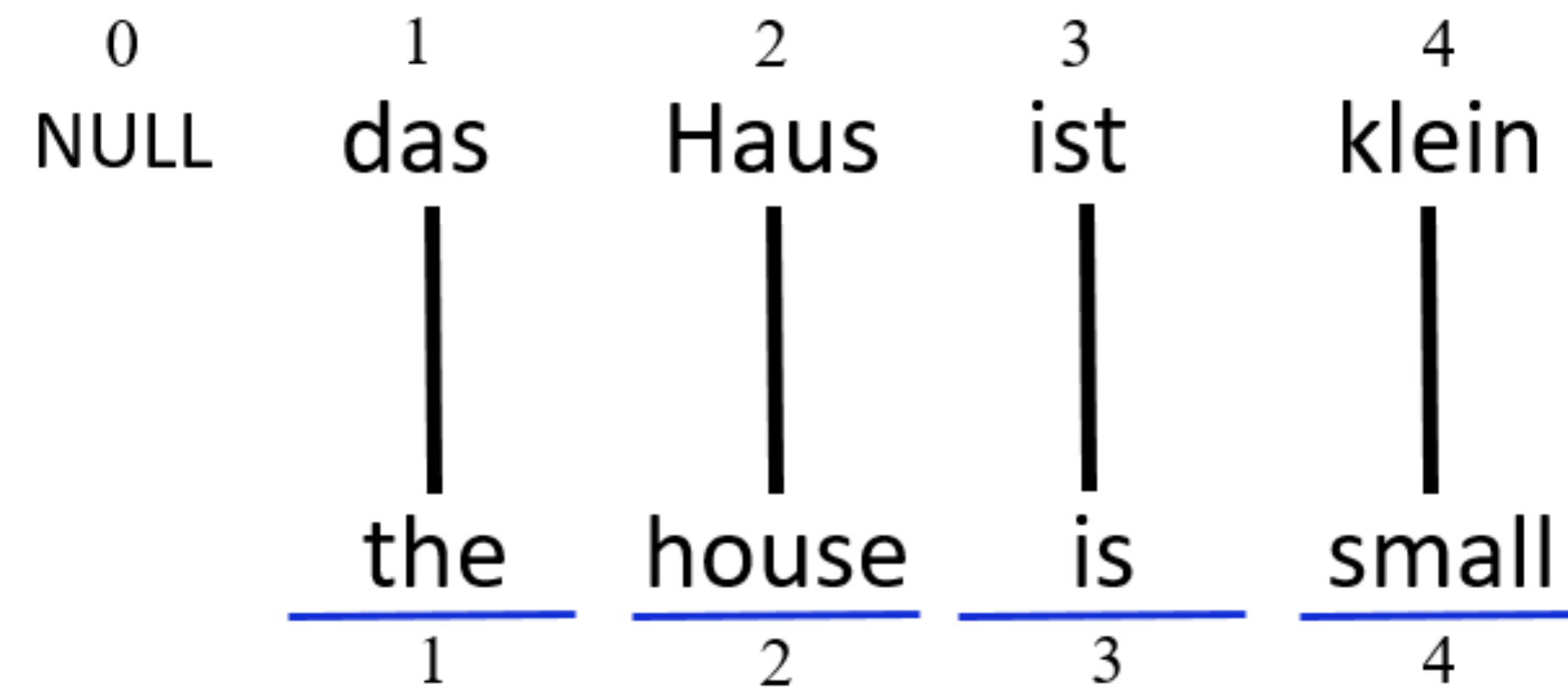
$$e_i \sim \text{Categorical}(\boldsymbol{\theta}_{f_{a_i}})$$

Translating with Model 1

0	1	2	3	4
NULL	das	Haus	ist	klein

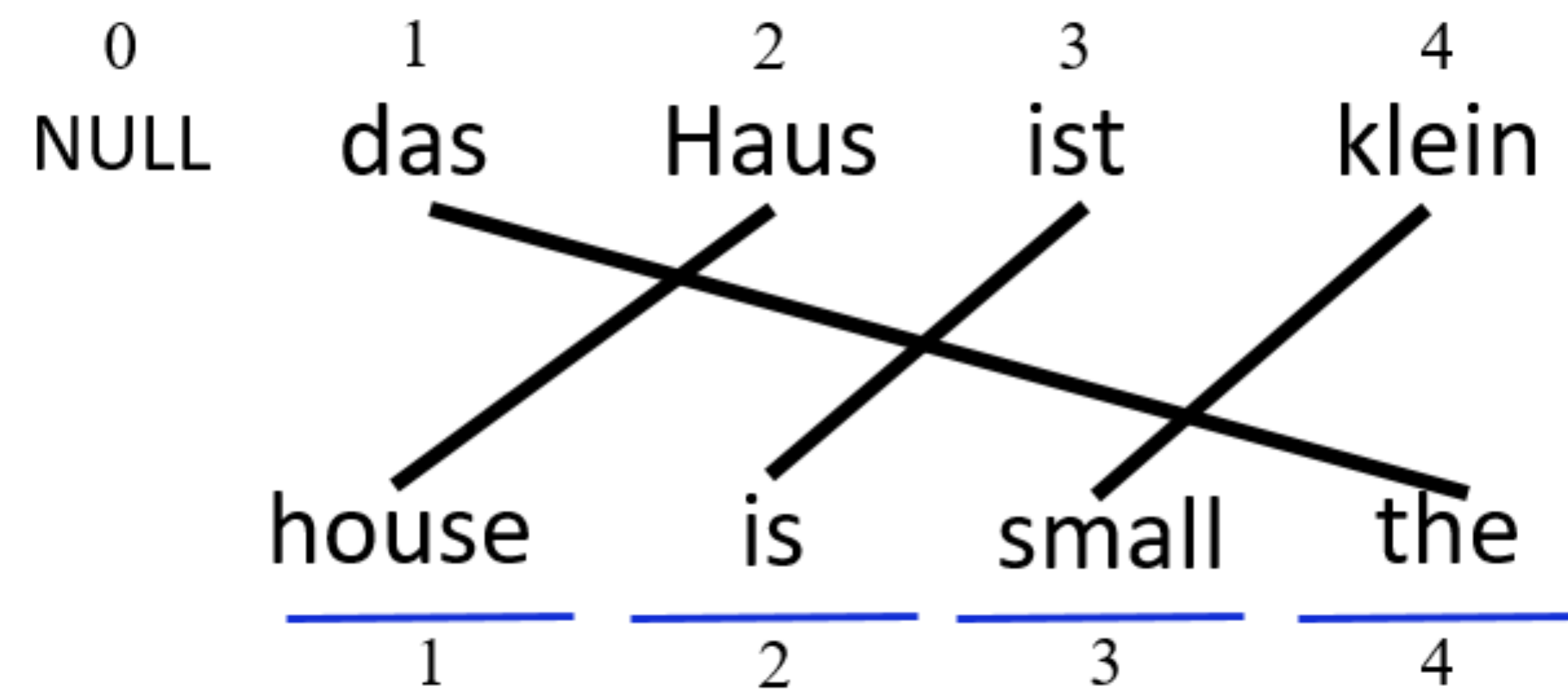
1	2	3	4
---	---	---	---

Translating with Model 1



Language model says: 😊

Translating with Model 1



Language model says: 😞

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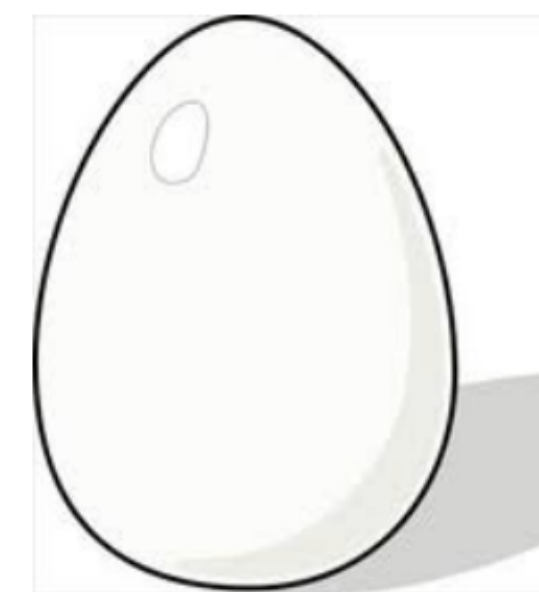
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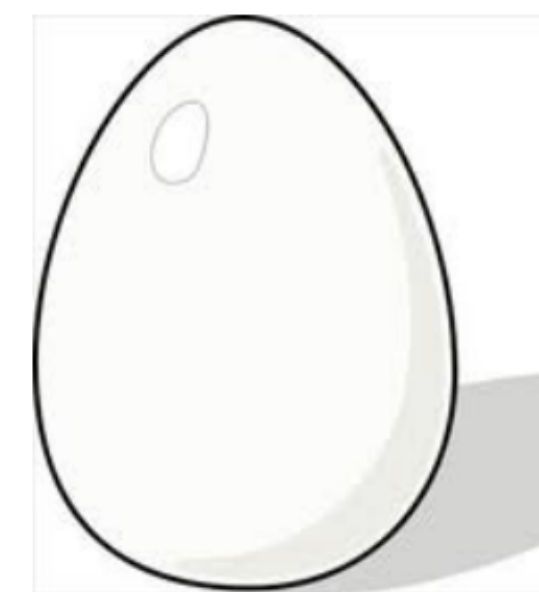
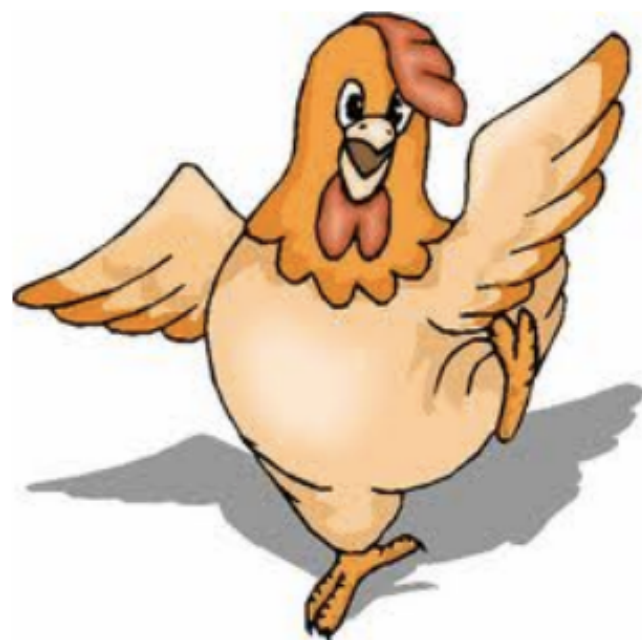
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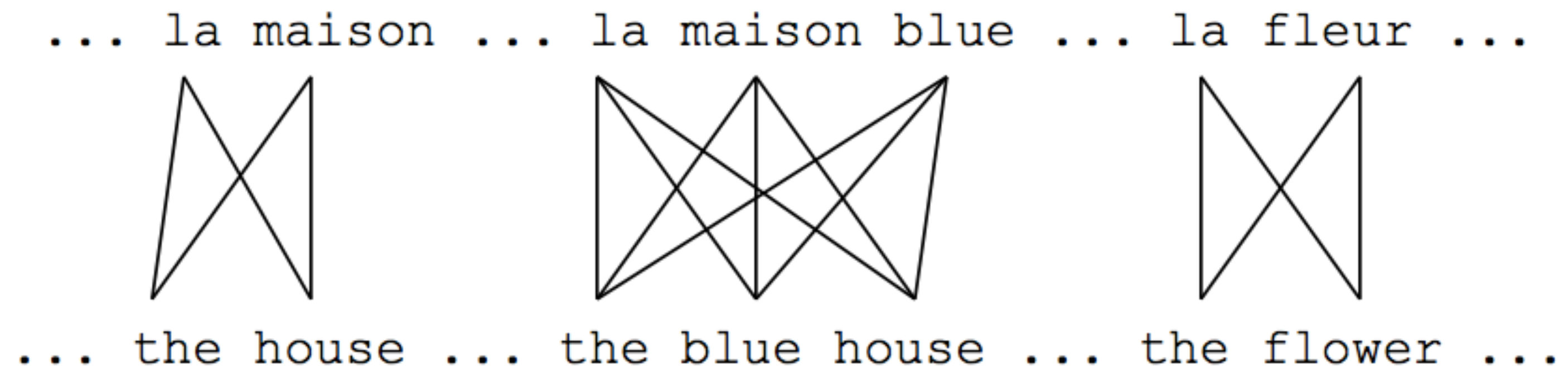
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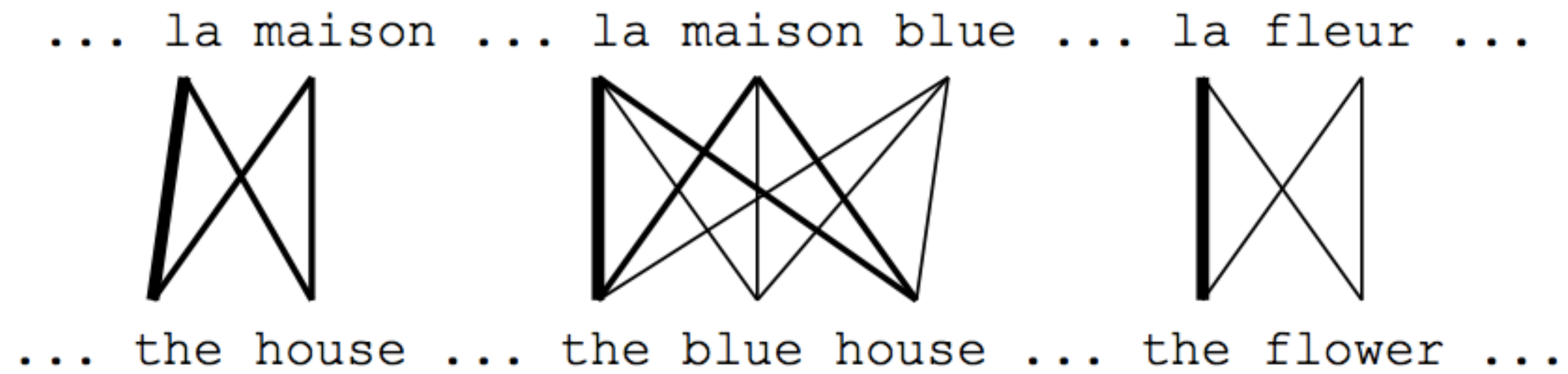
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 - Keep track of the expected number of times f translates into e throughout the whole corpus
 - Keep track of the number of times f is used in the source of any translation
 - Use these estimates in the standard MLE equation to get a better set of parameters

EM for Model 1



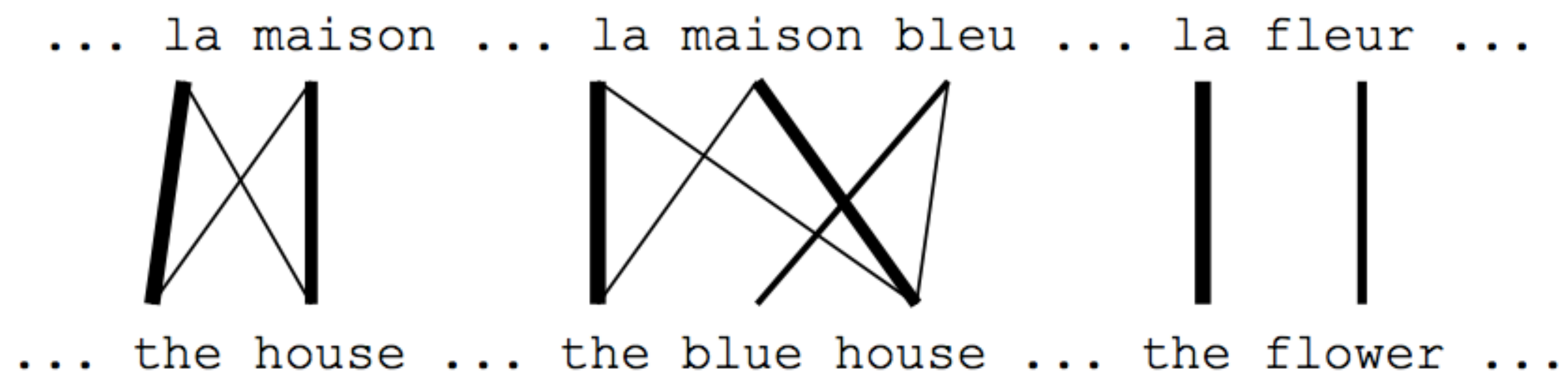
- Initial step: all alignments equally likely
- Model learns that, e.g., *la* is often aligned with *the*

EM for Model 1



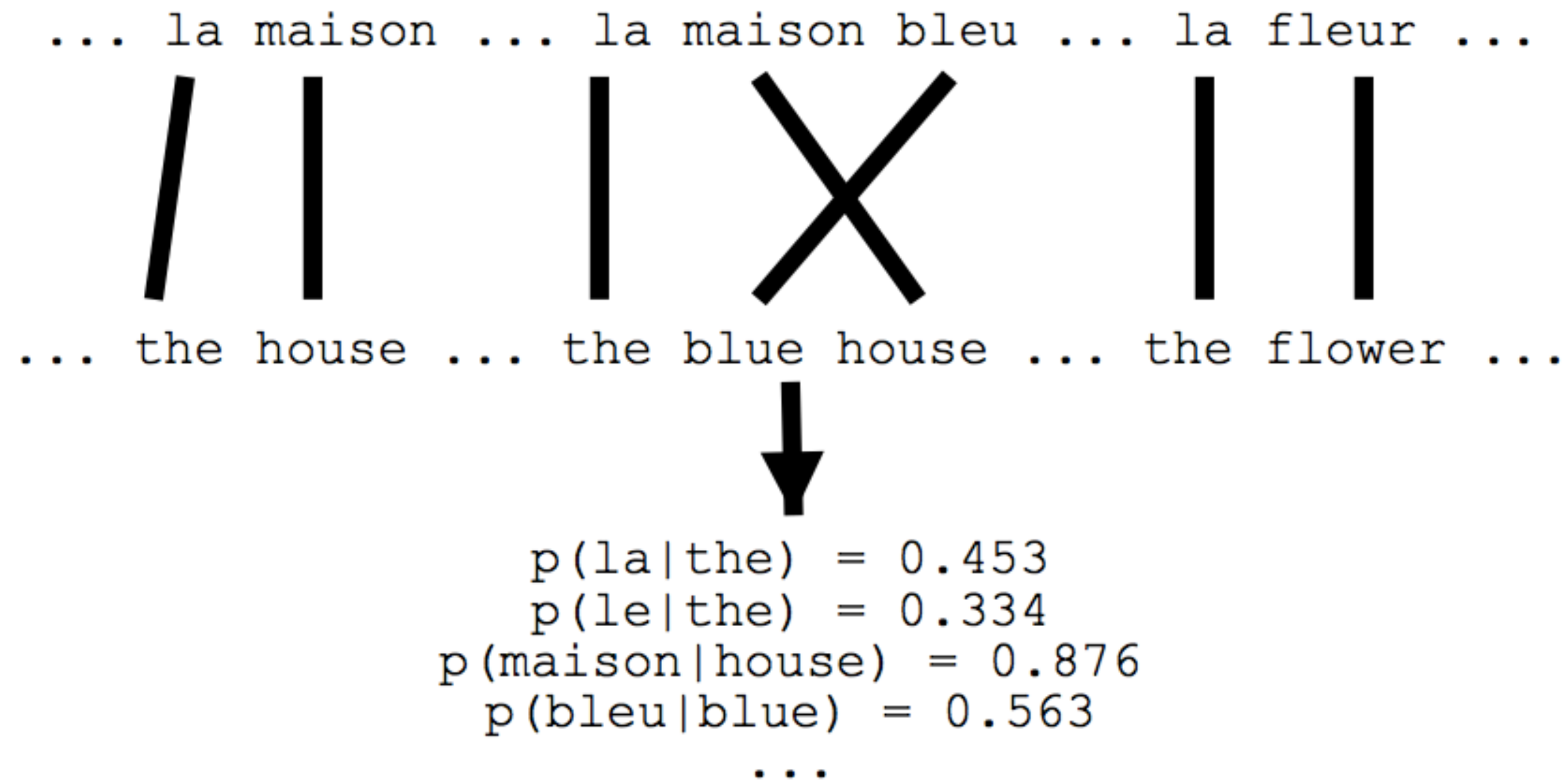
- After one iteration
- Alignments, e.g., between **la** and **the** are more likely

EM for Model 1



- After another iteration
- It becomes apparent that alignments, e.g., between fleur and flower are more likely (pigeon hole principle)

EM for Model 1



- Parameter estimation from the aligned corpus

Convergence

das Haus
the house

das Buch
the book

ein Buch
a book

<i>e</i>	<i>f</i>	initial	1st it.	2nd it.	3rd it.	...	final
the	das	0.25	0.5	0.6364	0.7479	...	1
book	das	0.25	0.25	0.1818	0.1208	...	0
house	das	0.25	0.25	0.1818	0.1313	...	0
the	buch	0.25	0.25	0.1818	0.1208	...	0
book	buch	0.25	0.5	0.6364	0.7479	...	1
a	buch	0.25	0.25	0.1818	0.1313	...	0
book	ein	0.25	0.5	0.4286	0.3466	...	0
a	ein	0.25	0.5	0.5714	0.6534	...	1
the	haus	0.25	0.5	0.4286	0.3466	...	0
house	haus	0.25	0.5	0.5714	0.6534	...	1

From words to phrases

Word Alignment

	michael	geht	davon	aus	,	dass	er	im	haus	bleibt
michael										
assumes										
that										
he										
will										
stay										
in										
the										
house										

Extracting Phrase Pairs

	michael	geht	davon	aus	,	dass	er	im	haus	bleibt
michael										
assumes										
that										
he										
will										
stay										
in										
the										
house										

extract phrase pair consistent with word alignment:

assumes that / geht davon aus , dass

Phrase Pair Extraction

	michael	geht	davon	aus	,	dass	er	im	haus	bleibt
michael										
assumes										
that										
he										
will										
stay										
in										
the										
house										

Smallest phrase pairs:

- michael — michael
- assumes — geht davon aus / geht davon aus ,
- that — dass / , dass
- he — er
- will stay — bleibt
- in the — im
- house — haus

unaligned words (here: German comma) lead to multiple translations

Larger Phrase Pairs

	michael	geht	davon	aus	,	dass	er	im	haus	bleibt
michael										
assumes										
that										
he										
will										
stay										
in										
the										
house										

michael assumes — michael geht davon aus / michael geht davon aus ,
assumes that — geht davon aus , dass ; assumes that he — geht davon aus , dass er
that he — dass er / , dass er ; in the house — im haus
michael assumes that — michael geht davon aus , dass
michael assumes that he — michael geht davon aus , dass er
michael assumes that he will stay in the house — michael geht davon aus , dass er im haus bleibt
assumes that he will stay in the house — geht davon aus , dass er im haus bleibt
that he will stay in the house — dass er im haus bleibt ; dass er im haus bleibt ,
he will stay in the house — er im haus bleibt ; will stay in the house — im haus bleibt

Extensions

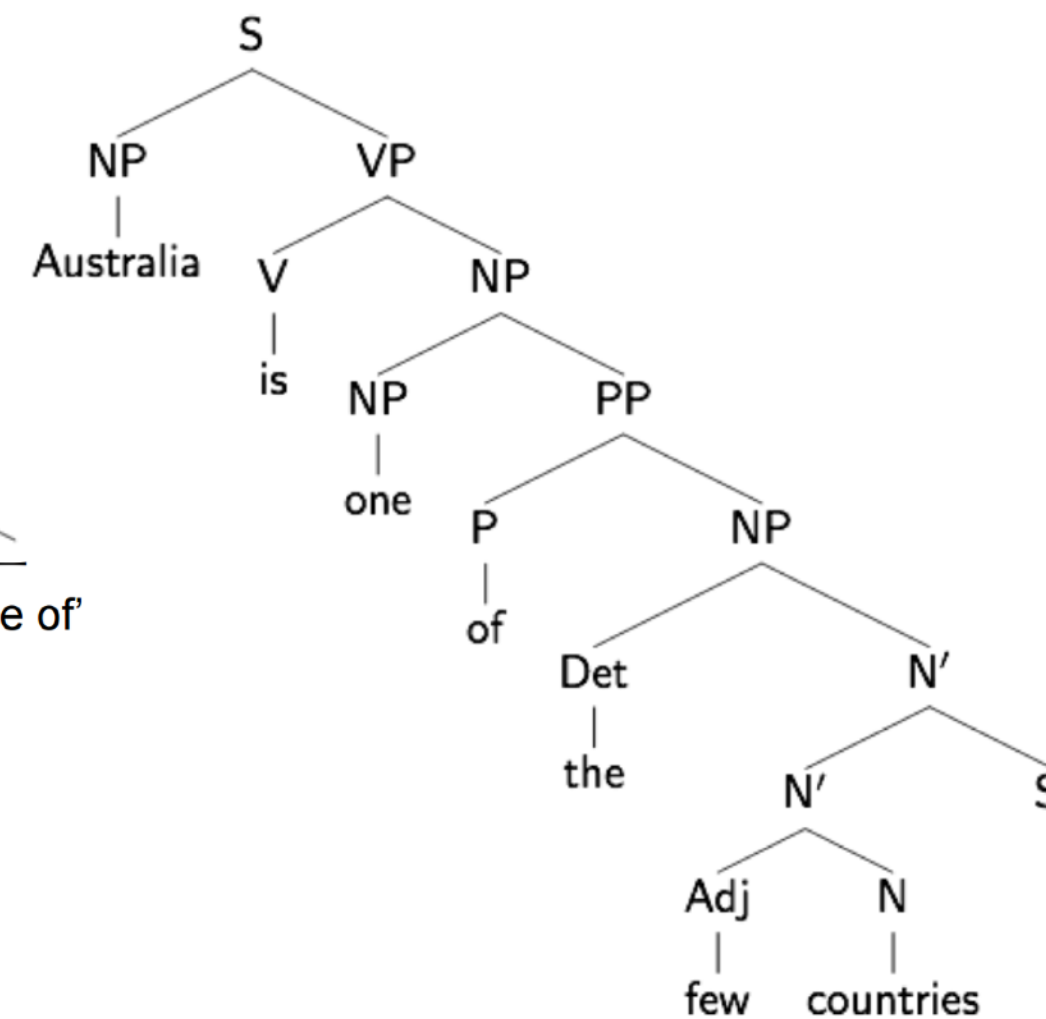
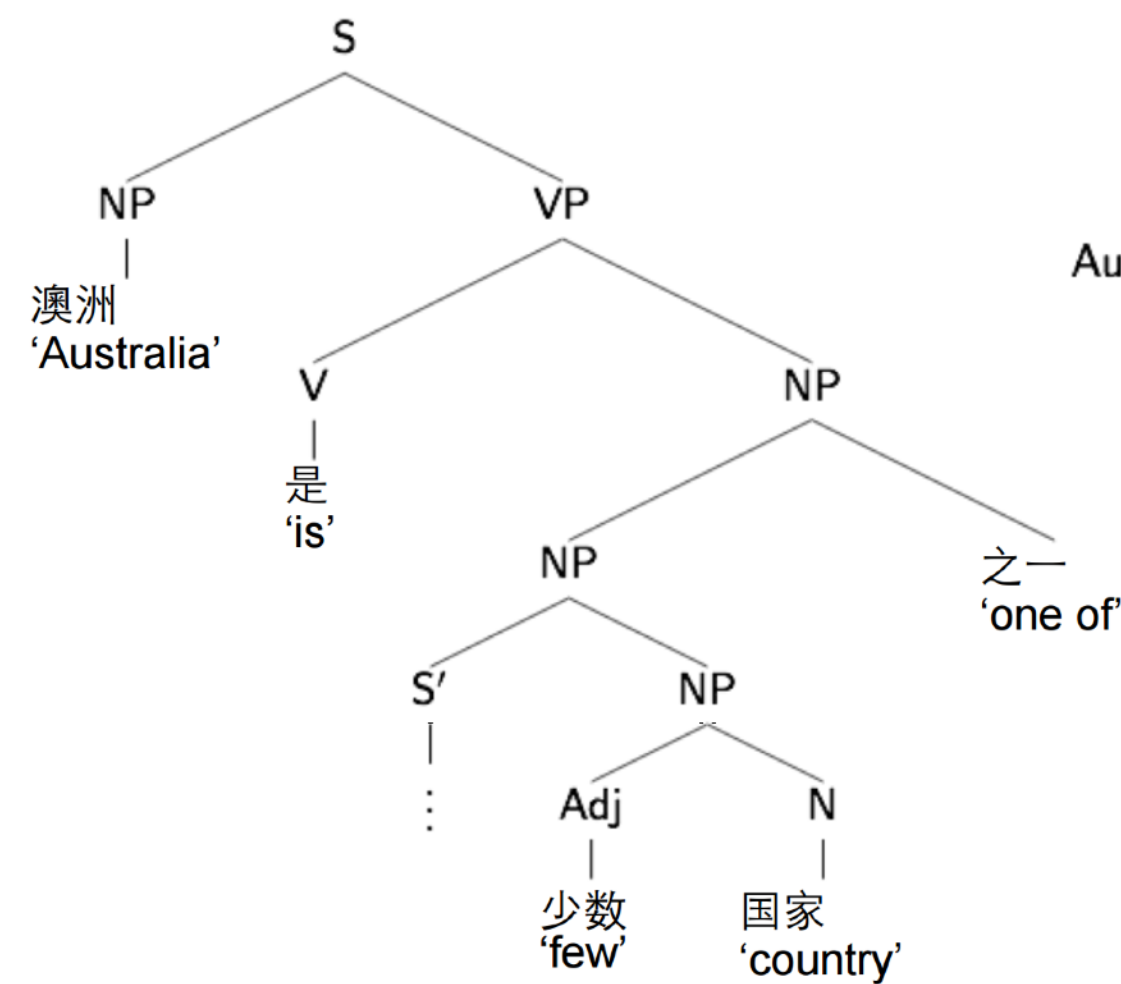
- Phrase-based MT:
 - Allow multiple words to translate as chunks (including many-to-one)
 - Introduce another latent variable, the source *segmentation*

Maria	no	dio	una	bofetada	a	la	bruja	verde
Mary	not	give	a	slap	to	the	witch	green
	did not		a slap	by		hag	bawdy	
	no	slap		to the		green witch		
	did not give			the				
				the witch				

Adapted from Koehn (2006)

Another Paradigm: Syntax-Based MT

- Syntactic structure
- Rules of the form:
- $X\text{之一} \rightarrow \text{one of the } X$



Chang (2005),
Galley et al. (2006)

2014

(dramatic reenactment)

2014

Neural Machine Translation

MT research

(dramatic reenactment)

What is Neural Machine Translation?

- Neural Machine Translation (NMT) is a way to do Machine Translation with a single neural network
- The neural network architecture is called sequence-to-sequence (aka seq2seq) and it involves two RNNs.

Conditional Language Models

$$P(Y|X) = \prod_{j=1}^J P(y_j \mid X, y_1, \dots, y_{j-1})$$

Conditional Language Models

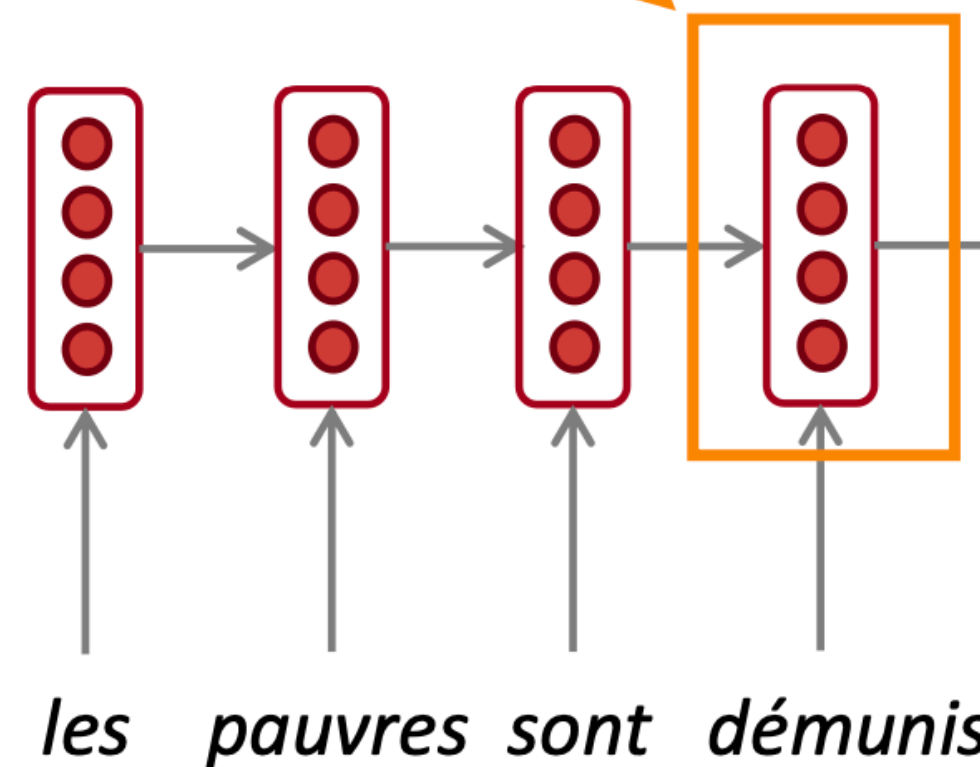
$$P(Y|X) = \prod_{j=1}^J P(y_j \mid \overbrace{X, y_1, \dots, y_{j-1}}^{\text{Added Context!}})$$

Neural Machine Translation (NMT)

The sequence-to-sequence model

Encoding of the source sentence.
Provides initial hidden state
for Decoder RNN.

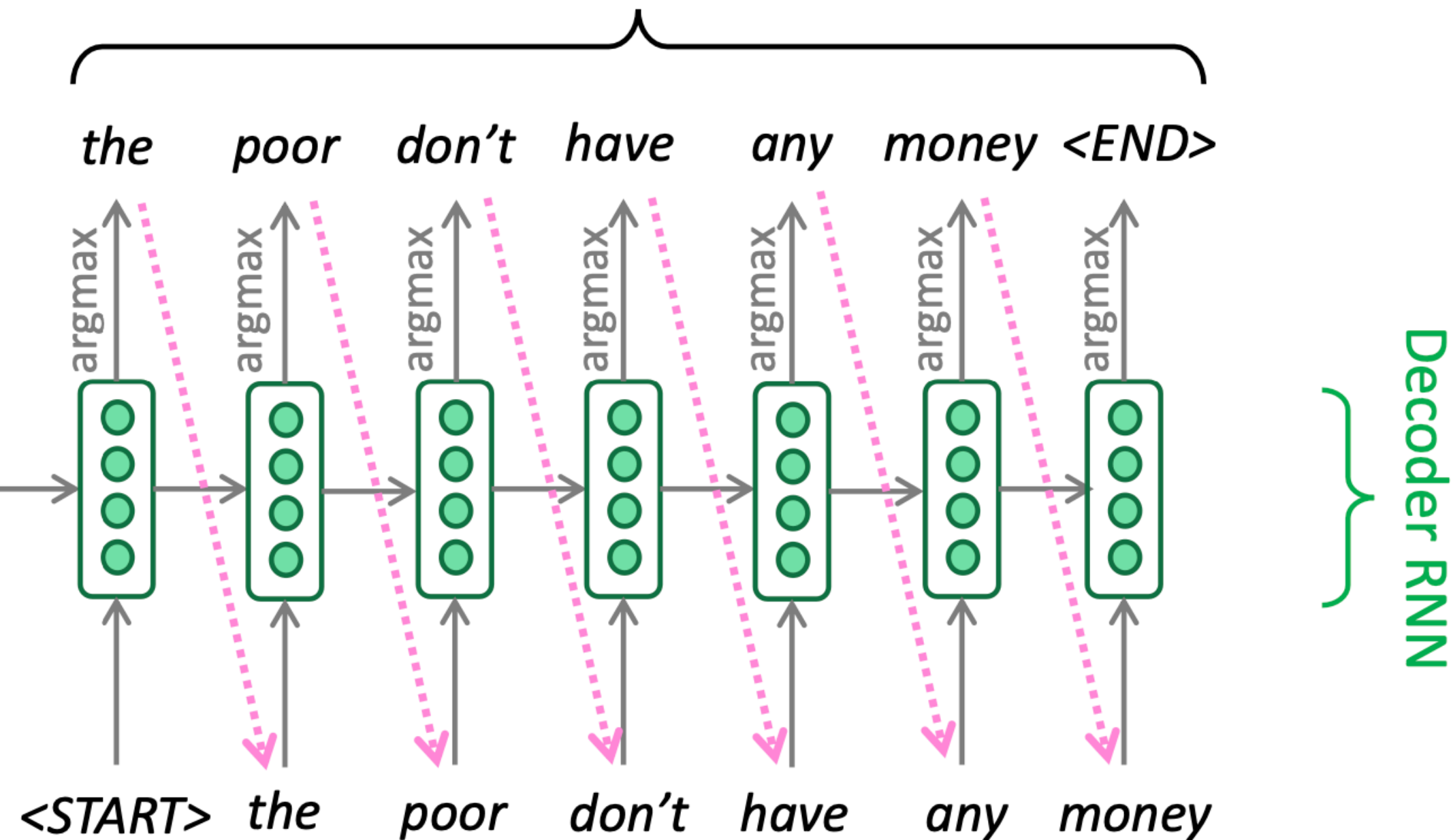
Encoder RNN



Source sentence (input)

Encoder RNN produces
an **encoding** of the
source sentence.

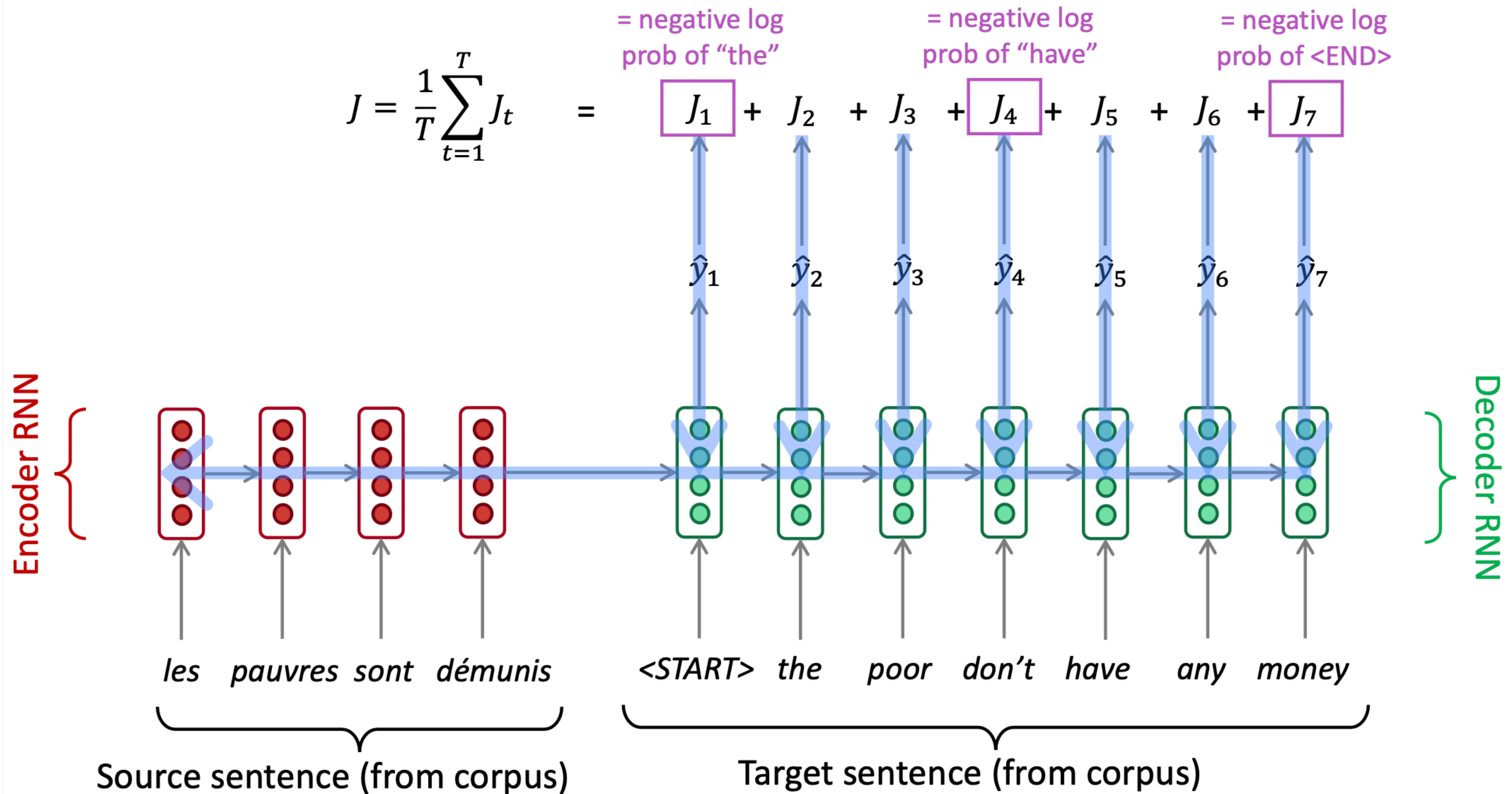
Target sentence (output)



Decoder RNN is a Language Model that generates
target sentence conditioned on **encoding**.

Note: This diagram shows **test time** behavior:
decoder output is fed in➤ as next step's input

Neural Machine Translation (NMT)



Seq2seq is optimized as a single system.
Backpropagation operates "*end to end*".

Advantages of NMT

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- Compared to SMT, NMT has many advantages:

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- Better performance
 - More fluent
 - Better use of context
 - Better use of phrase similarities
- A single neural network to be optimized end-to-end
 - No subcomponents to be individually optimized
- Requires much less human engineering effort
 - No feature engineering
 - Same method for all language pairs

Disadvantages of NMT?

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Compared to SMT:

Disadvantages of NMT?

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- NMT is less interpretable

Disadvantages of NMT?

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 - Hard to debug

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- NMT is difficult to control
 - For example, can't easily specify rules or guidelines for translation

Disadvantages of NMT?

Compared to SMT:

- NMT is less interpretable
 - Hard to debug
- NMT is difficult to control
 - For example, can't easily specify rules or guidelines for translation
 - Safety concerns!

Generation

Can we find the best (most likely) translation?

Generation through Sampling

No but we can approximate it!

Generating New Sentences

Generating New Sentences

- Generate sentences:

while didn't choose end-of-sentence symbol:

calculate probability of $P(x_t | x_1, \dots, x_{t-1})$

sample a new word from the probability
distribution

Greedy Decoding

- Generate next word conditioned on the context (i.e., the previously generated words)
- “Greedy”: always pick the most probable next word
$$x_t = \operatorname{argmax}_{\hat{x}} P(\hat{x} | x_1, \dots, x_{t-1})$$

Greedy Decoding

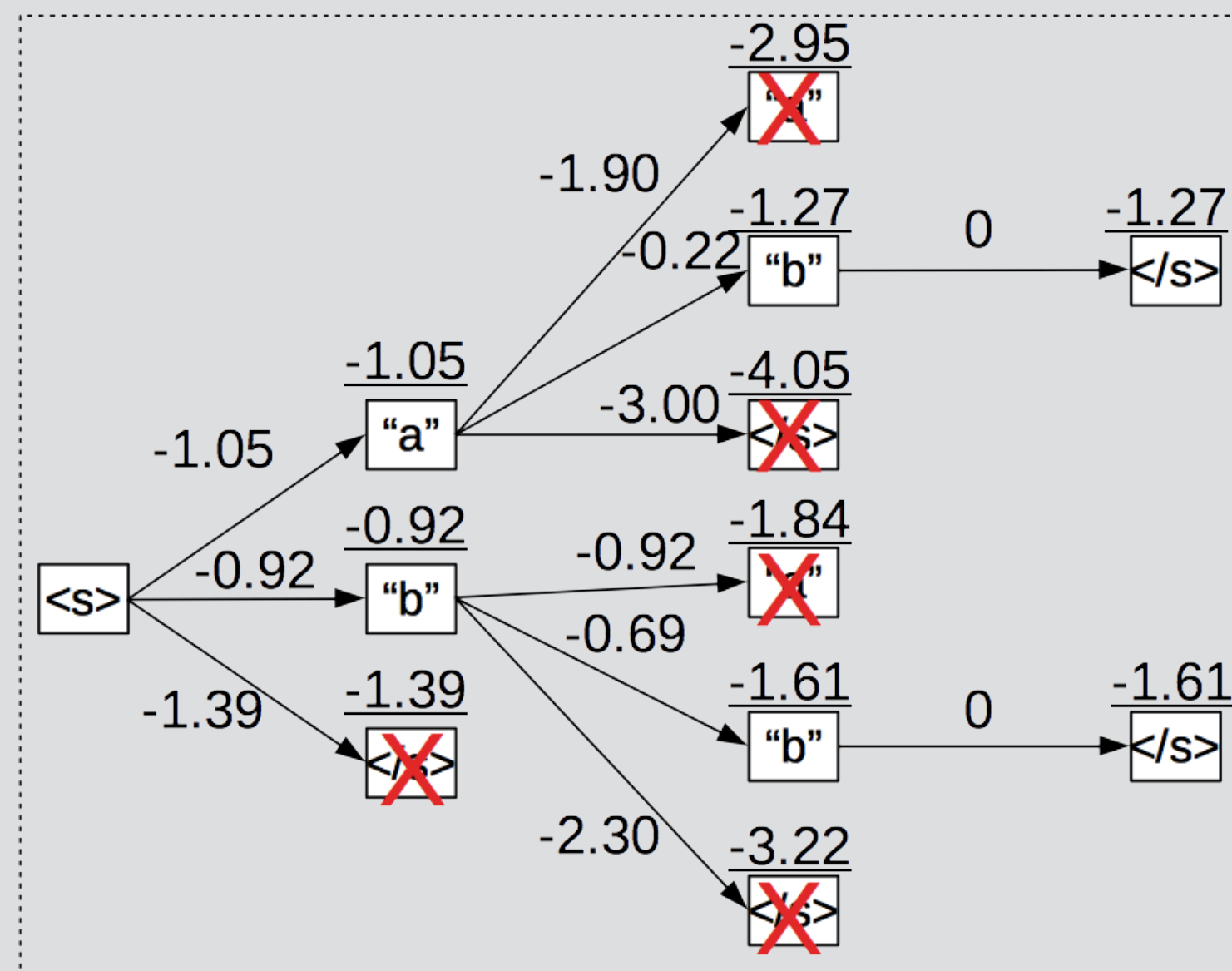
- Generate next word conditioned on the context (i.e., the previously generated words)
- “Greedy”: always pick the most probable next word
$$x_t = \operatorname{argmax}_{\hat{x}} P(\hat{x} | x_1, \dots, x_{t-1})$$
- Problem:
 - The most probable next word does not always lead to the most probable sentence;
 - We should be able to generate a diverse set of sentences!

Beam Search

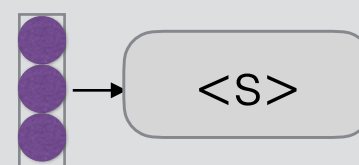
- Beam search: instead of picking one high-probability word, maintain several paths

Beam Search

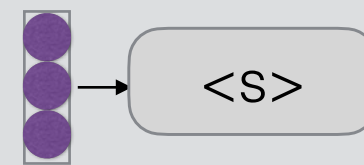
- Beam search: instead of picking one high-probability word, maintain several paths



Beam Search



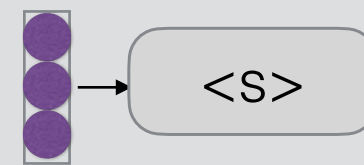
Beam Search



a	0.001
the	0.0002
I	0.12
you	0.04
cat	0.0004
movie	0.01
this	0.02
...	

Beam Search

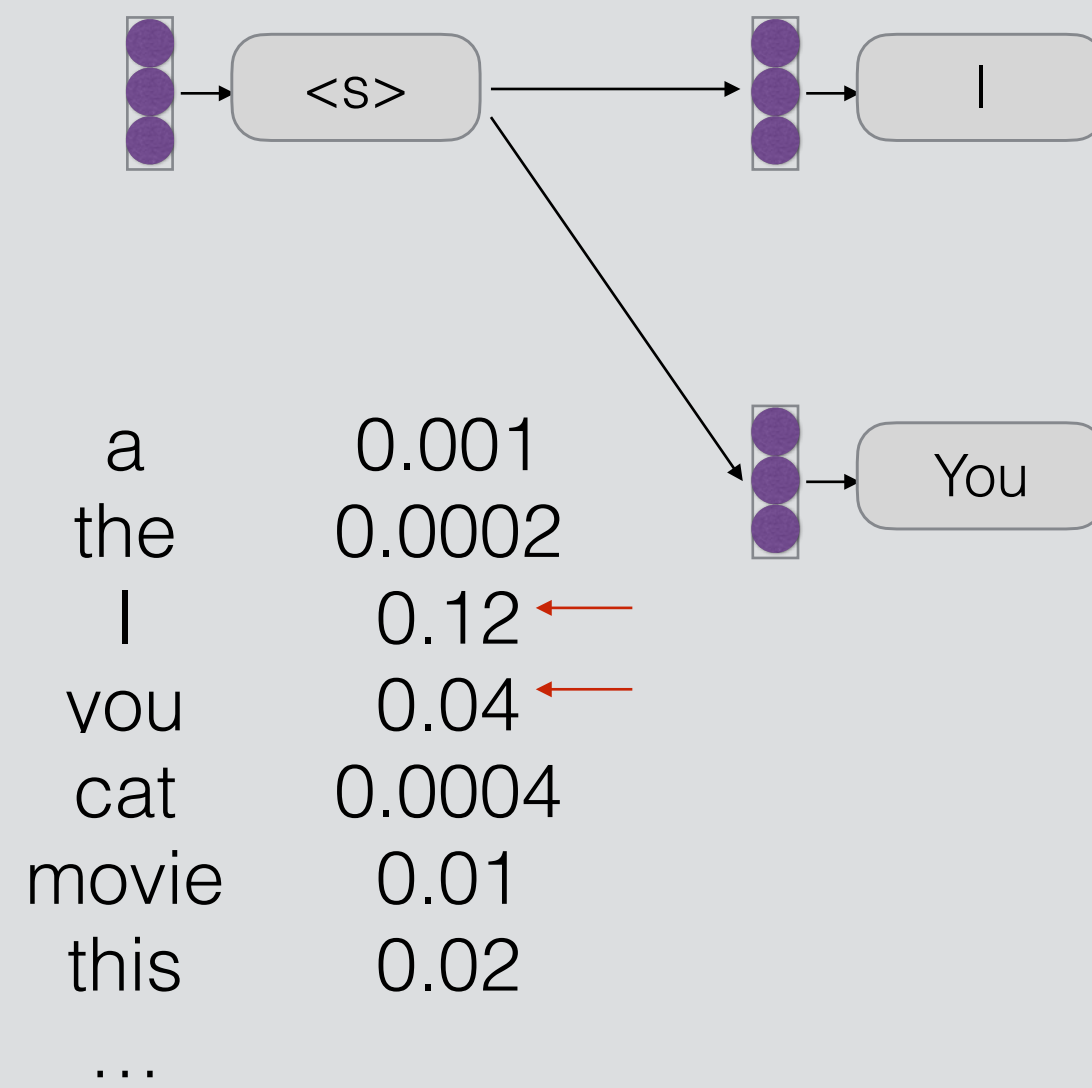
$k=2$



a	0.001
the	0.0002
I	0.12 ←
you	0.04 ←
cat	0.0004
movie	0.01
this	0.02
...	

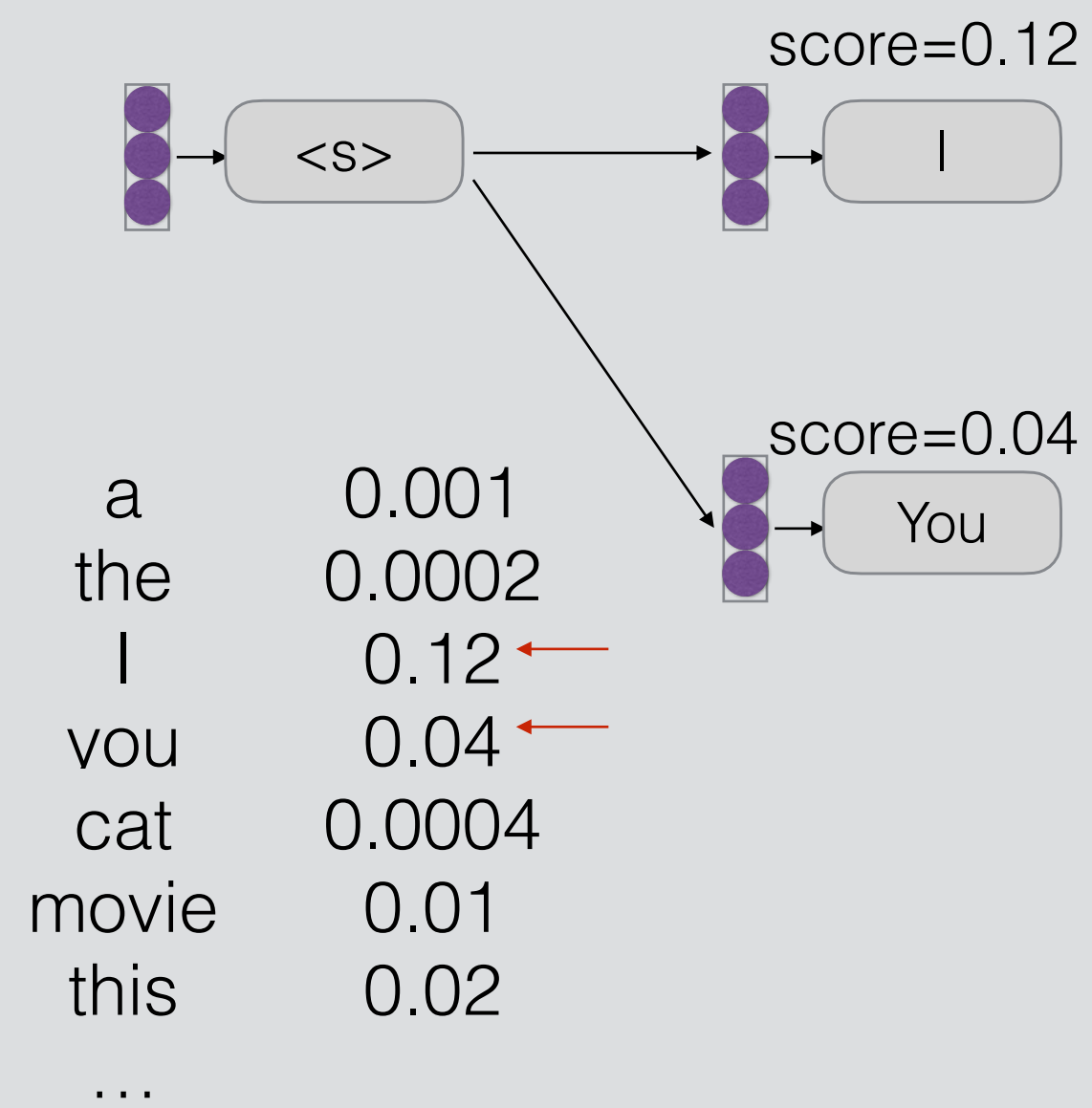
Beam Search

$k=2$



Beam Search

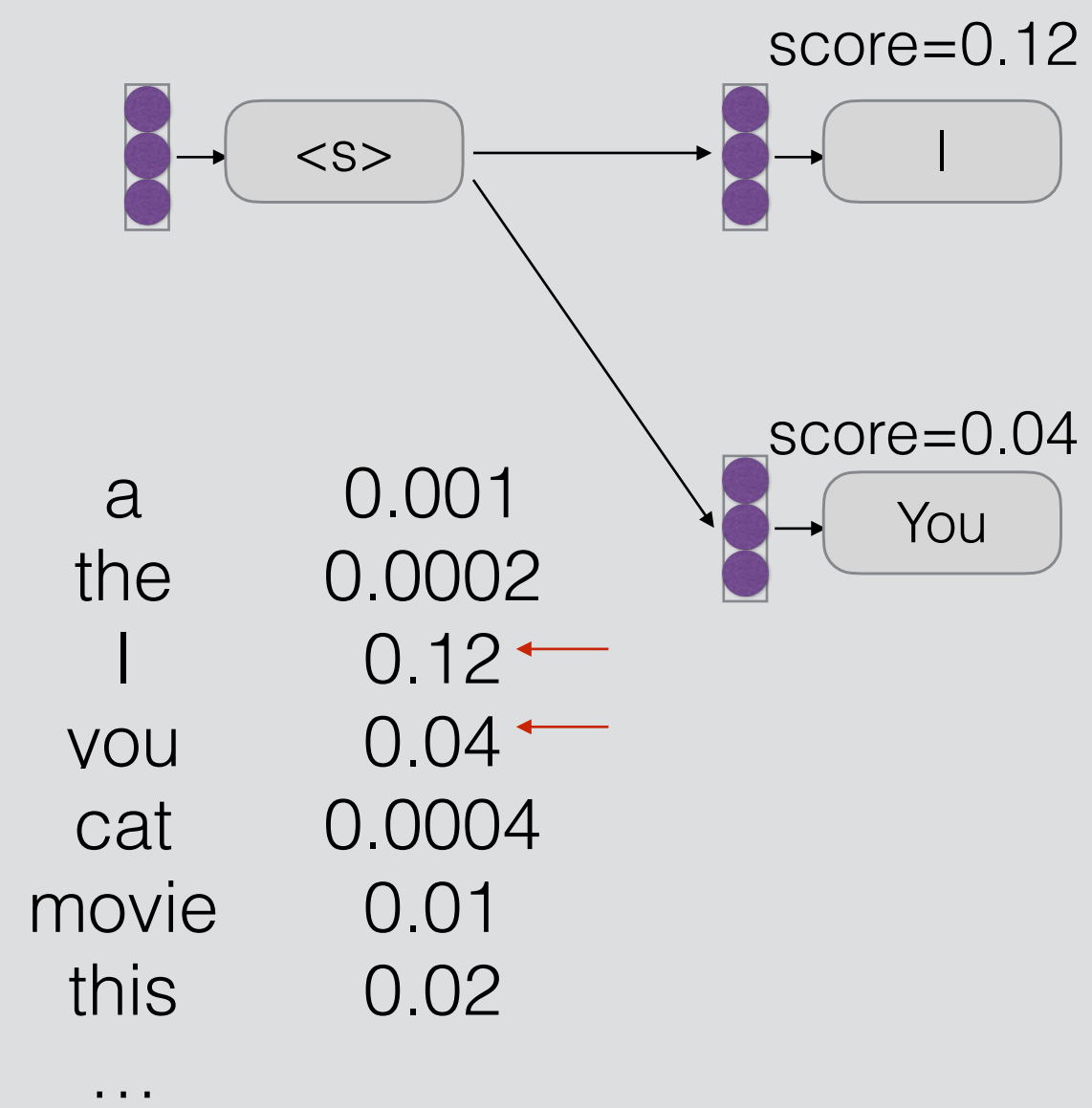
k=2



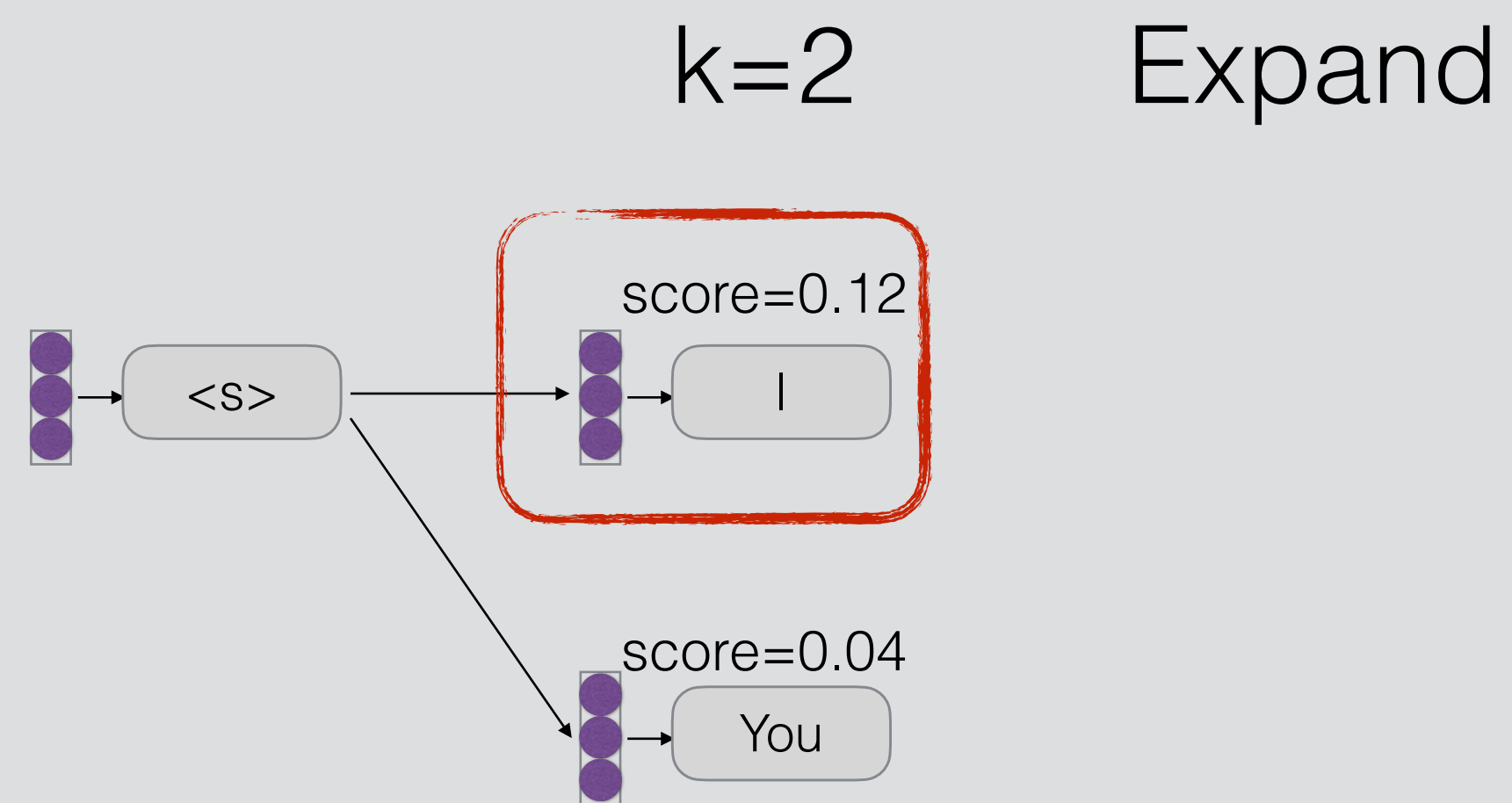
Beam Search

k=2

Expand

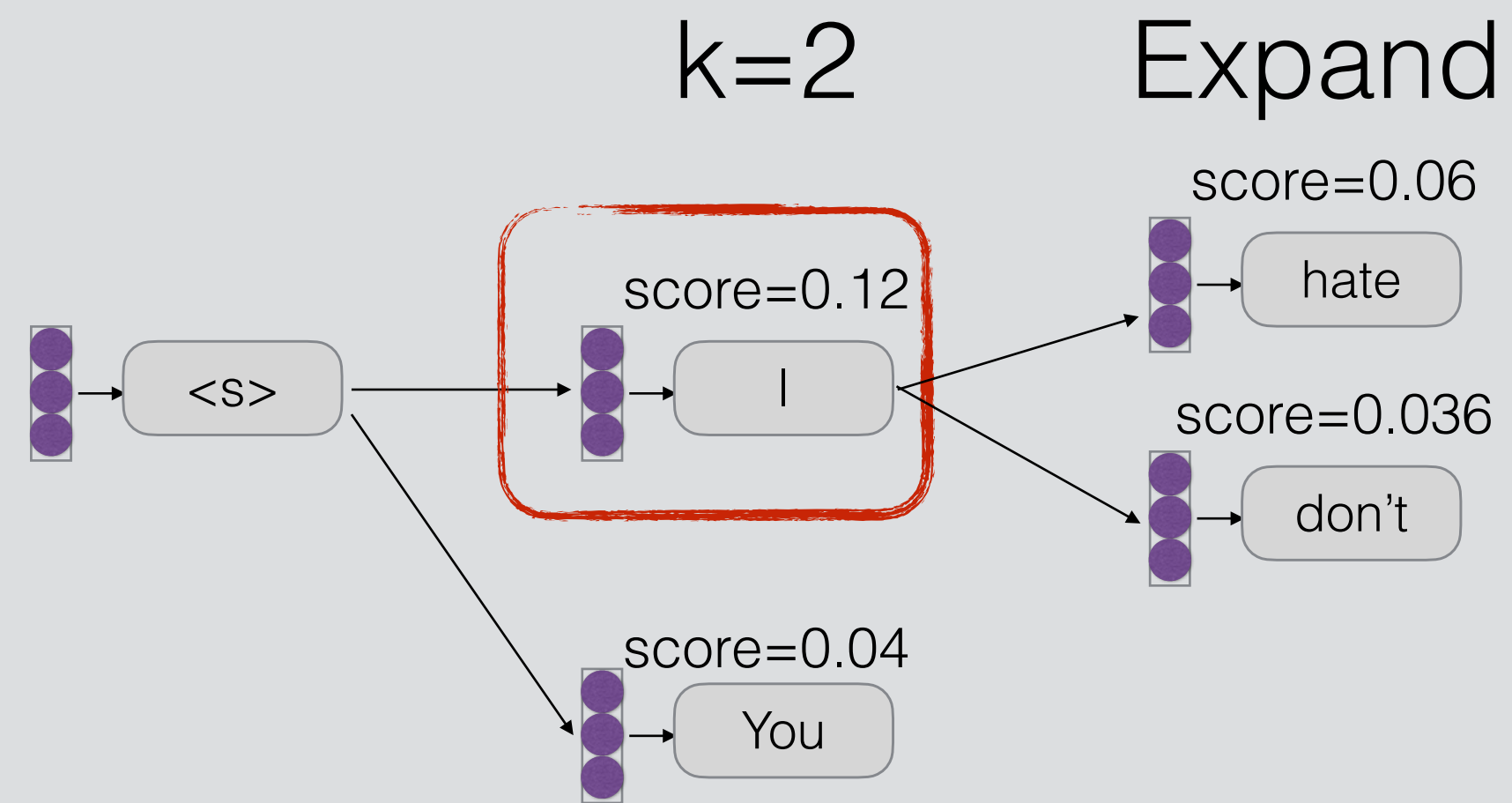


Beam Search



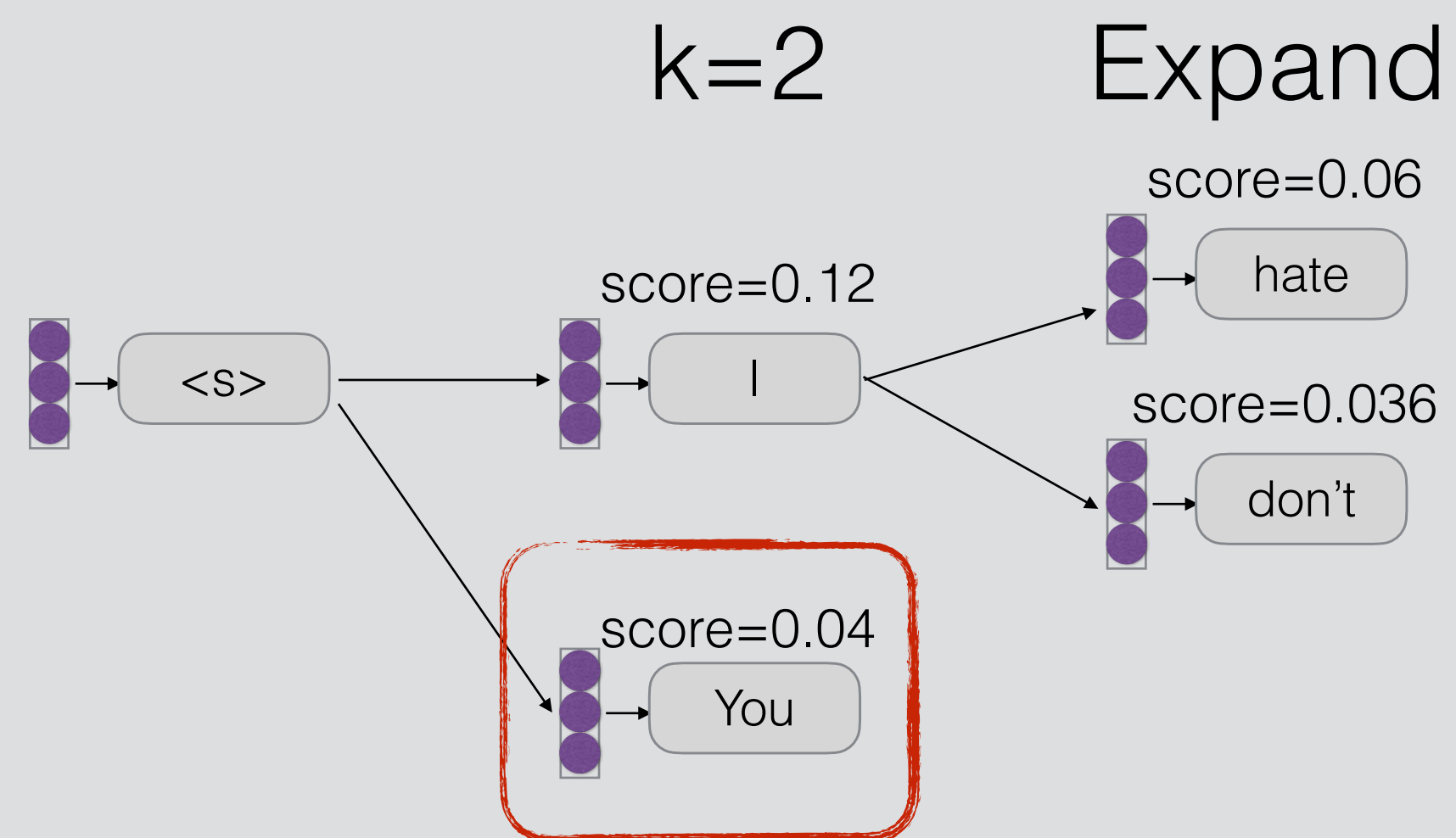
a	0.001
the	0.0002
hate	0.5 ←
this	0.001
cat	0.003
movie	0.07
don't	0.3 ←
...	...

Beam Search



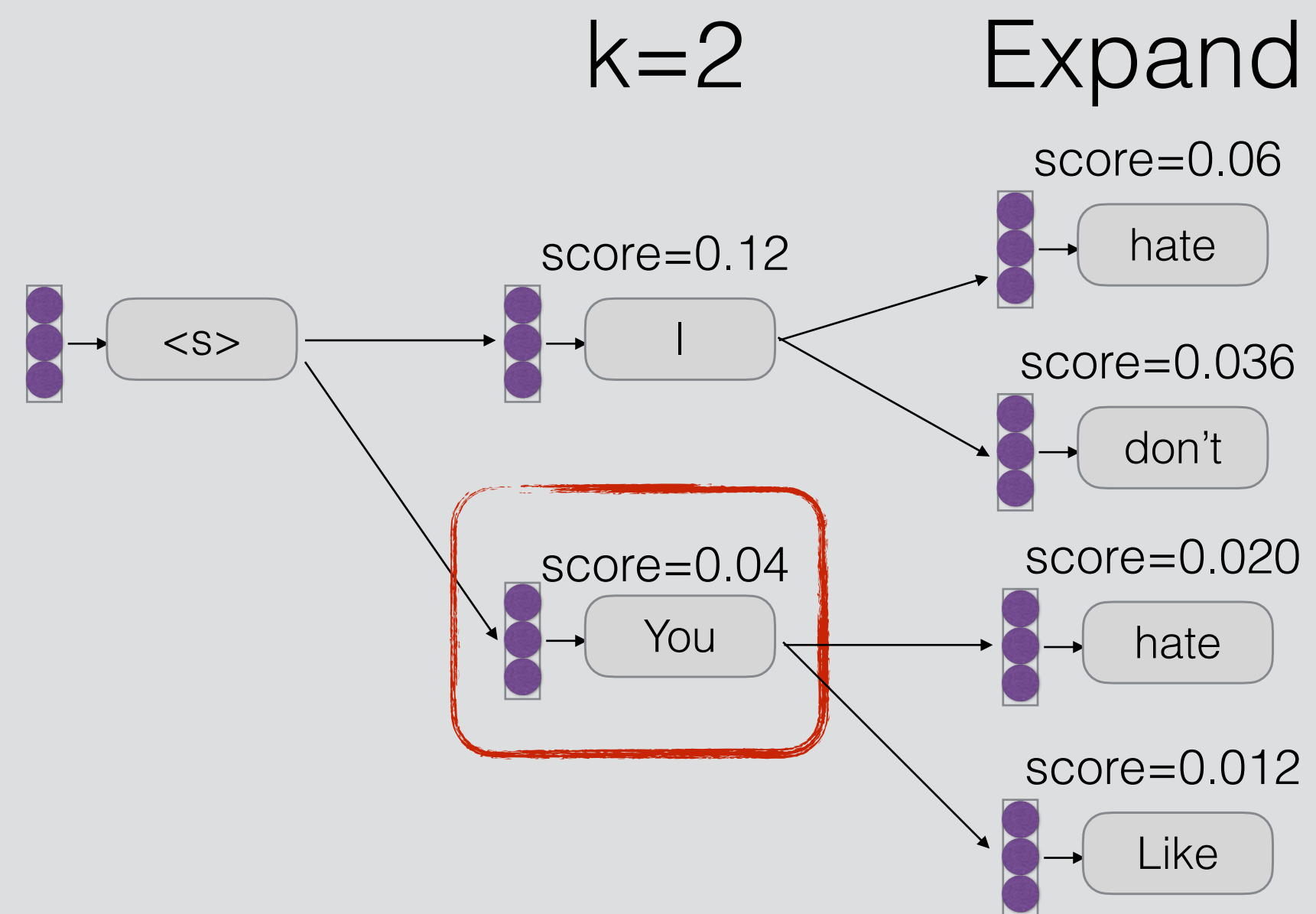
a	0.001	
the	0.0002	
hate	0.5	←
this	0.001	
cat	0.003	
movie	0.07	
don't	0.3	←
...	...	

Beam Search



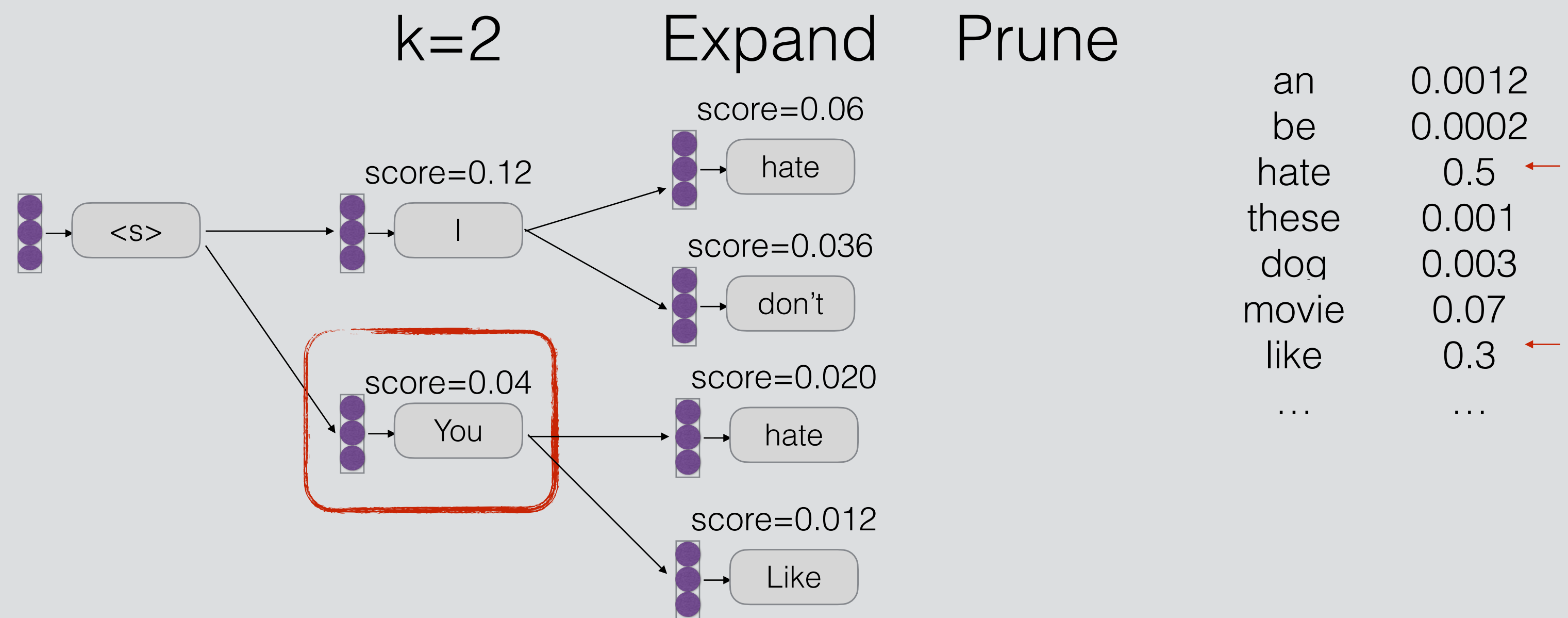
an	0.0012
be	0.0002
hate	0.5 ←
these	0.001
dog	0.003
movie	0.07
like	0.3 ←
...	...

Beam Search

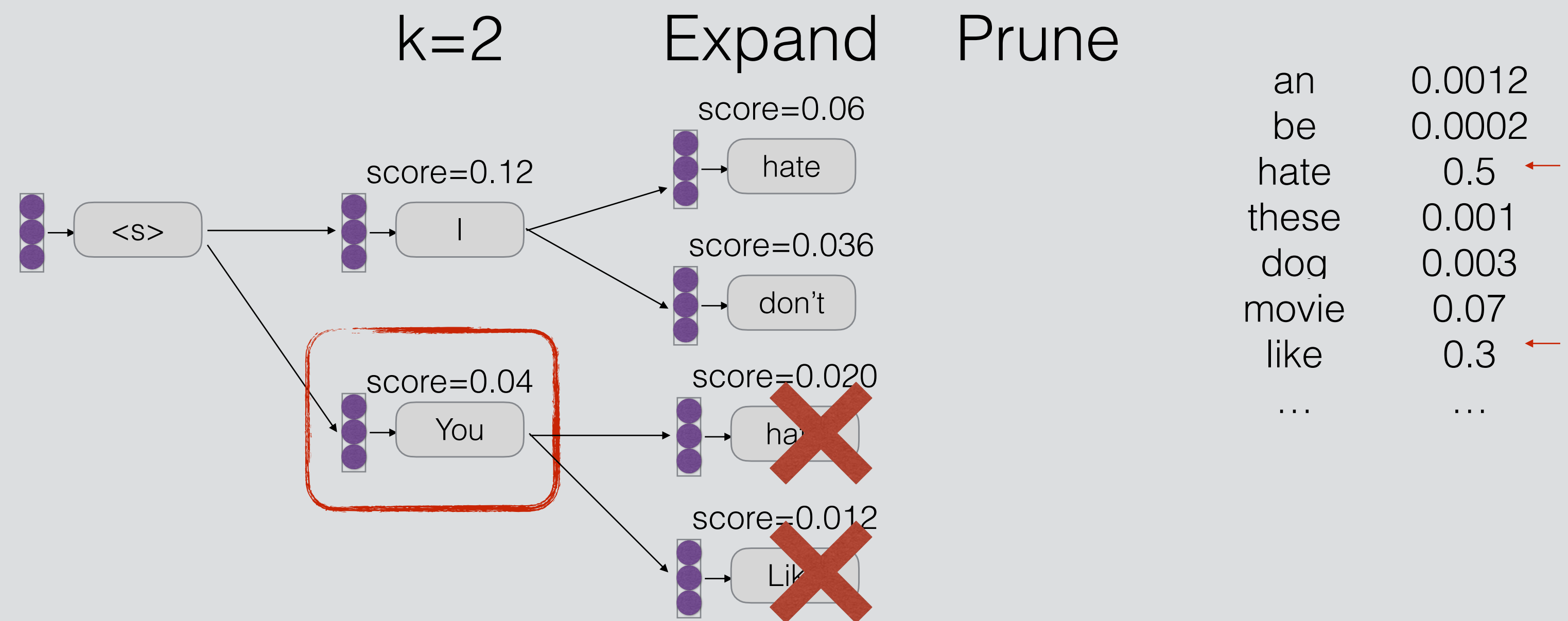


an	0.0012
be	0.0002
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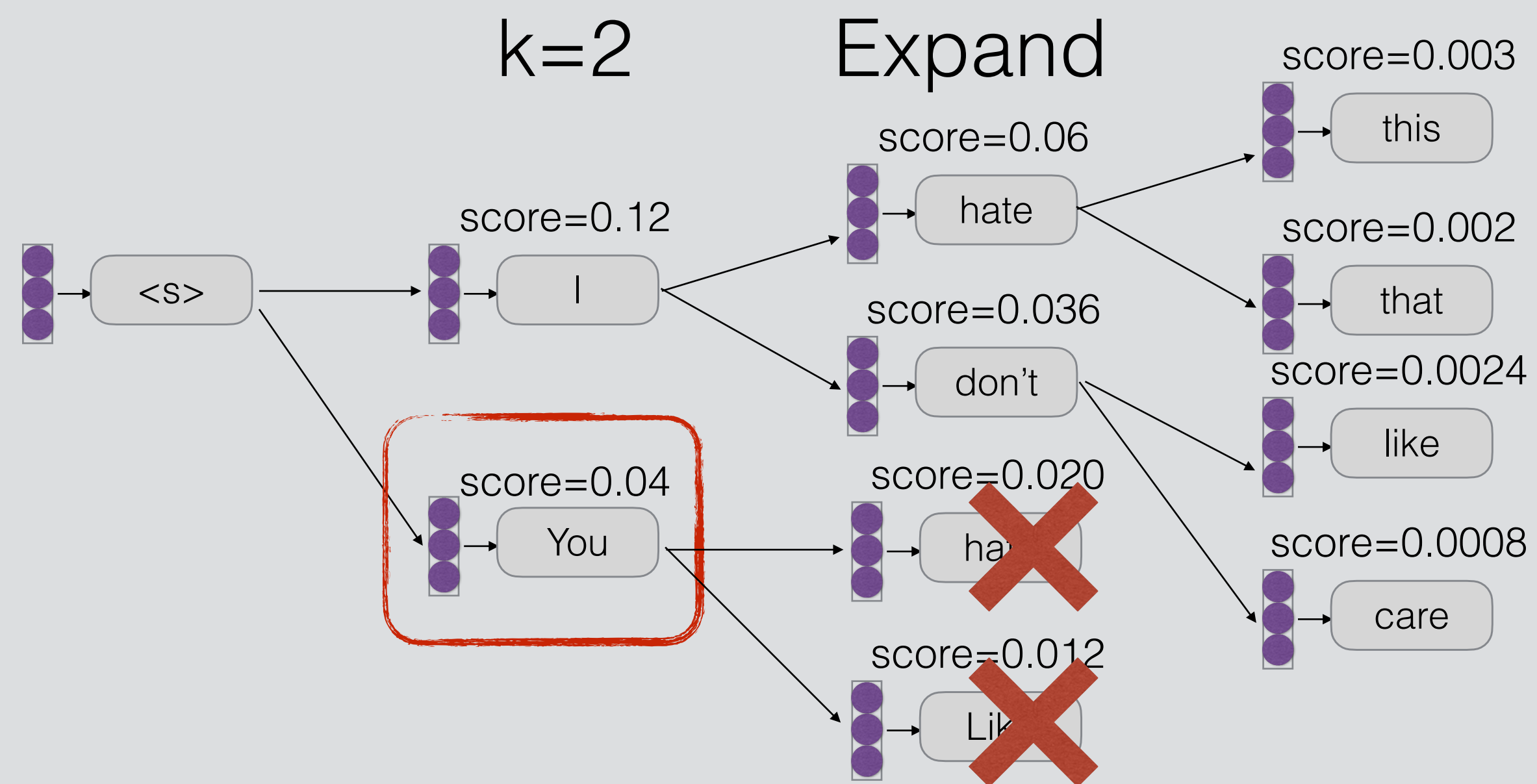
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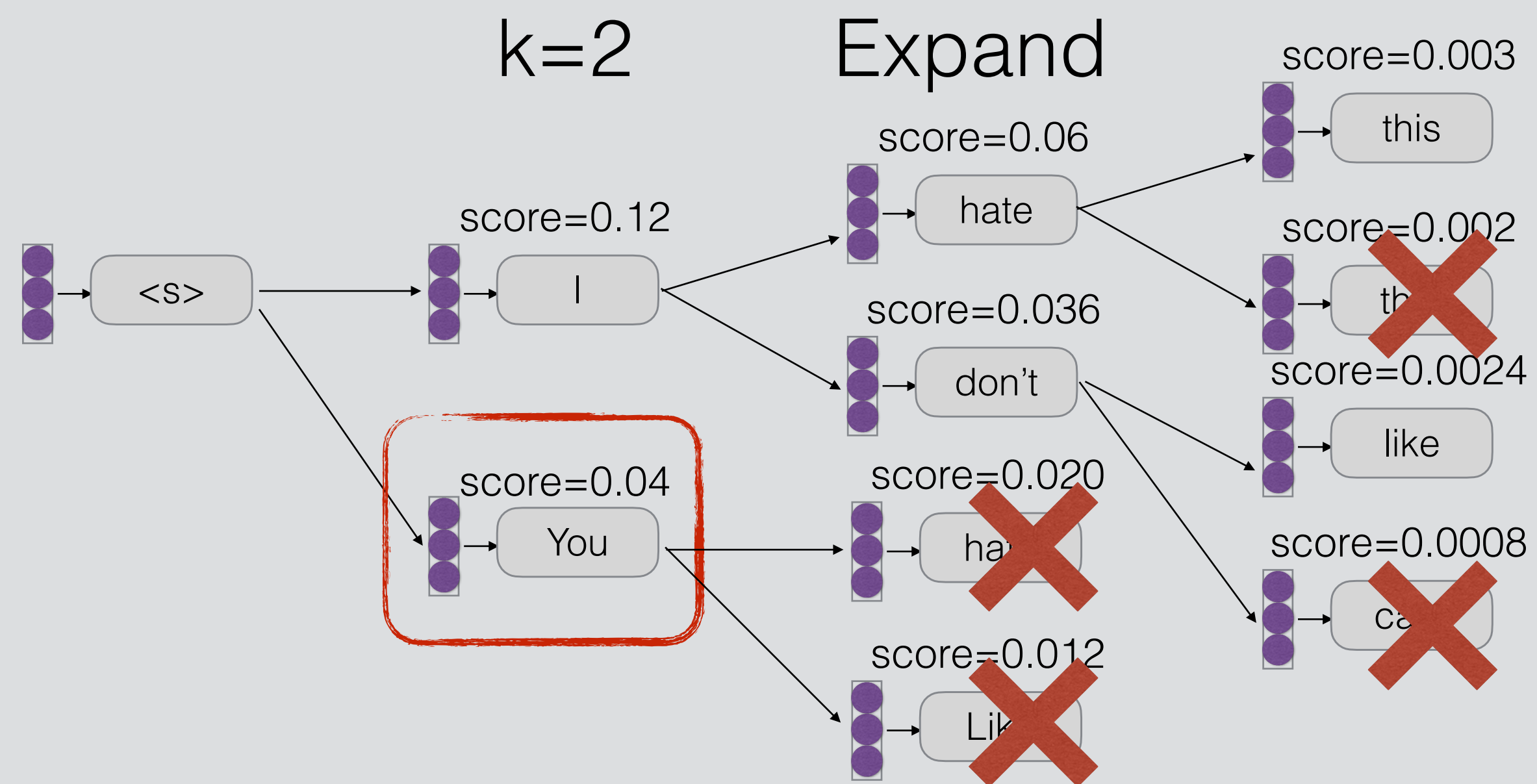
Beam Search



Beam Search



Beam Search



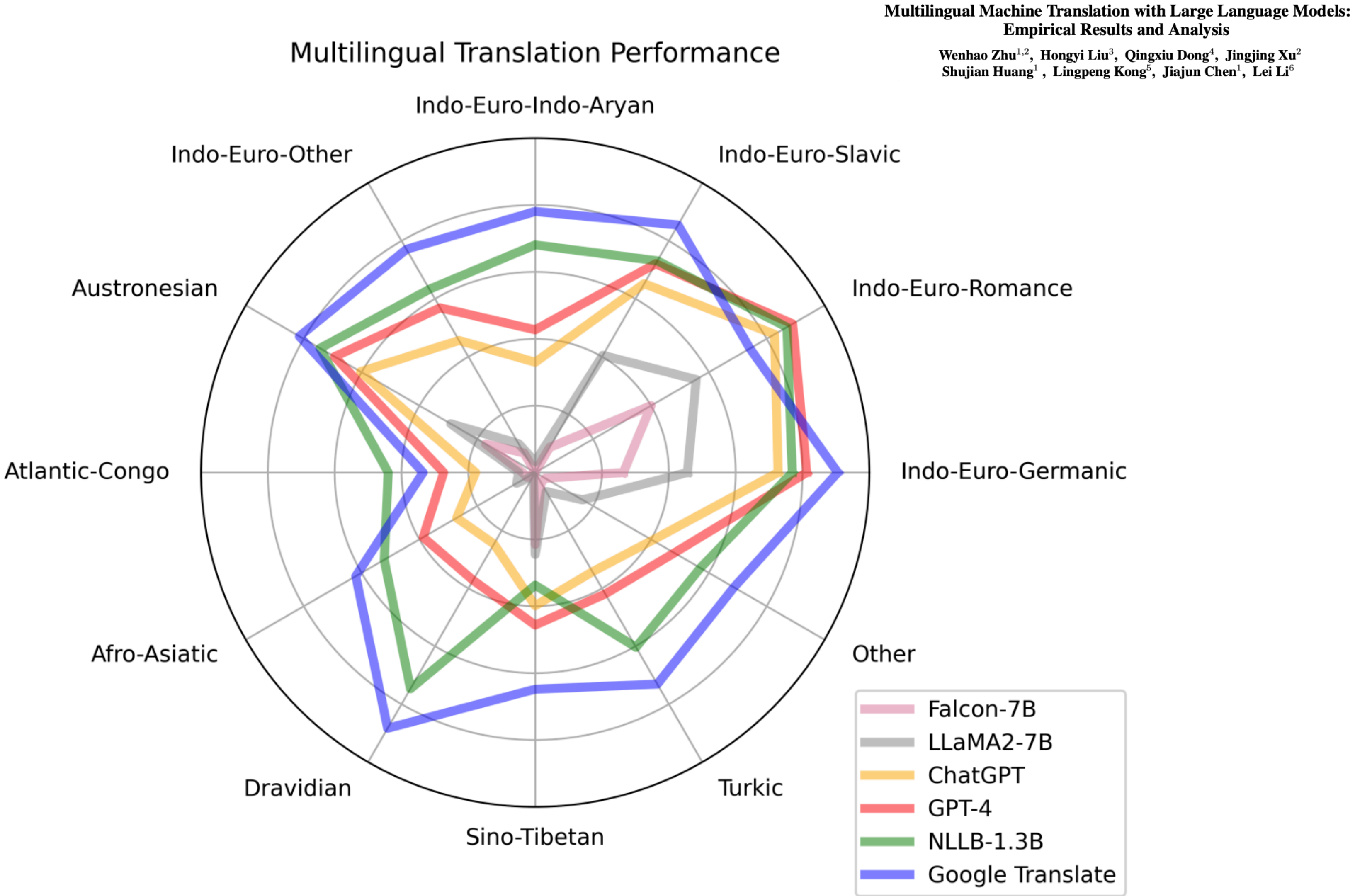
2022



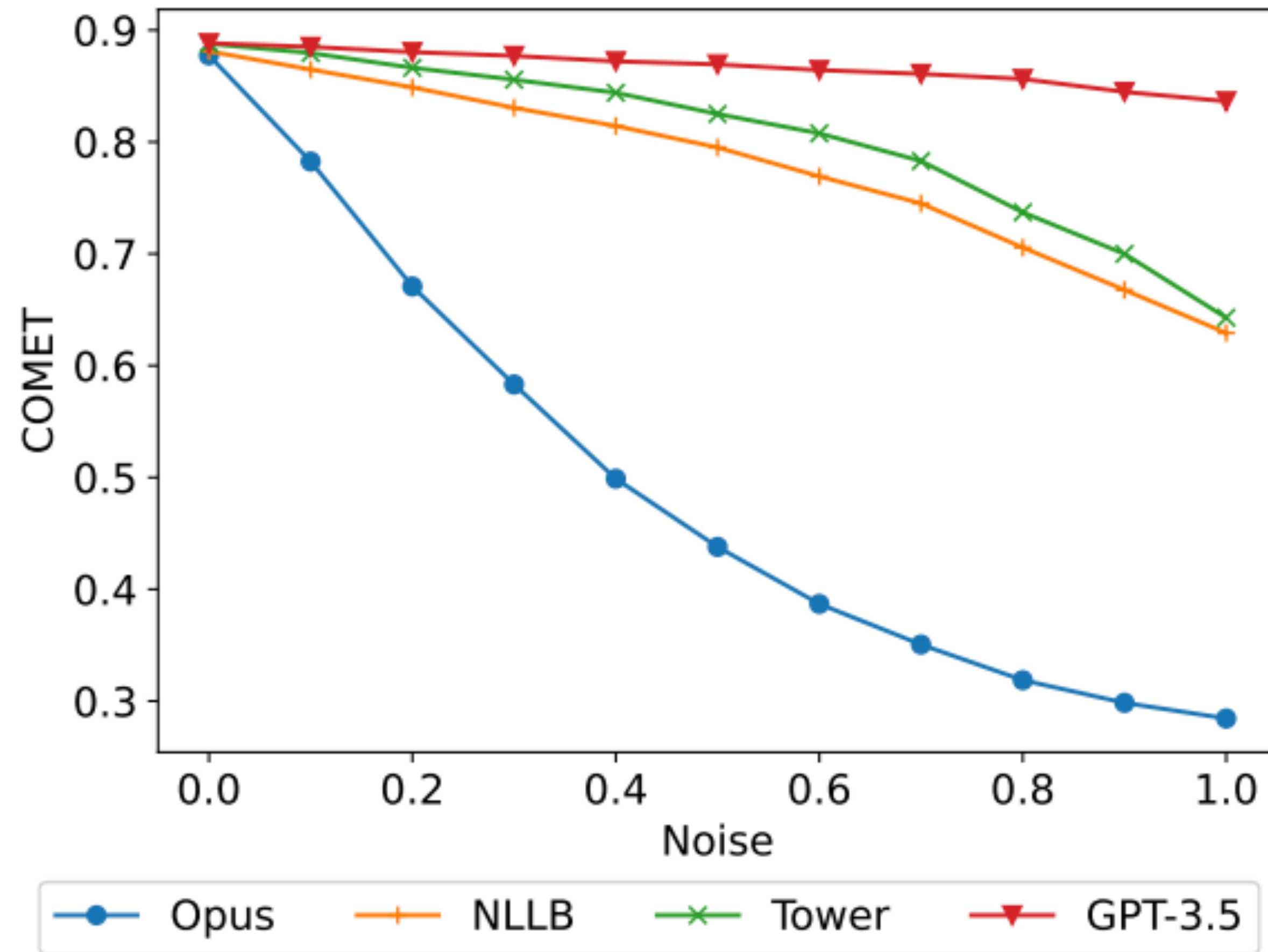
~~MT research~~ NLP

(dramatic reenactment)

Benchmarking LLMs for Translation



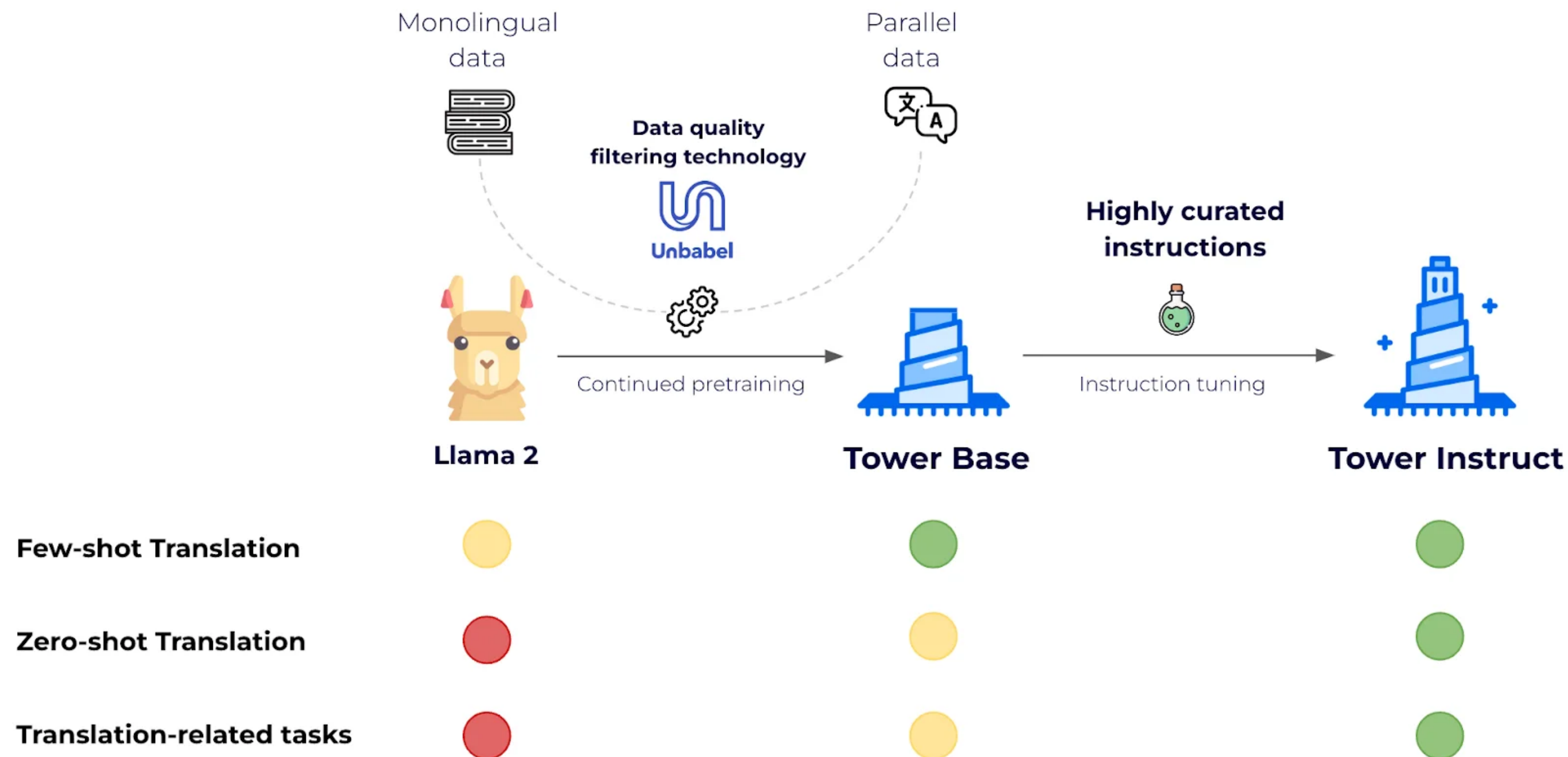
LLMs for Translation: Robustness



Did Translation Models Get More Robust Without Anyone Even Noticing?

Ben Peters* and André F. T. Martins*^{†‡◊}

The Tower models



LLM-powered translation

Simulate a translation pipeline with noisy inputs.



User prompt

Fix all errors in the following sentence.

If the sentence does not contain any errors, just copy it.

Do not change the meaning of the sentence or insert stylistic changes.

Sentence: Ontme, a meu amiga foi ao supermrcado mas estava fechado. Queria comprsr legumes e fruta.

Corrected sentence:



TowerInstruct

Ontem, a minha amiga foi ao supermercado mas estava fechado. Queria comprar legumes e fruta.



User prompt

Now, **translate** the corrected sentence from Portuguese into English.

English:



TowerInstruct

Yesterday, my friend went to the supermarket but it was closed. She wanted to buy vegetables and fruit.

[correct pronoun resolution](#)

LLM-powered translation

Introduce terminology/glossary terms on the prompt for terminology-aware translation

Without terminology terms:

 User prompt *Translate the source text from Portuguese to English.
Portuguese: Eu vou de férias este verão para Itália.
English:*

 TowerInstruct `I'm going on holiday this summer to Italy.`

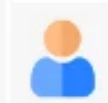
With terminology terms:

 User prompt *Consider the following translation rule:
férias -> vacation
Translate the source text from Portuguese to English following the provided translation rules.
Portuguese: Eu vou de férias este verão para Itália.
English:*

 TowerInstruct `I'm going on vacation this summer to Italy.`

LLM-powered translation

Translate into multiple languages on the same conversation



User prompt

Translate the following text from Portuguese into English.

Portuguese: Ontem, a minha amiga foi ao supermercado mas estava fechado. Queria comprar legumes e fruta.

English:



TowerInstruct

Yesterday, my friend went to the supermarket but it was closed. She wanted to buy vegetables and fruit.



User prompt

Can you now translate into Spanish?



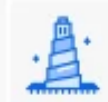
TowerInstruct

Ayer mi amiga fue al supermercado, pero estaba cerrado. Quería comprar verduras y fruta.



User prompt

Now translate to Italian.



TowerInstruct

Ieri la mia amica è andata al supermercato, ma era chiuso. Voleva comprare verdure e frutta.

Evaluation

Machine Translation (reference based)

Mi piacerebbe un
cappuccino freddo.

MT Model

I like one cold cappuccino.

Machine Translation (reference based)

Mi piacerebbe un
cappuccino freddo.

MT Model

I like one cold cappuccino.

reference: *I would like a cold cappuccino.*

Machine Translation (reference based)

Mi piacerebbe un
cappuccino freddo.

MT Model

I like one cold cappuccino.

reference: *I would like a cold cappuccino.*

Compare the output with the reference!

How do we evaluate MT?

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BLEU (Bilingual Evaluation Understudy)

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- BLEU compares the machine-written translation to one or several human-written translation(s), and computes a similarity score based on:

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 - Penalty for too-short system translations

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- BLEU is useful but imperfect

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 - Penalty for too-short system translations
- BLEU is useful but imperfect
 - There are many valid ways to translate a sentence

How do we evaluate MT?

BLEU (Bilingual Evaluation Understudy)

- BLEU compares the machine-written translation to one or several human-written translation(s), and computes a similarity score based on:
 - n-gram precision (usually up to 3 or 4-grams)
 - Penalty for too-short system translations
- BLEU is useful but imperfect
 - There are many valid ways to translate a sentence
 - So a good translation can get a poor BLEU score because it has low n-gram overlap with the human translation 😞

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I like one cold cappuccino

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I like one cold cappuccino

Unigrams	4/5
----------	-----

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I like one cold cappuccino

Unigrams	4/5
Bigrams	1/4

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I like one cold cappuccino

Unigrams	4/5
Bigrams	1/4
3-grams	0/3

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reference: I would like a cold cappuccino

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Bigrams	1/4
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Unigrams	4/5	→ average
Bigrams	1/4	
3-grams	0/3	

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reference: I would like a cold cappuccino

hypothesis: I like one cold cappuccino

Unigrams	4/5	→ average
Bigrams	1/4	
3-grams	0/3	

Can we cheat?

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I like like like like one cold cappuccino

Unigrams	7/8
Bigrams	1/7
3-grams	0/6

Can we cheat?

Solution: Only count each word once.

Machine Translation: BLEU

reference: I would like a cold cappuccino

hypothesis: I would like

Unigrams	3/3
Bigrams	2/2
3-grams	1/1

Can we cheat?

Solution: Brevity Penalty.

MT: Problems with BLEU

reference: I would like a cold cappuccino

hypothesis 1: *I would like one cold cappuccino*

These three hypotheses have the same BLEU score!

Solution: Use paraphrases, synonyms, etc (Meteor)

MT: Problems with BLEU

reference: I would like a cold cappuccino

hypothesis 1: *I would like one cold cappuccino*

hypothesis 2: *I would like a cold espresso*

These three hypotheses have the same BLEU score!

Solution: Use paraphrases, synonyms, etc (Meteor)

MT: Problems with BLEU

reference: I would like a cold cappuccino

hypothesis 1: *I would like one cold cappuccino*

hypothesis 2: *I would like a cold espresso*

hypothesis 3: *I would like a cold monk*

These three hypotheses have the same BLEU score!

Solution: Use paraphrases, synonyms, etc (Meteor)

MT: Problems with BLEU

source: *behaving as if you are among those whom we could not civilize*

reference: *uygarlatıramadıklarımızdanmı,ssınızcasına*

Languages with Rich Morphology: How do we even evaluate this?

Solution: Use subwords, character-Fscore — chrF

MT: Human Evaluation

It is almost always better to ask humans!
e.g. in MT, we ask translators

Way 1:

We show system outputs to the annotators, and they provide a score (e.g. 1-5 Likert scale, or 0-100 score)

Way 2:

We show **2** system outputs to the annotators, and they annotate which one of the two they think is better.

Evaluation of Evaluation Metrics

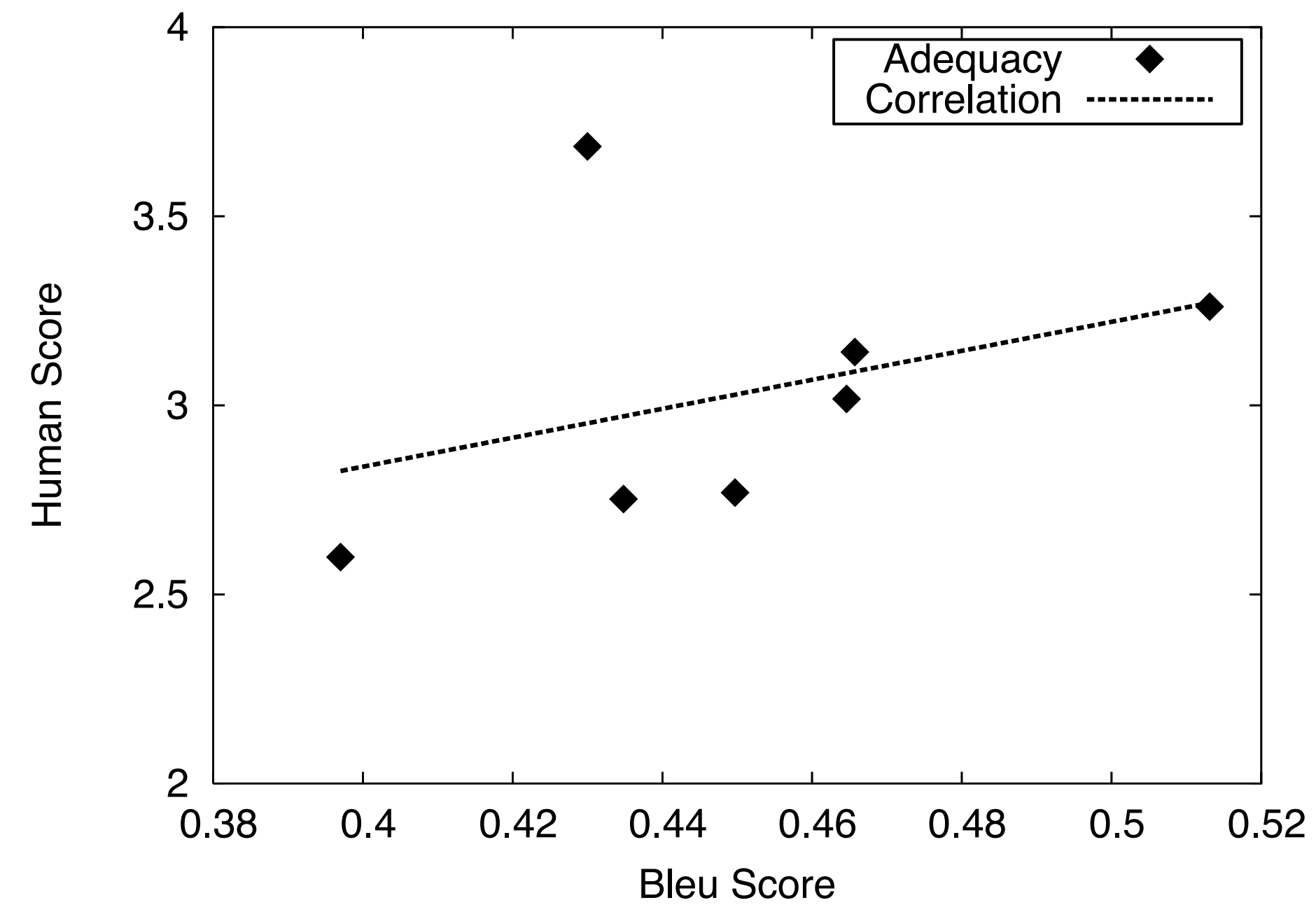


- Automatic metrics are low cost, tunable, consistent
 - But are they correct?
- Yes, if they correlate with human judgement

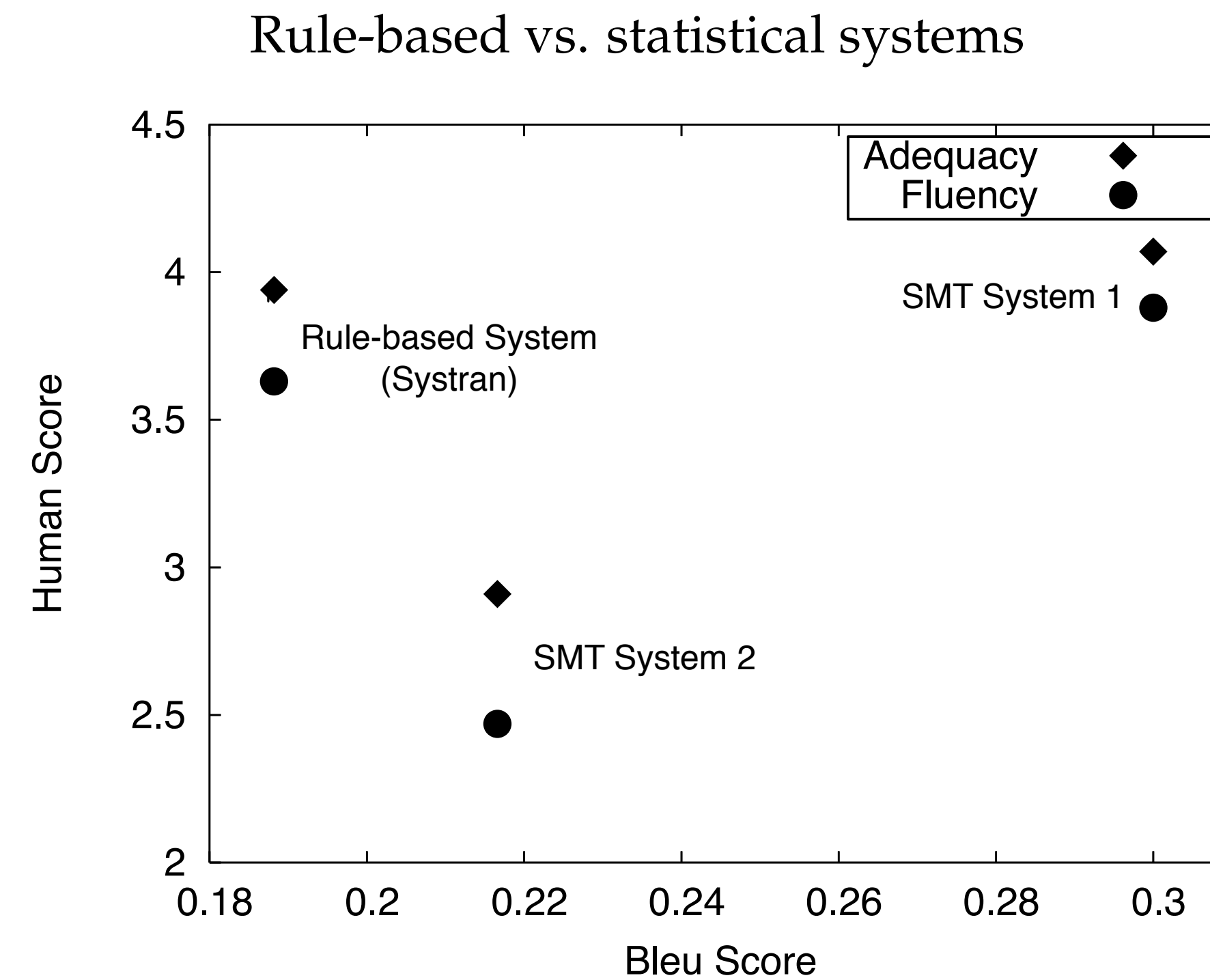
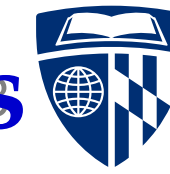
Evidence of Shortcomings of Automatic Metrics



Post-edited output vs. statistical systems (NIST 2005)



Evidence of Shortcomings of Automatic Metrics



WMT Metrics Shared Task

- Annual event to evaluate metrics
- Piggy-backs on the WMT General Translation Task
 - new test set every year
 - research systems and commercial systems
 - lately also large language models
 - human evaluation of automatic evaluations
- New metrics proposed
- Evaluation by correlation with human judgments

(WMT 2023)

Metric		avg corr
XCOMET-Ensemble	1	0.825
XCOMET-QE-Ensemble*	2	0.808
MetricX-23	2	0.808
GEMBA-MQM*	2	0.802
MetricX-23-QE*	2	0.800
mbr-metricx-qe*	3	0.788
MaTESe	3	0.782
CometKiwi*	3	0.782
COMET	3	0.779
BLEURT-20	3	0.776
KG-BERTScore*	3	0.774
sescoreX	3	0.772
cometoid22-wmt22*	4	0.772
docWMT22CometDA	4	0.768
docWMT22CometKiwiDA*	4	0.767
Calibri-COMET22	4	0.767
Calibri-COMET22-QE*	4	0.755
YiSi-1	4	0.754
MS-COMET-QE-22*	5	0.744
prismRef	5	0.744
mre-score-labse-regular	5	0.743
BERTscore	5	0.742
XLsim	6	0.719
f200spBLEU	7	0.704
MEE4	7	0.704
tokengram_F	7	0.703
embed_llama	7	0.701
BLEU	7	0.696
chrF	7	0.694
eBLEU	7	0.692
Random-sysname*	8	0.529
prismSrc*	9	0.455

Trained Metrics: COMET



- Two decades of evaluation campaigns for machine translation metrics
→ a lot of human judgment data
- Goal: automatic metric that correlates with human judgment
- Make it a machine learning problem
 - input: machine translation, reference translation
 - output: human annotation score
- COMET: Trained neural model for evaluation

Reference-Free Evaluation



- We have data in the form

input, translation, human reference → human judgment

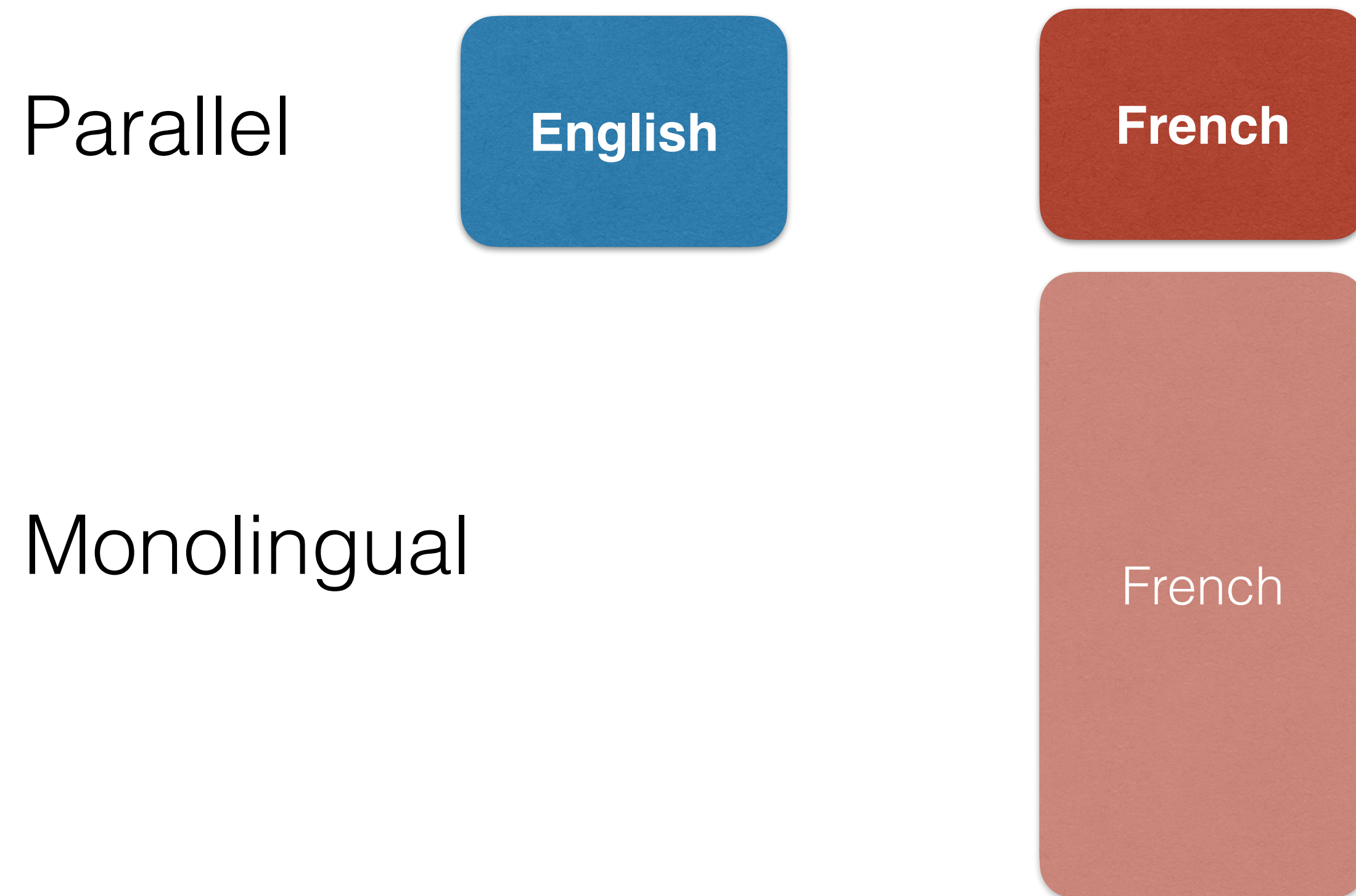
- We can also train a model on

input, translation → human judgment

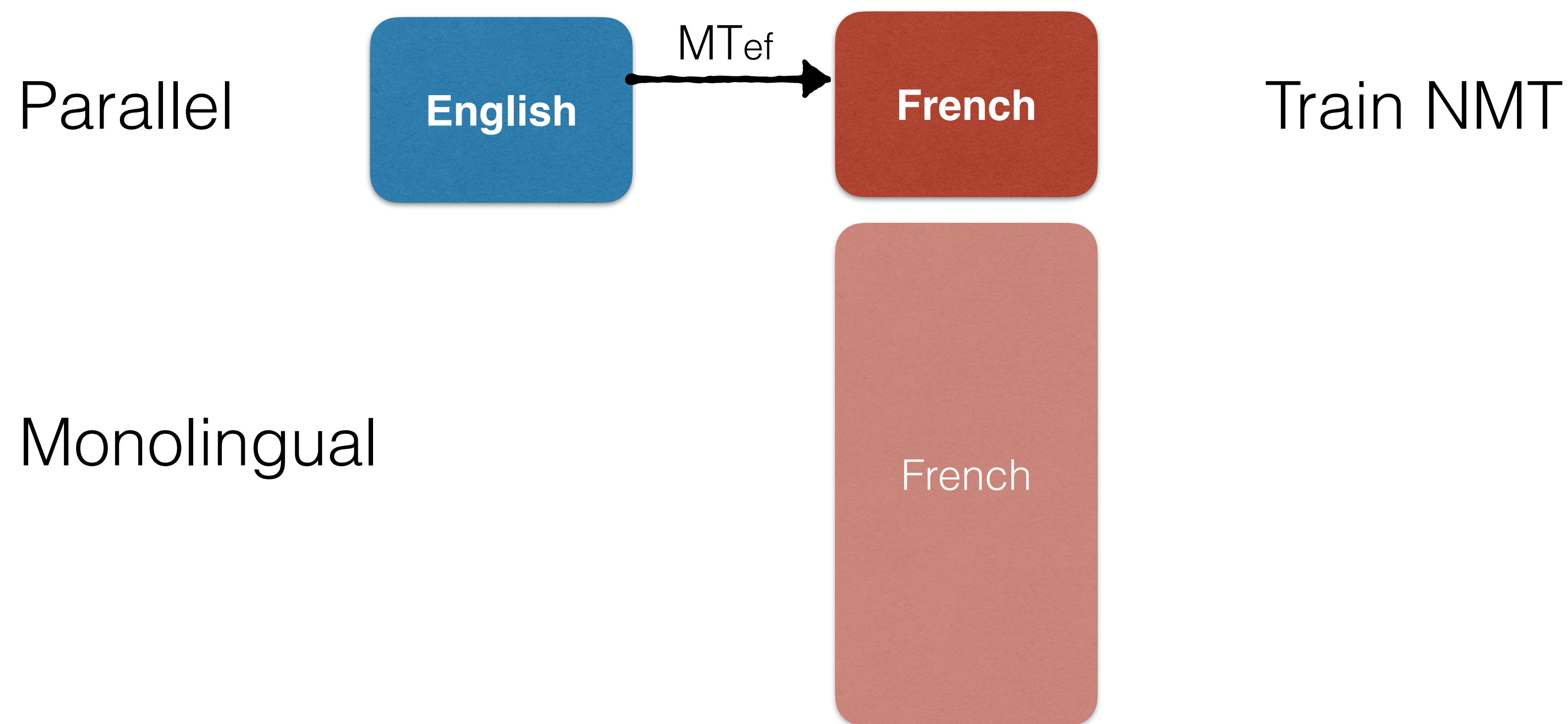
- CometKiwi: trained evaluation model without references
- Also called **quality estimation** or **confidence estimation**

Semisupervised and Unsupervised Methods

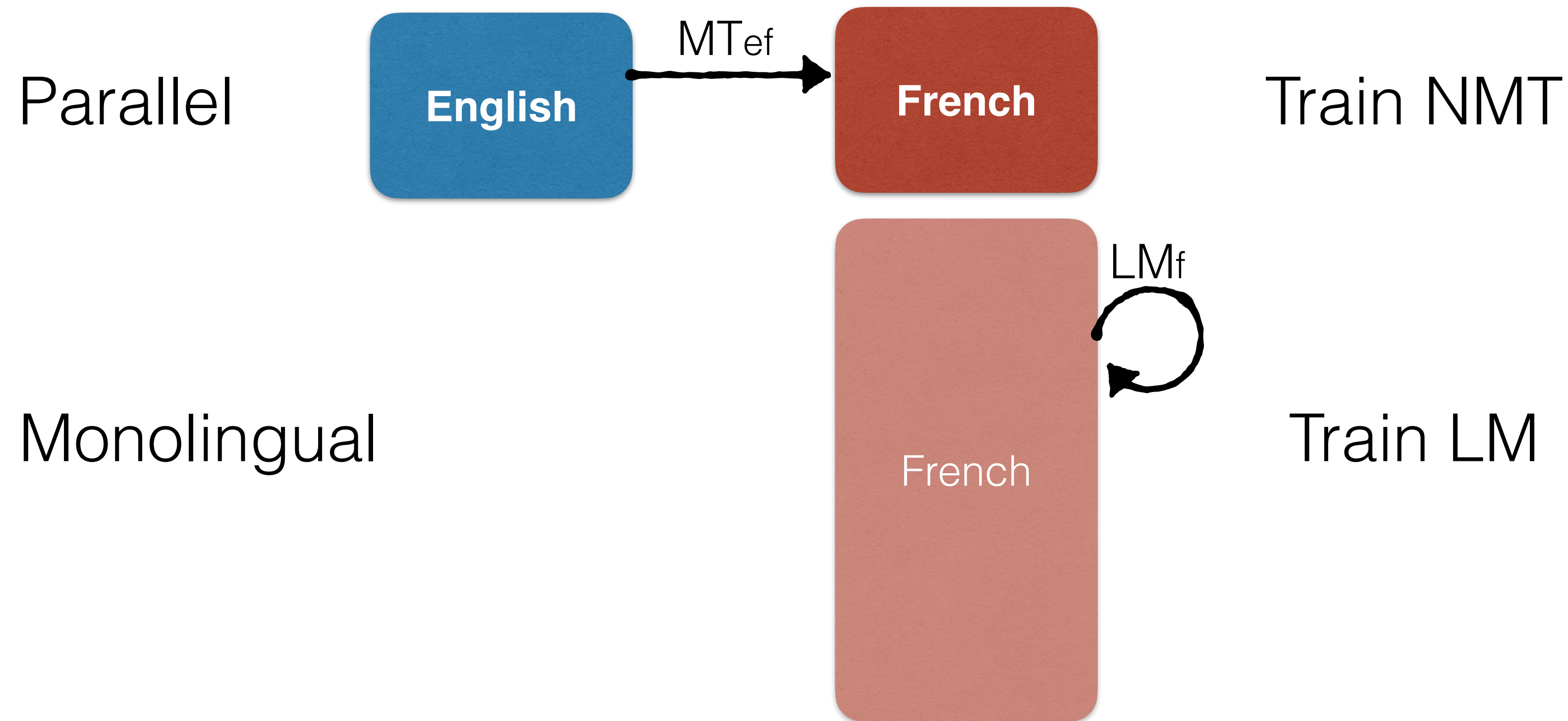
On Using Monolingual Corpora in Neural Machine Translation (Gulcehre et al. 2015)



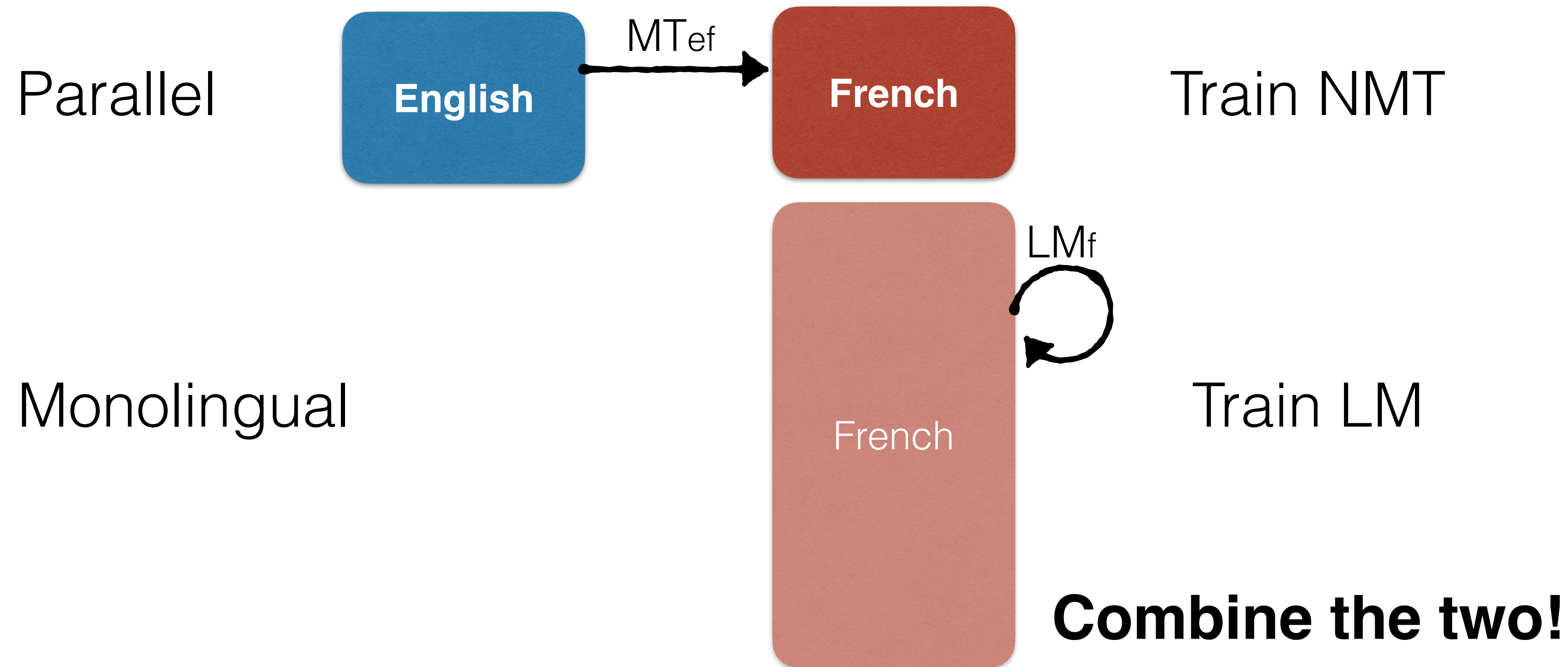
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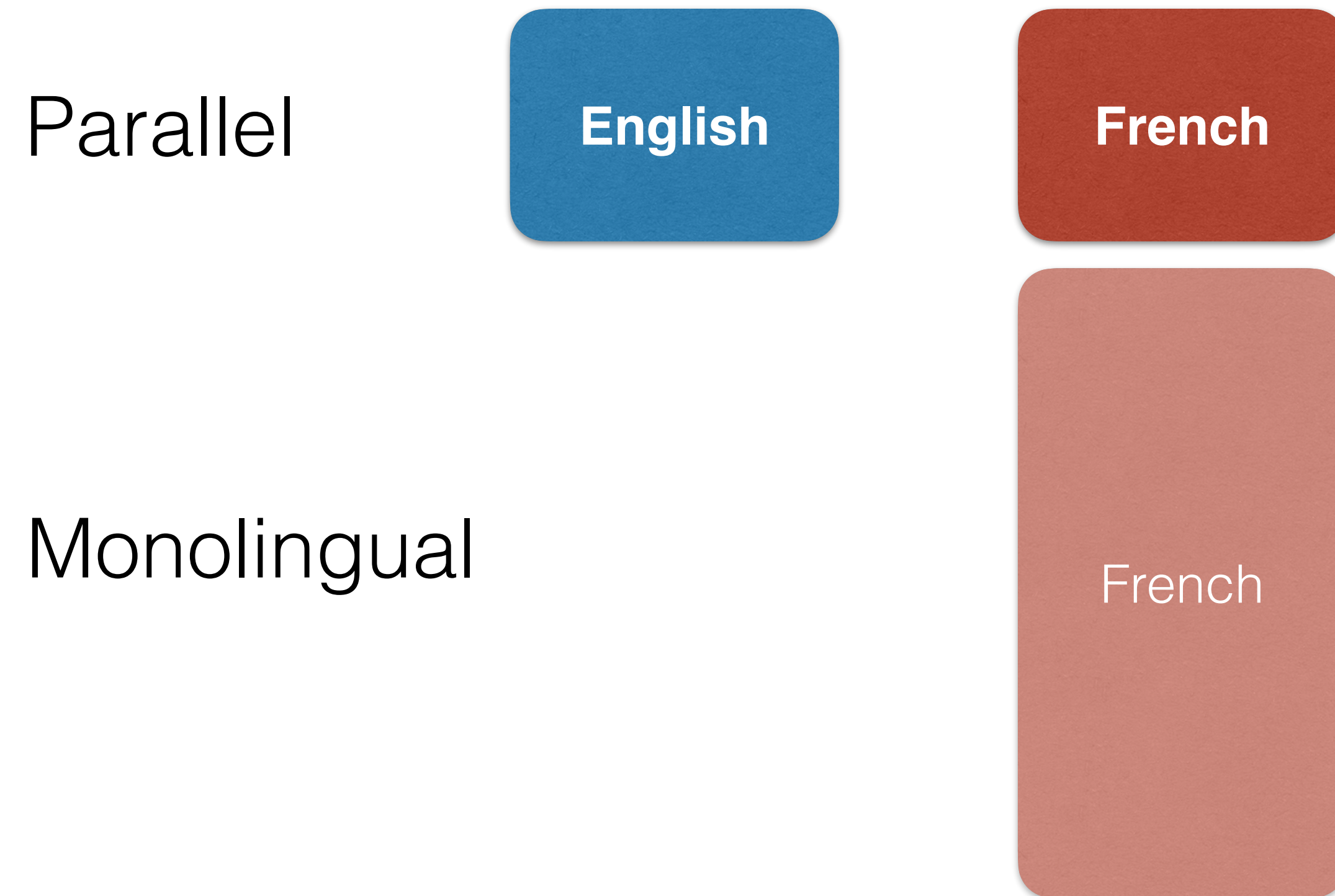
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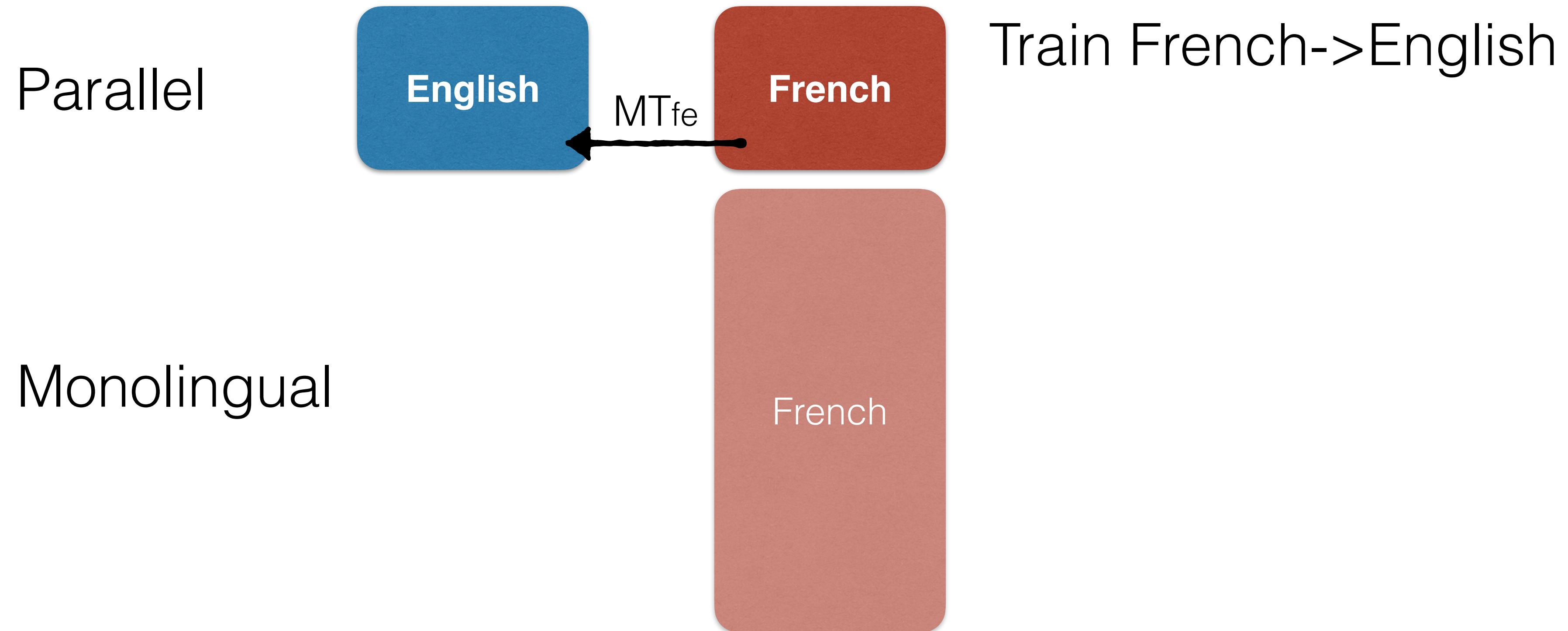
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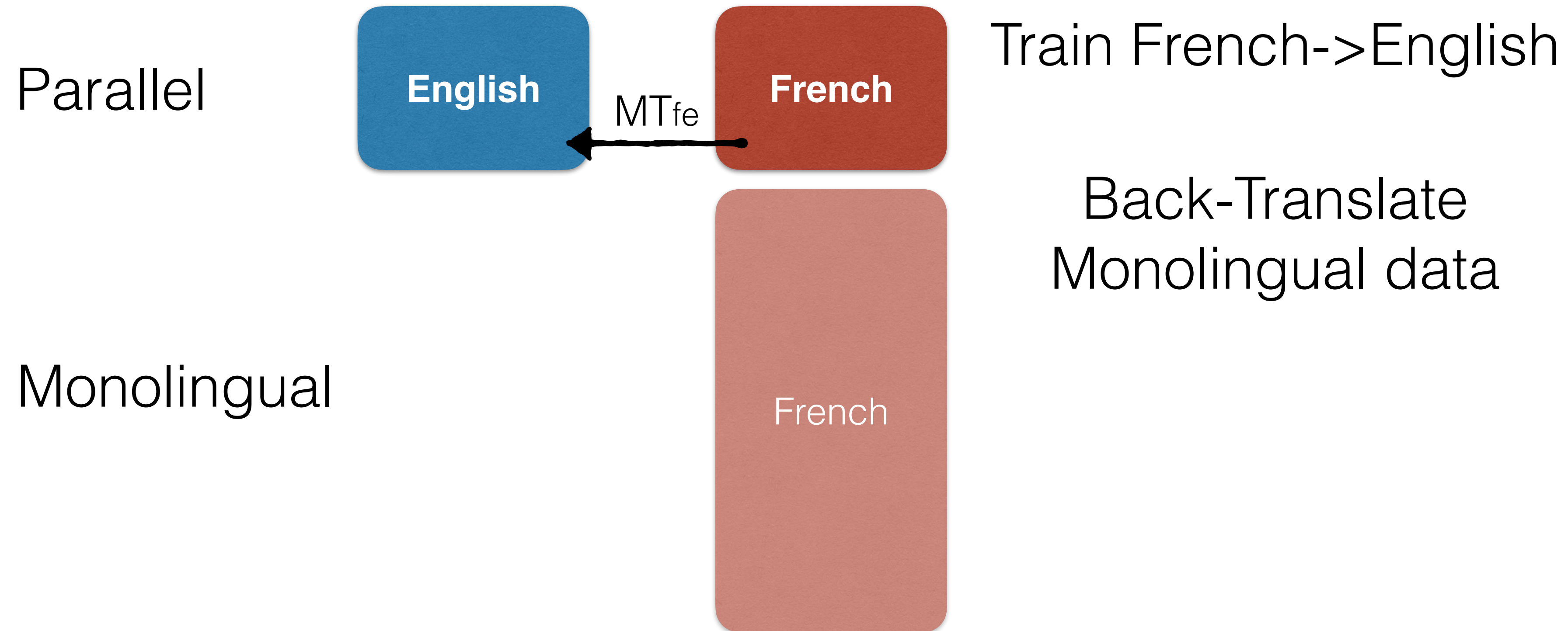
Back-translation (Sennrich et al. 2016)



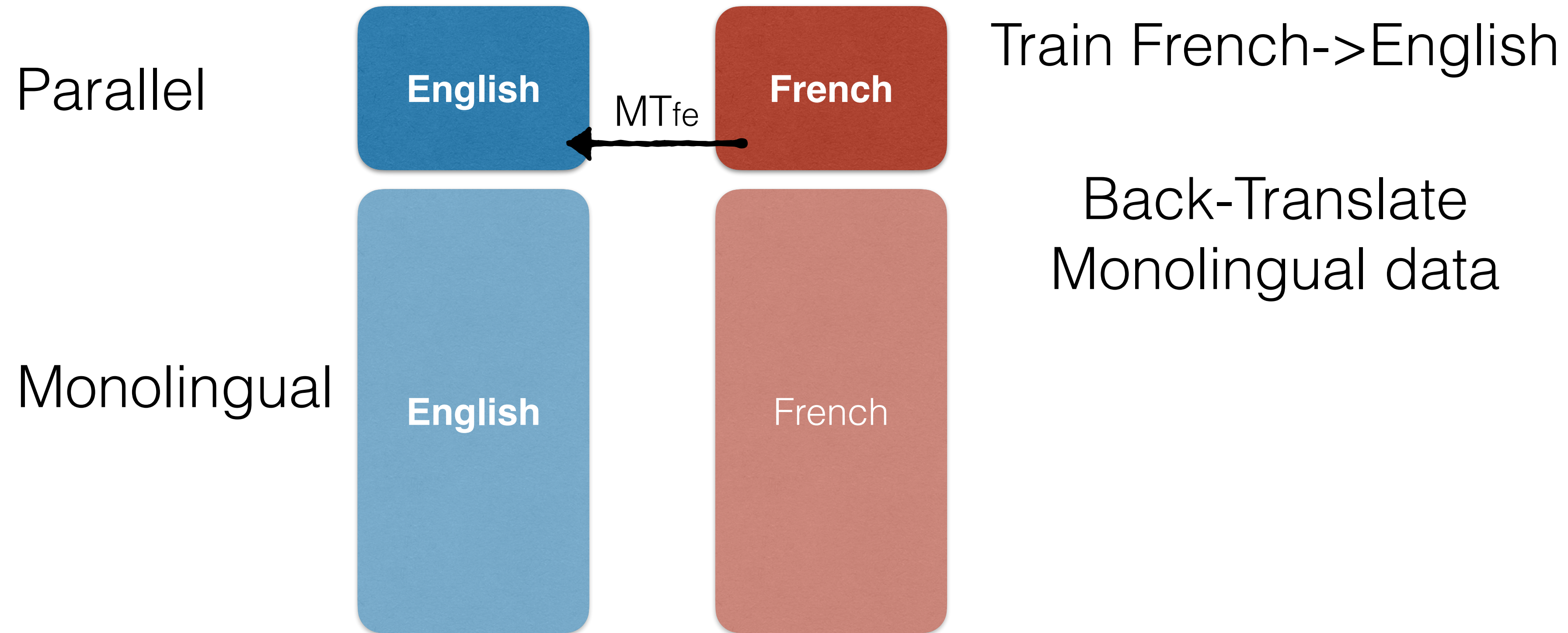
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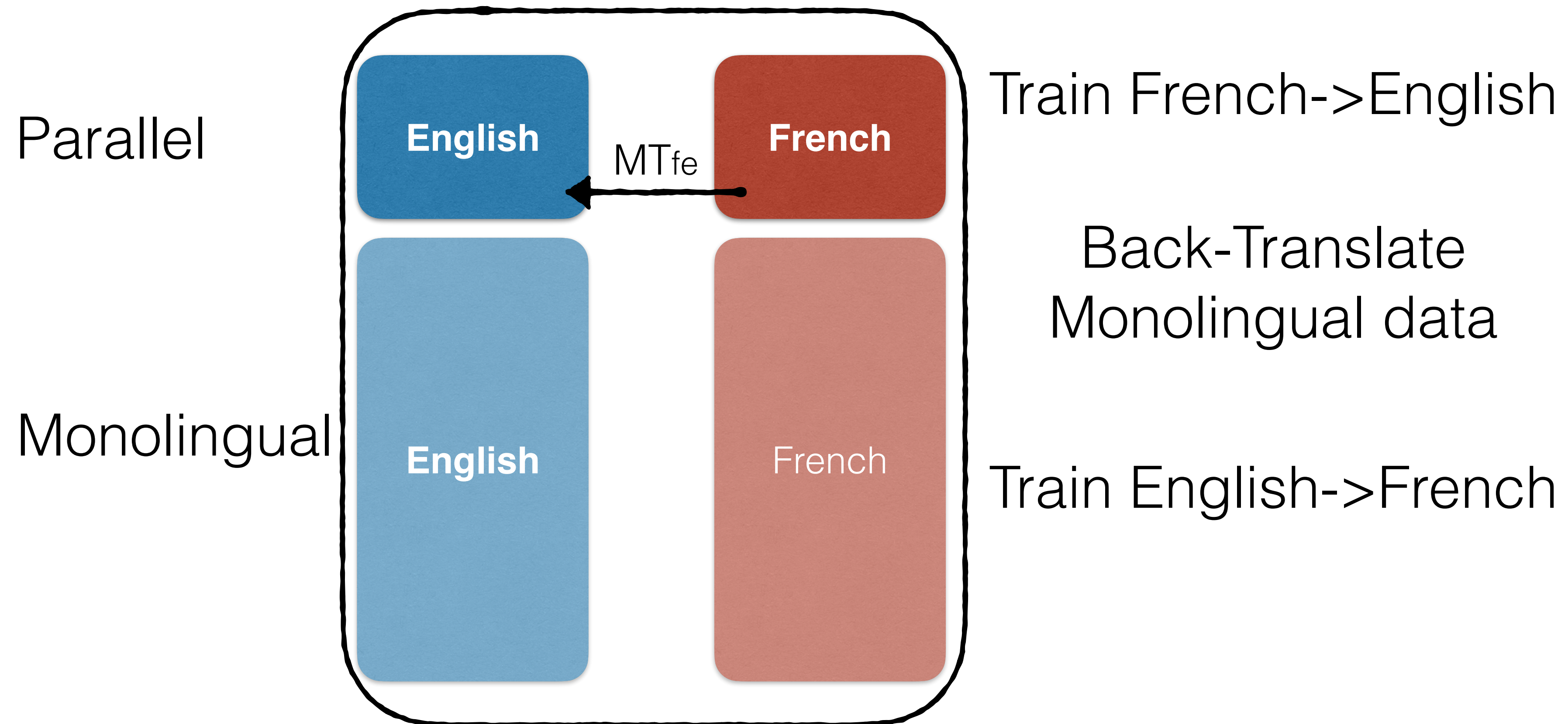
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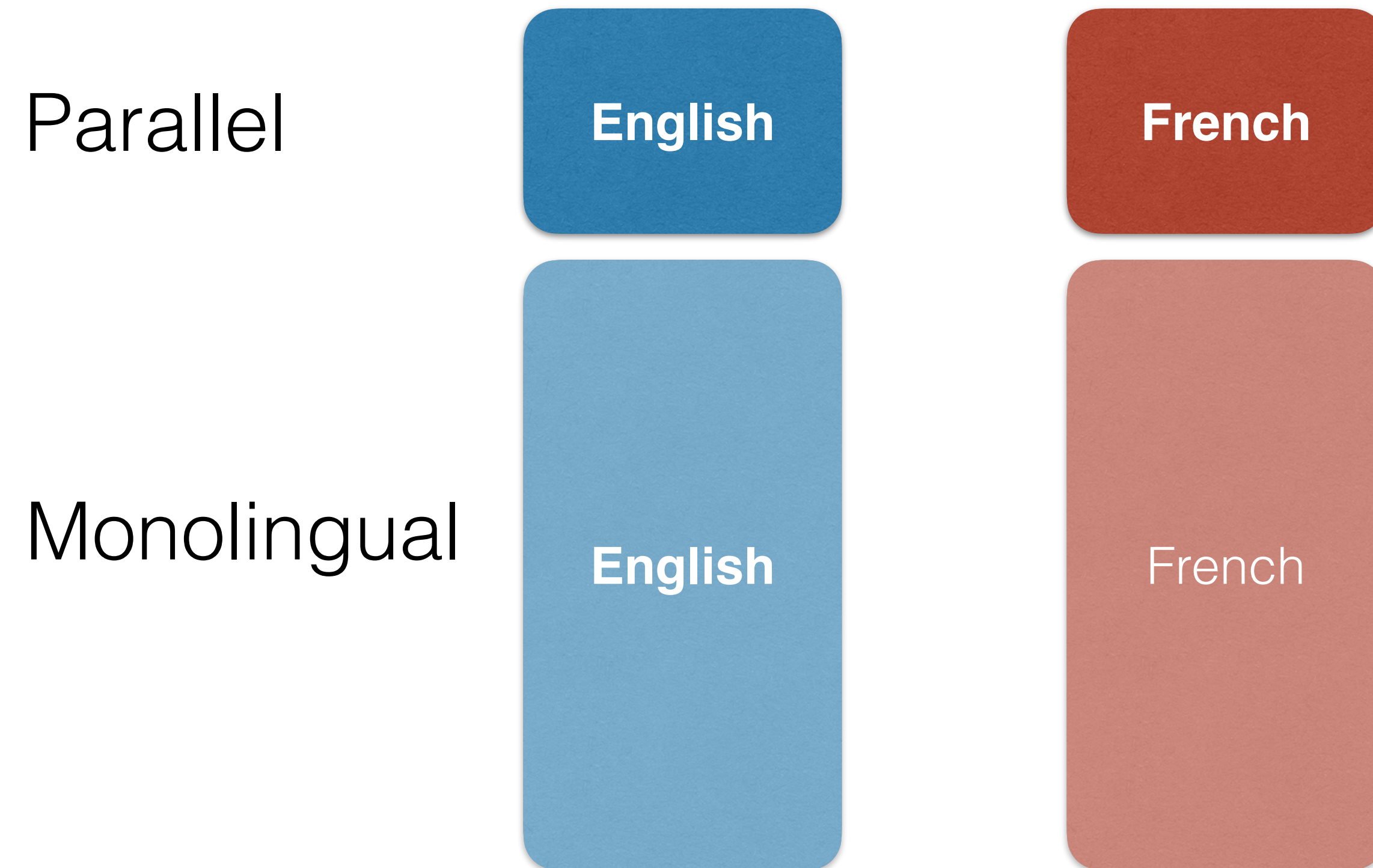
Back-translation (Sennrich et al. 2016)



Back-translation (Sennrich et al. 2016)



Dual Learning (He et al. 2016)



Dual Learning (He et al. 2016)

Assume MT_{ef} , MT_{fe} , LM_e , LM_f

Parallel

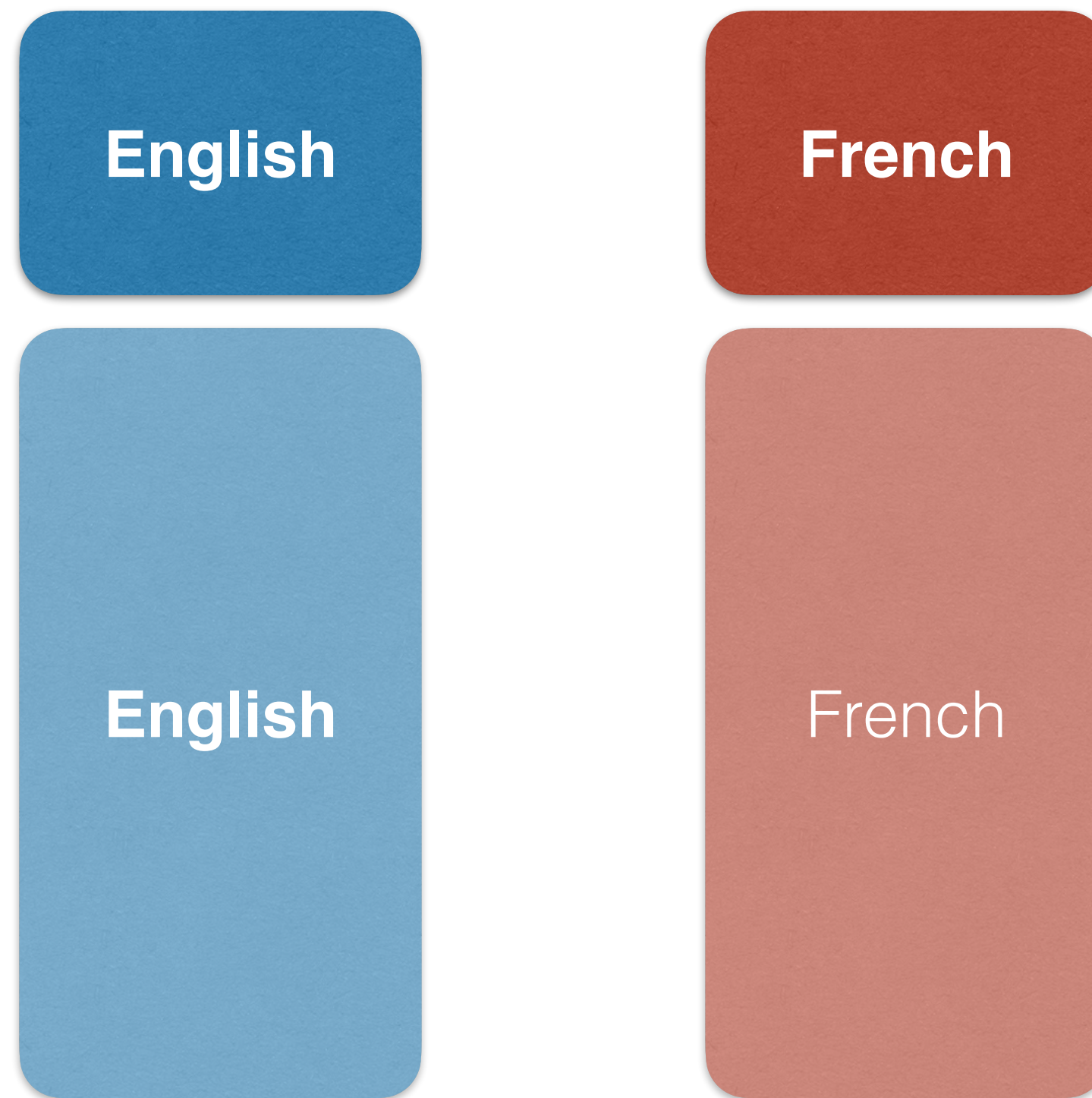
English

French

Monolingual

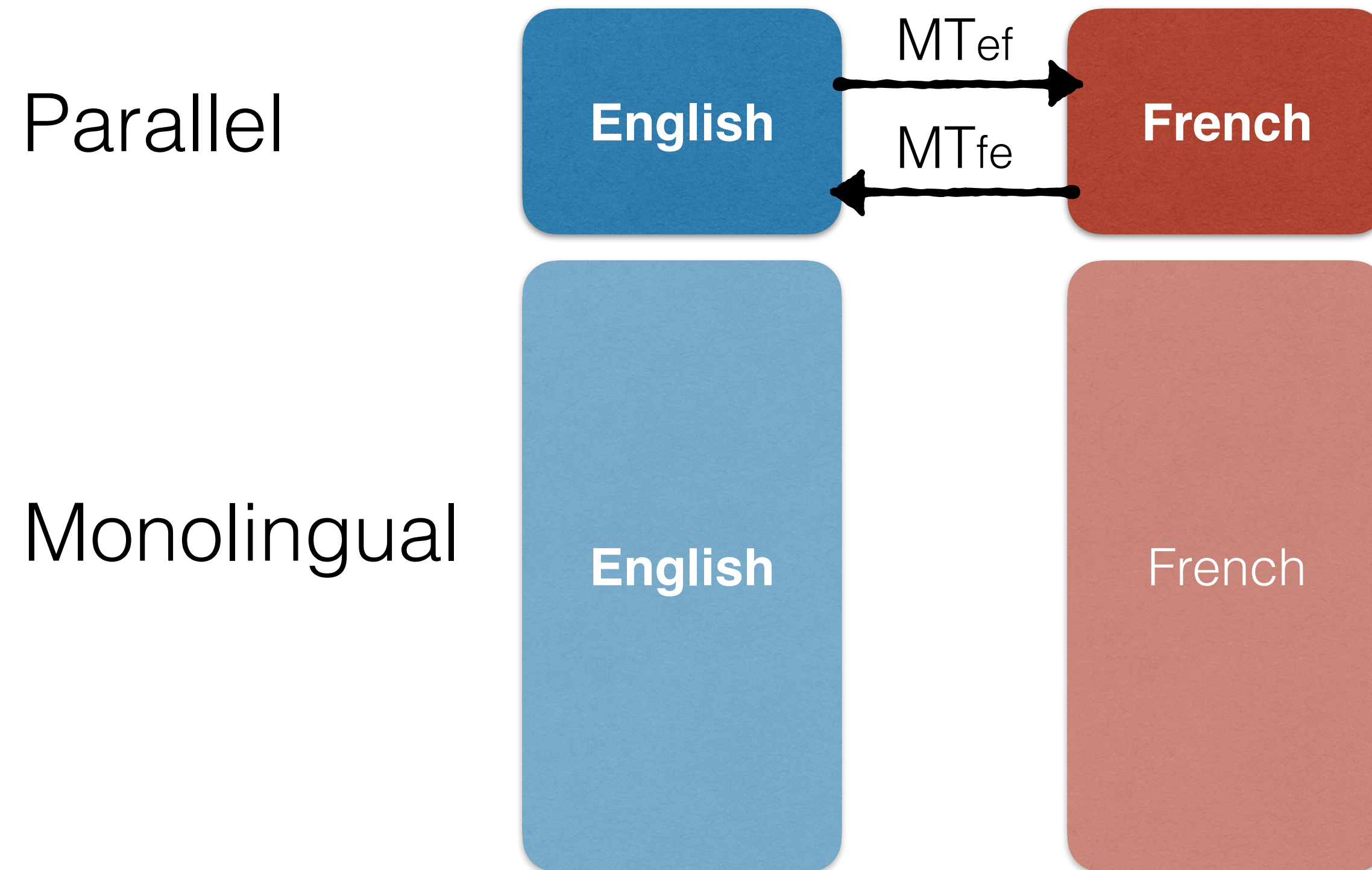
English

French



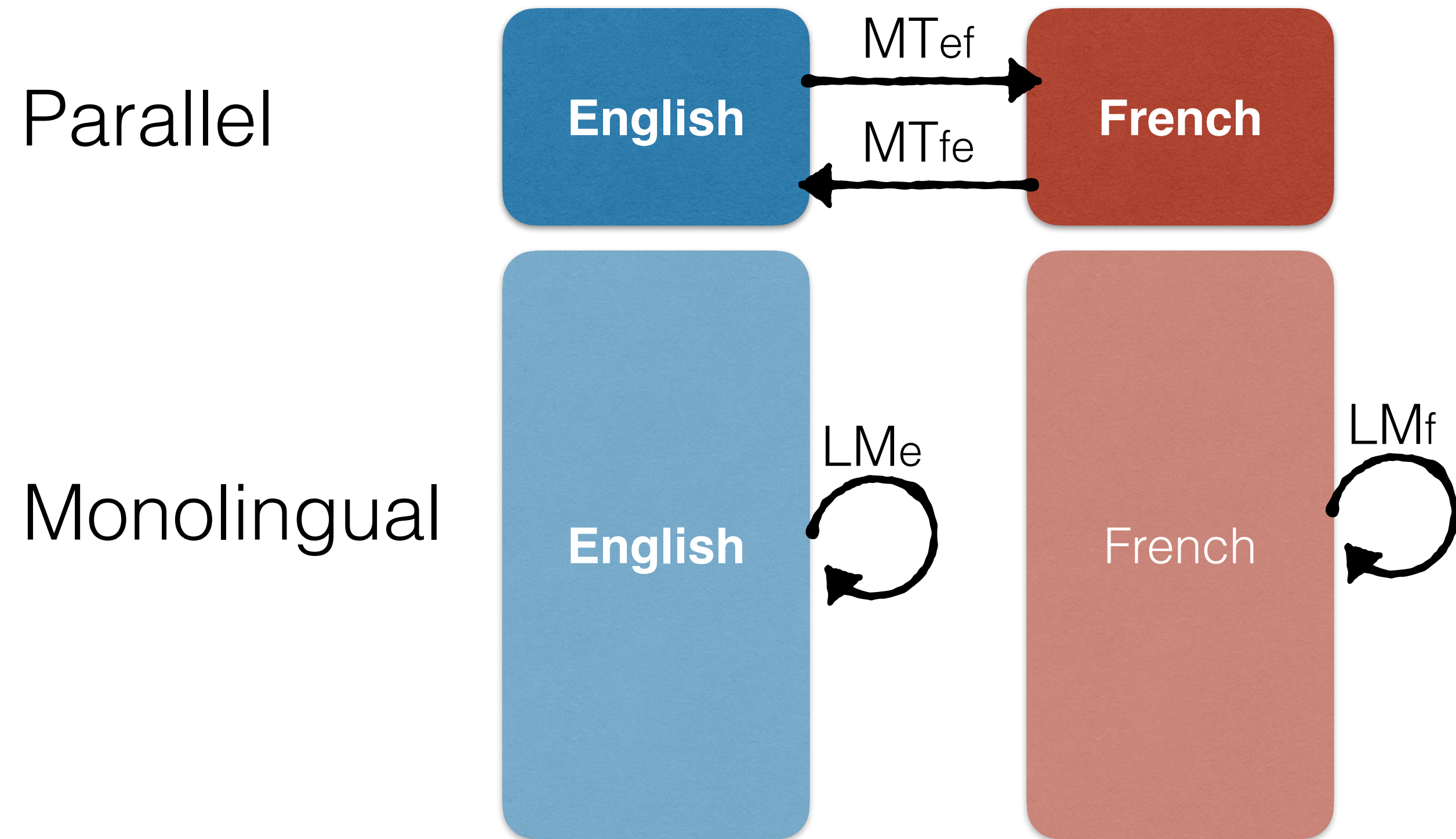
Dual Learning (He et al. 2016)

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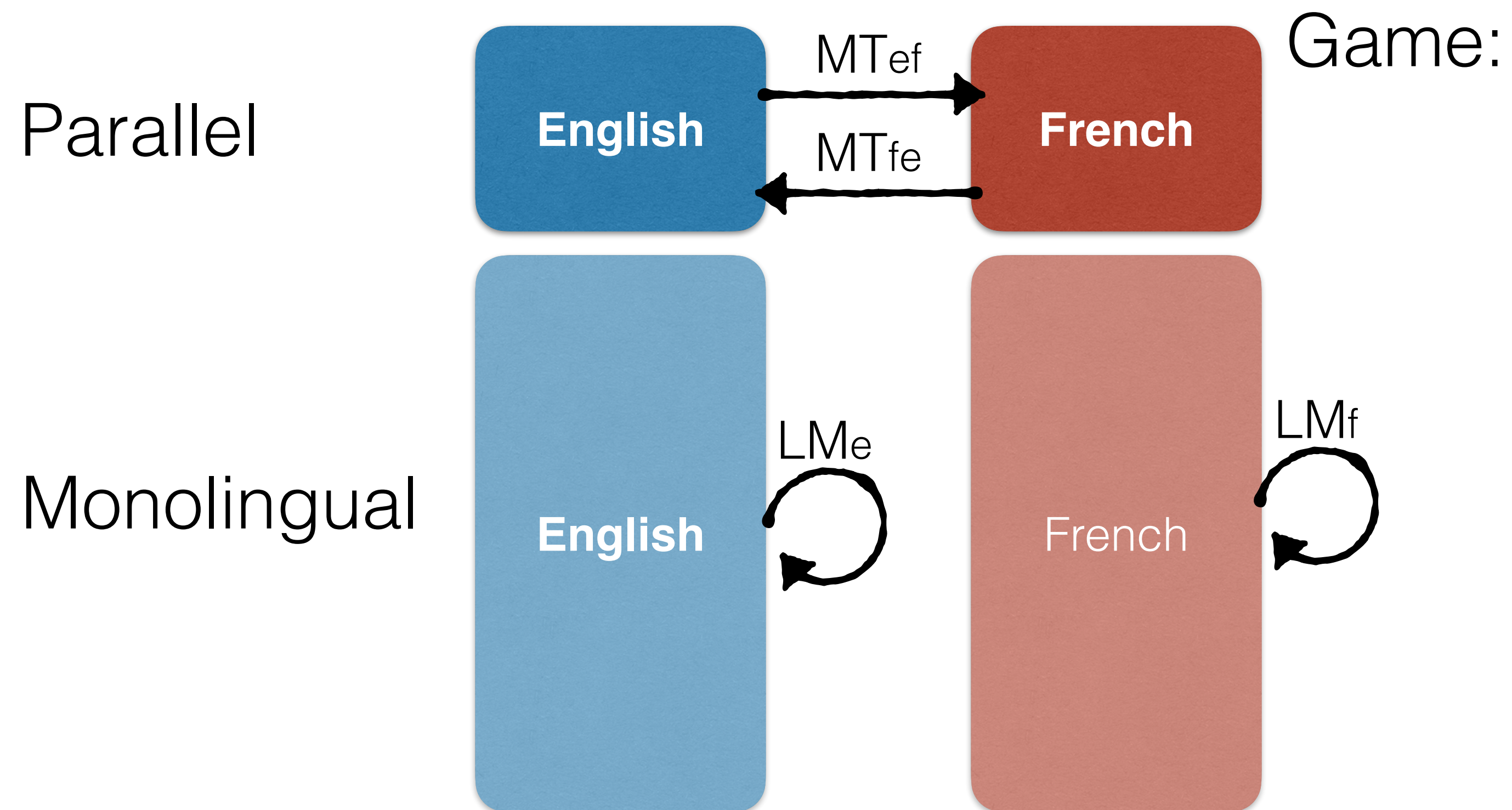
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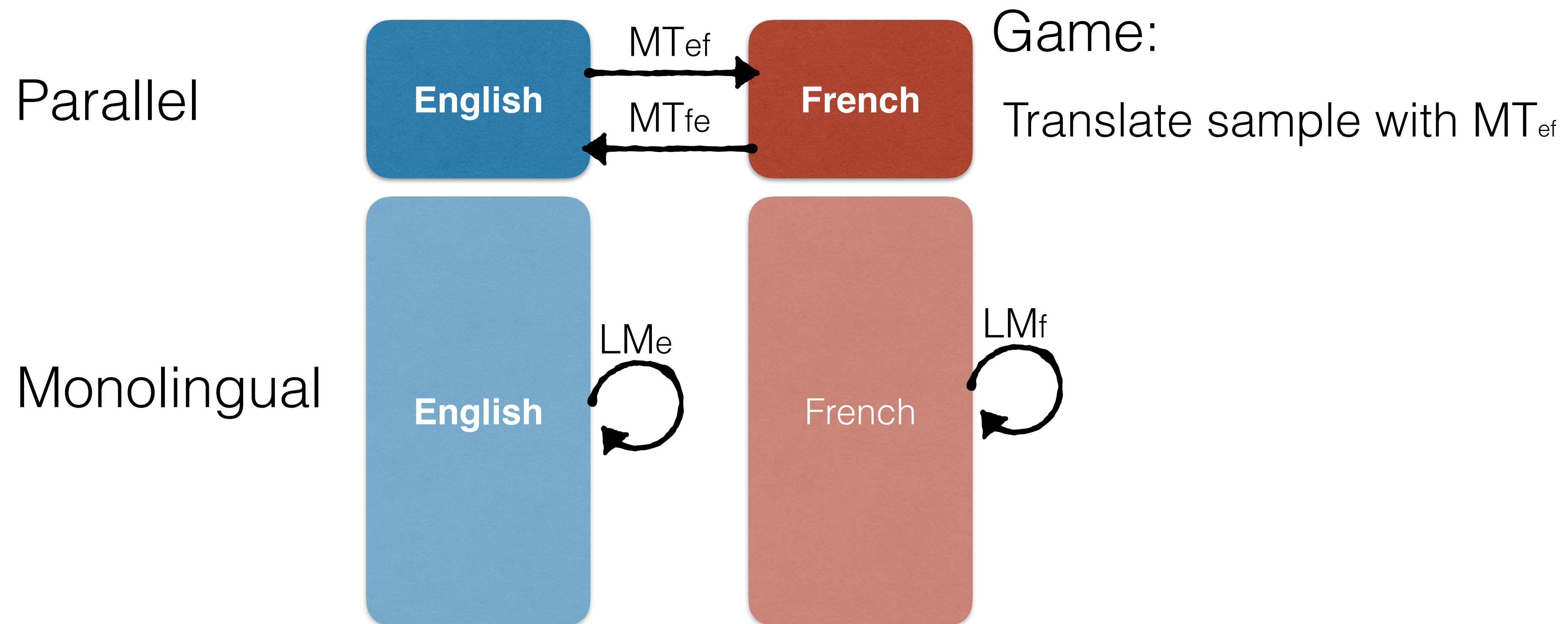
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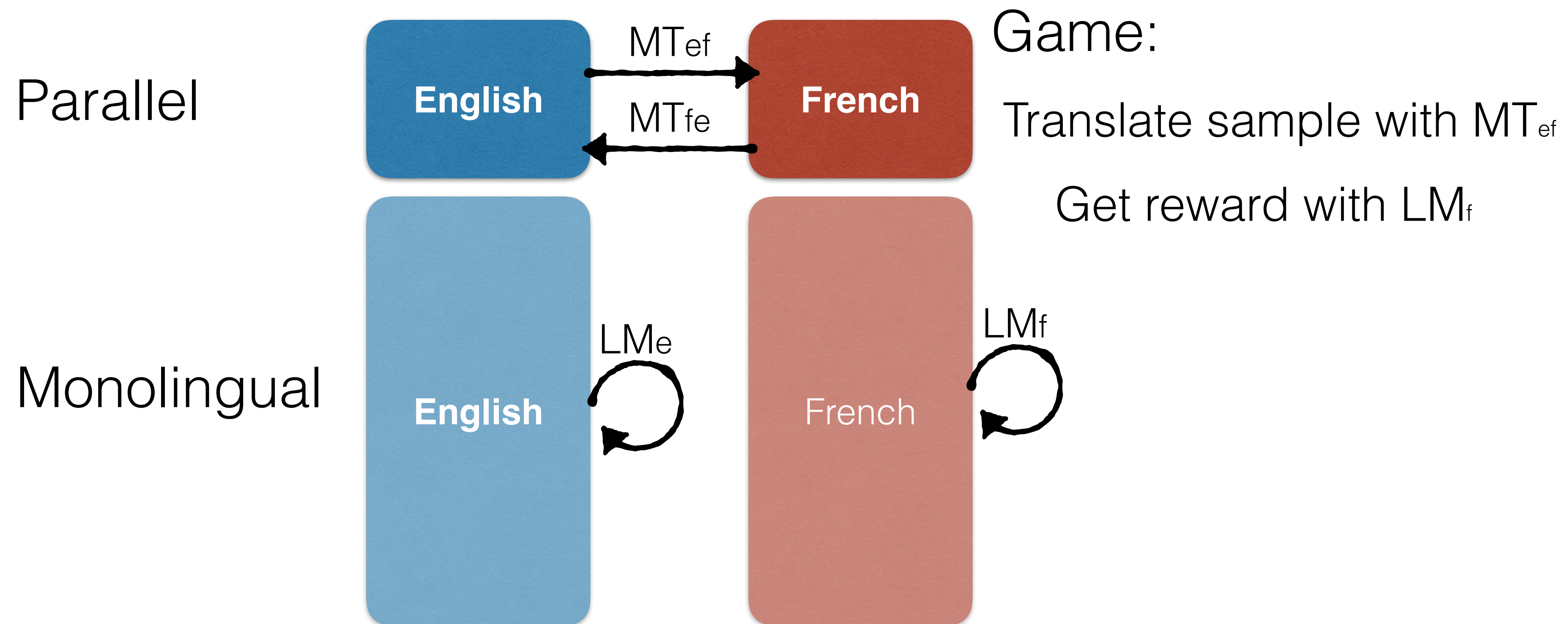
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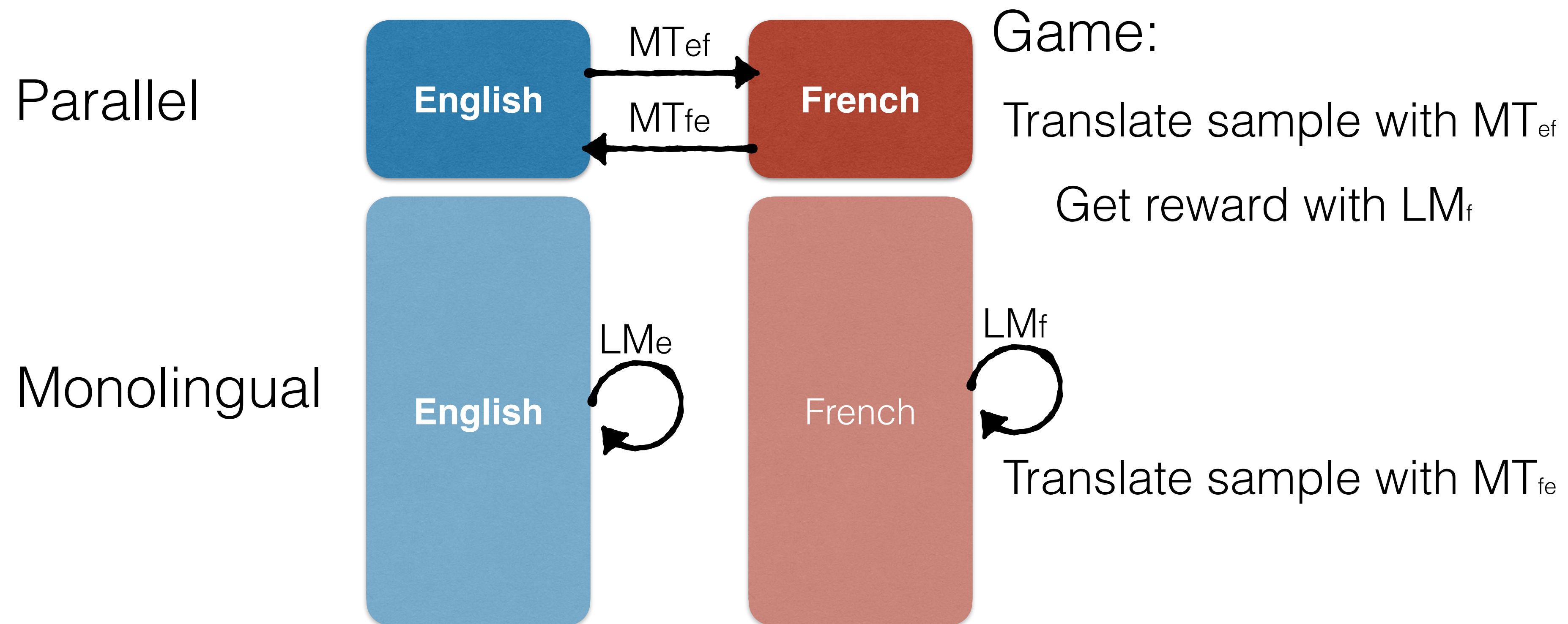
Dual Learning (He et al. 2016)

Assume MT_{ef} , MT_{fe} , LM_e , LM_f



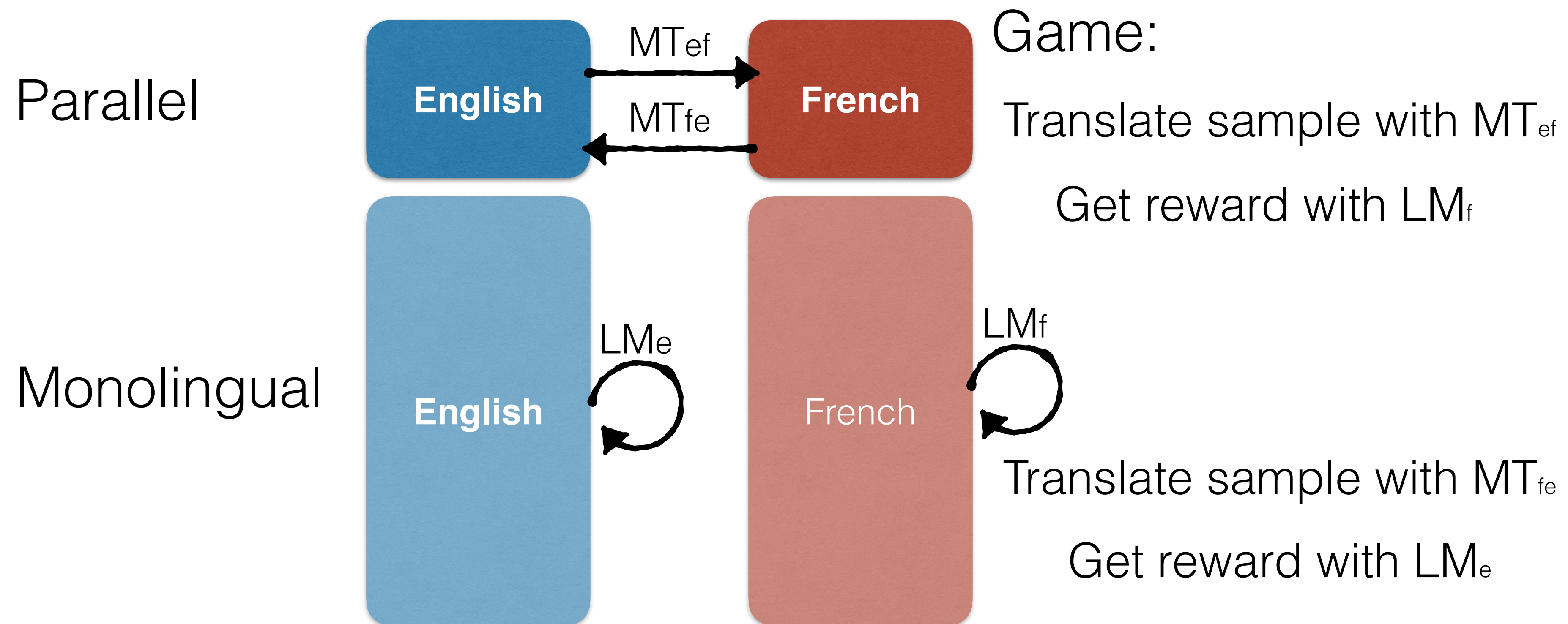
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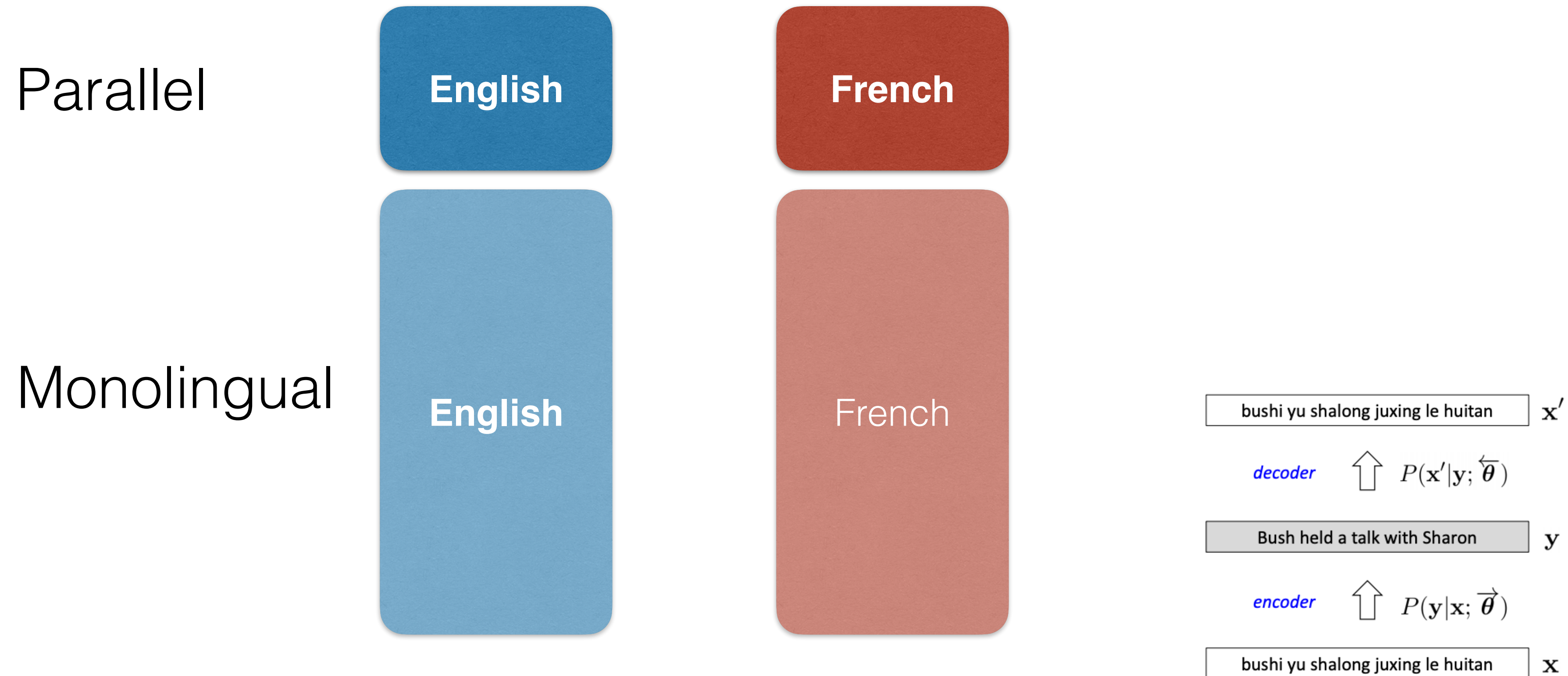


Dual Learning (He et al. 2016)

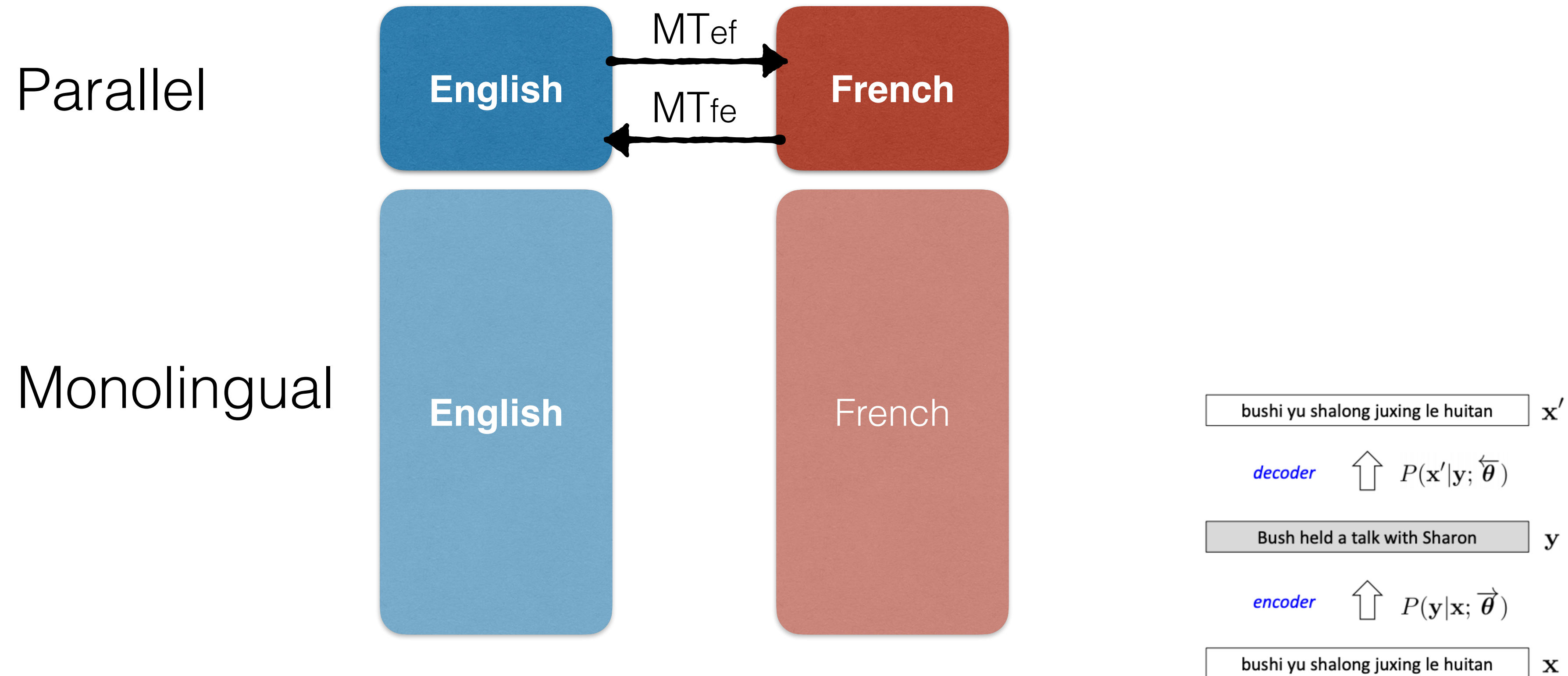
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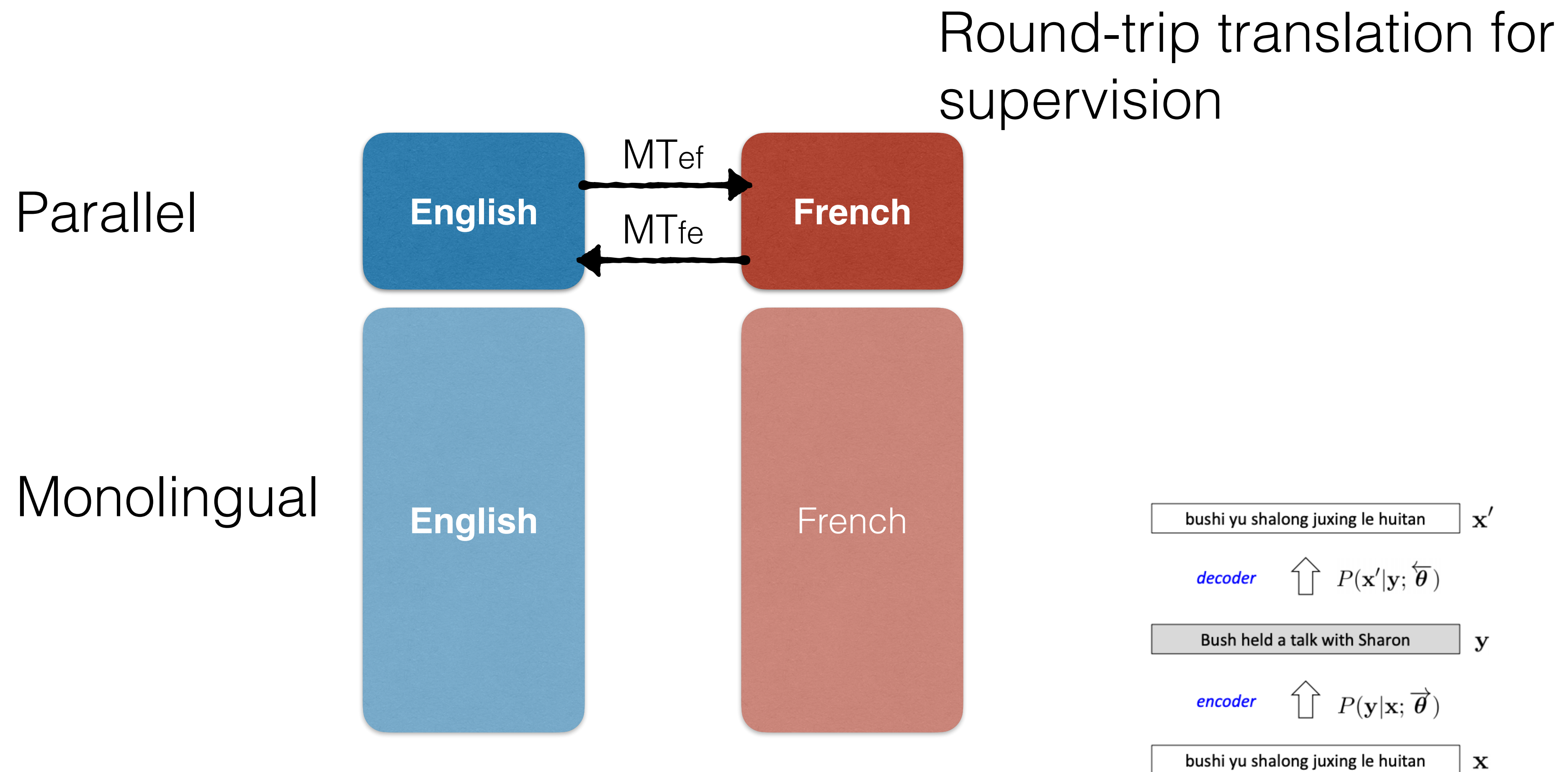
Semi-Supervised Learning for MT (Cheng et al. 2016)



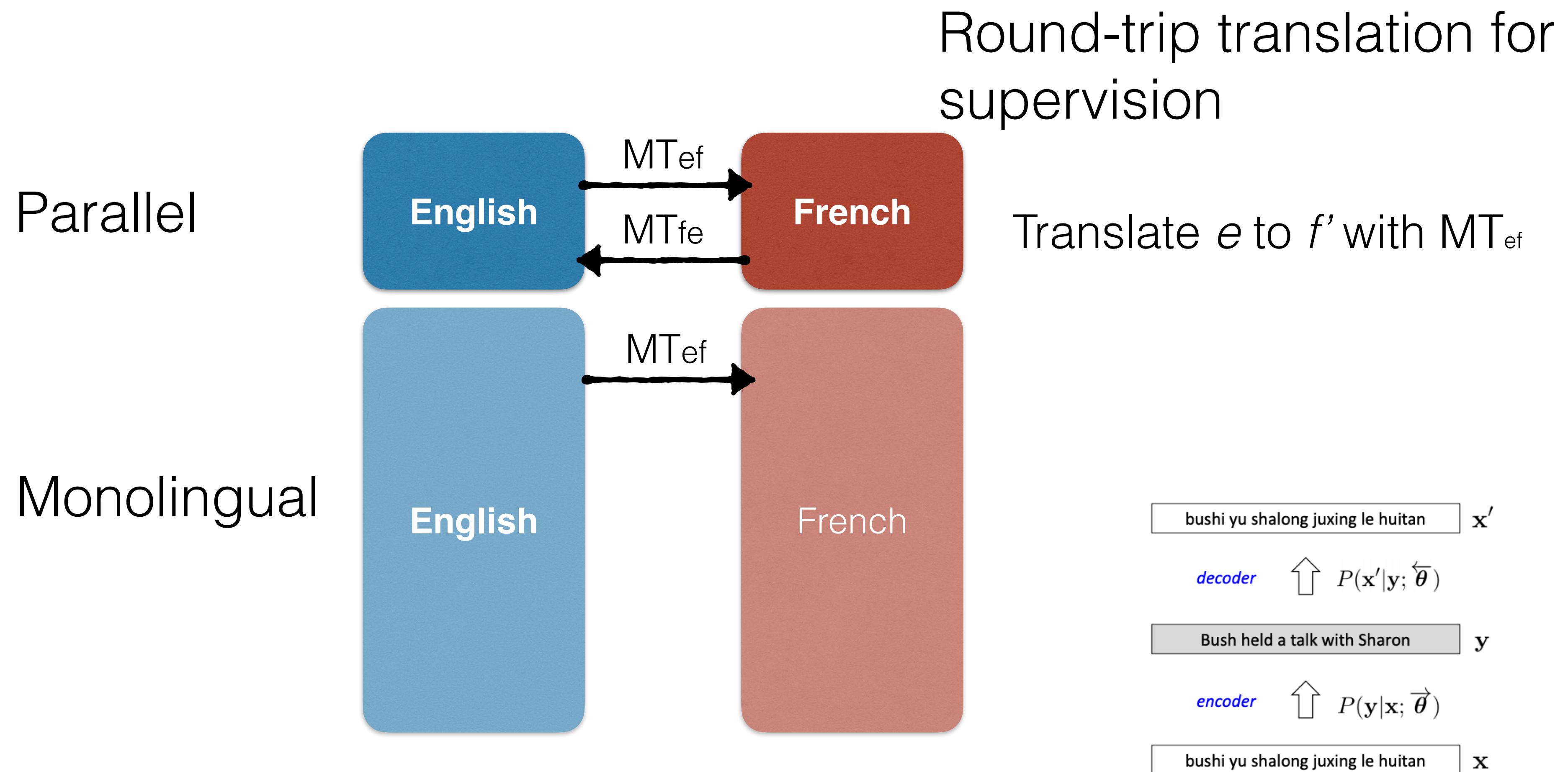
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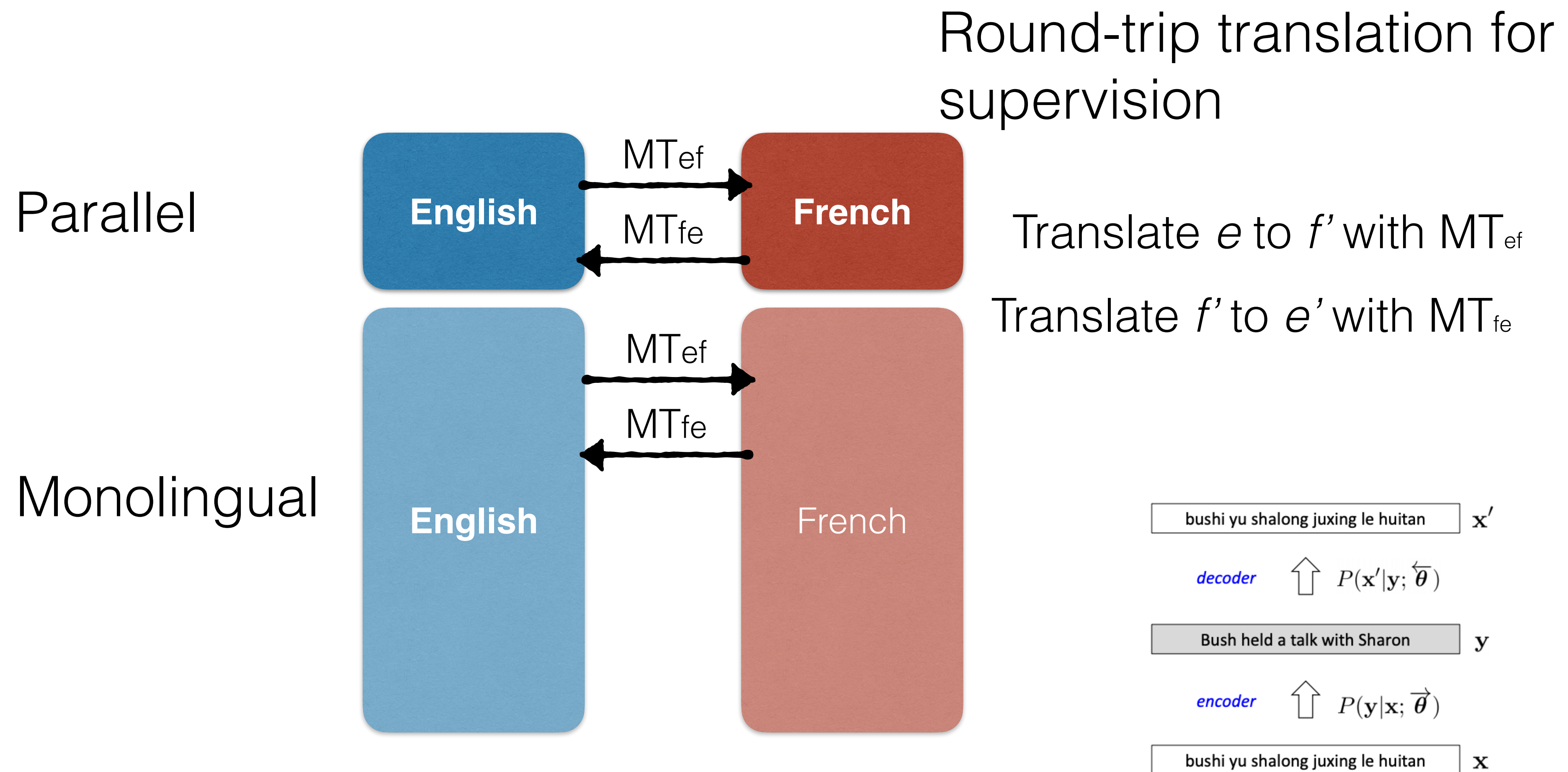
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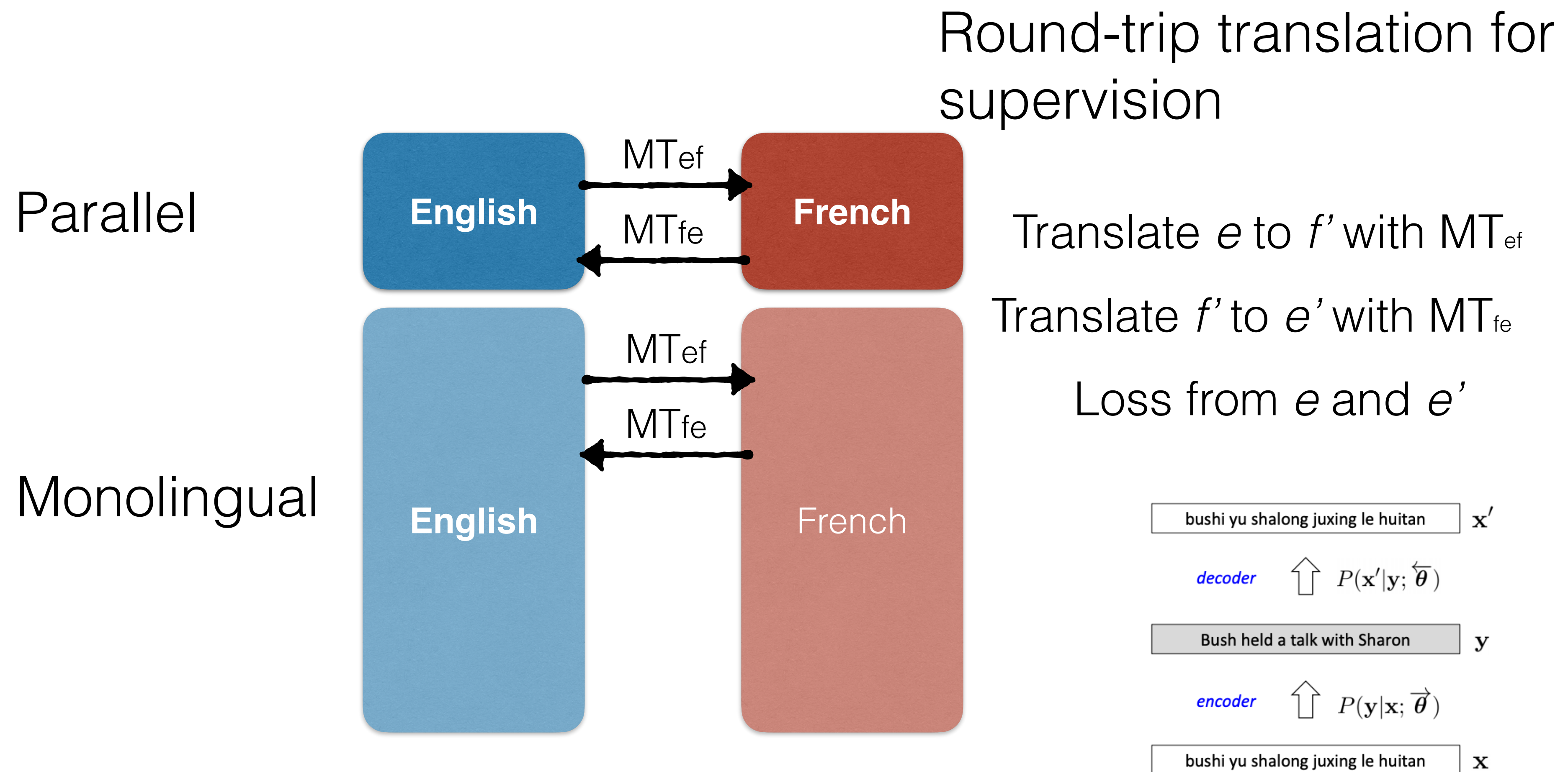
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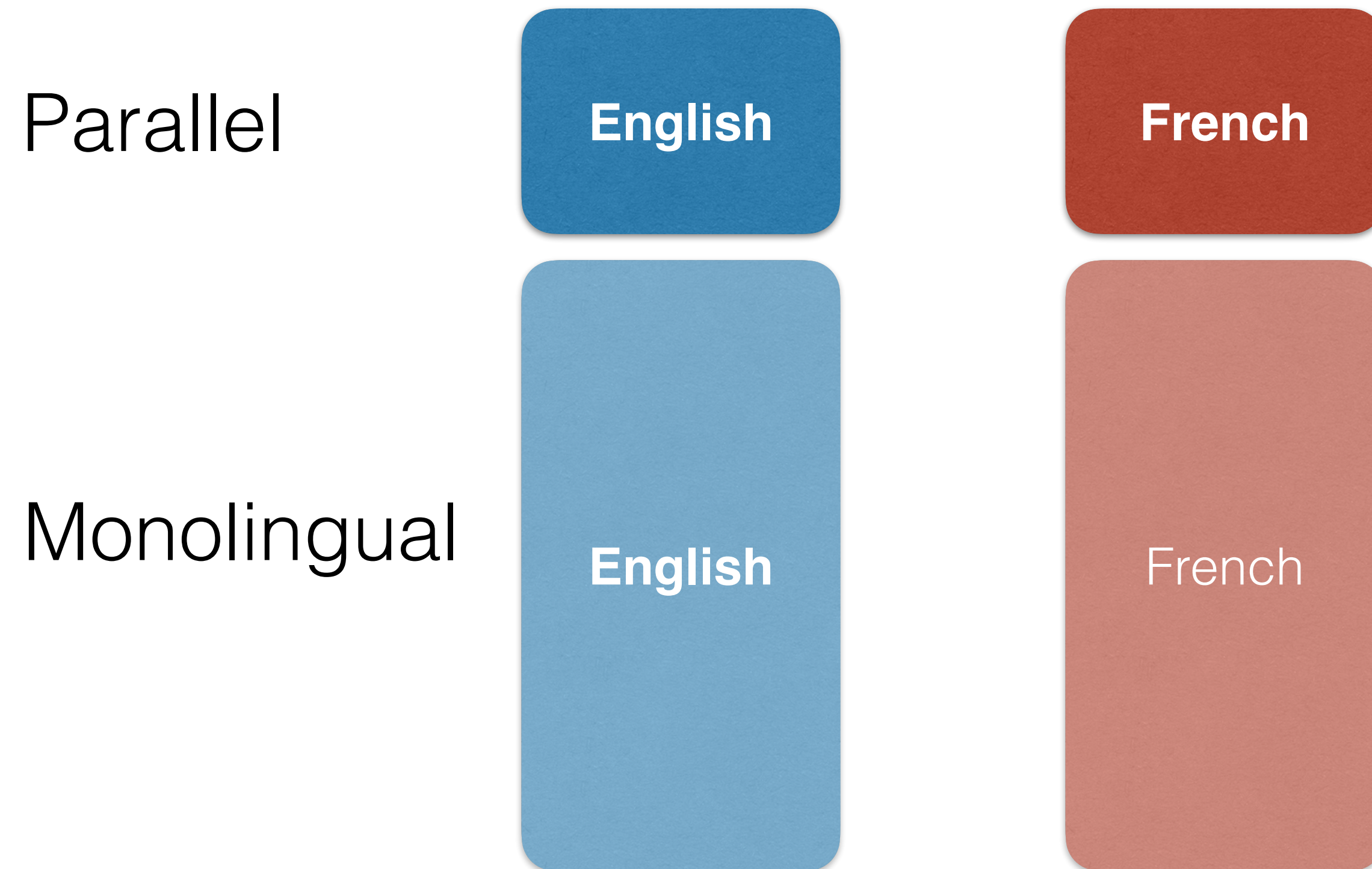
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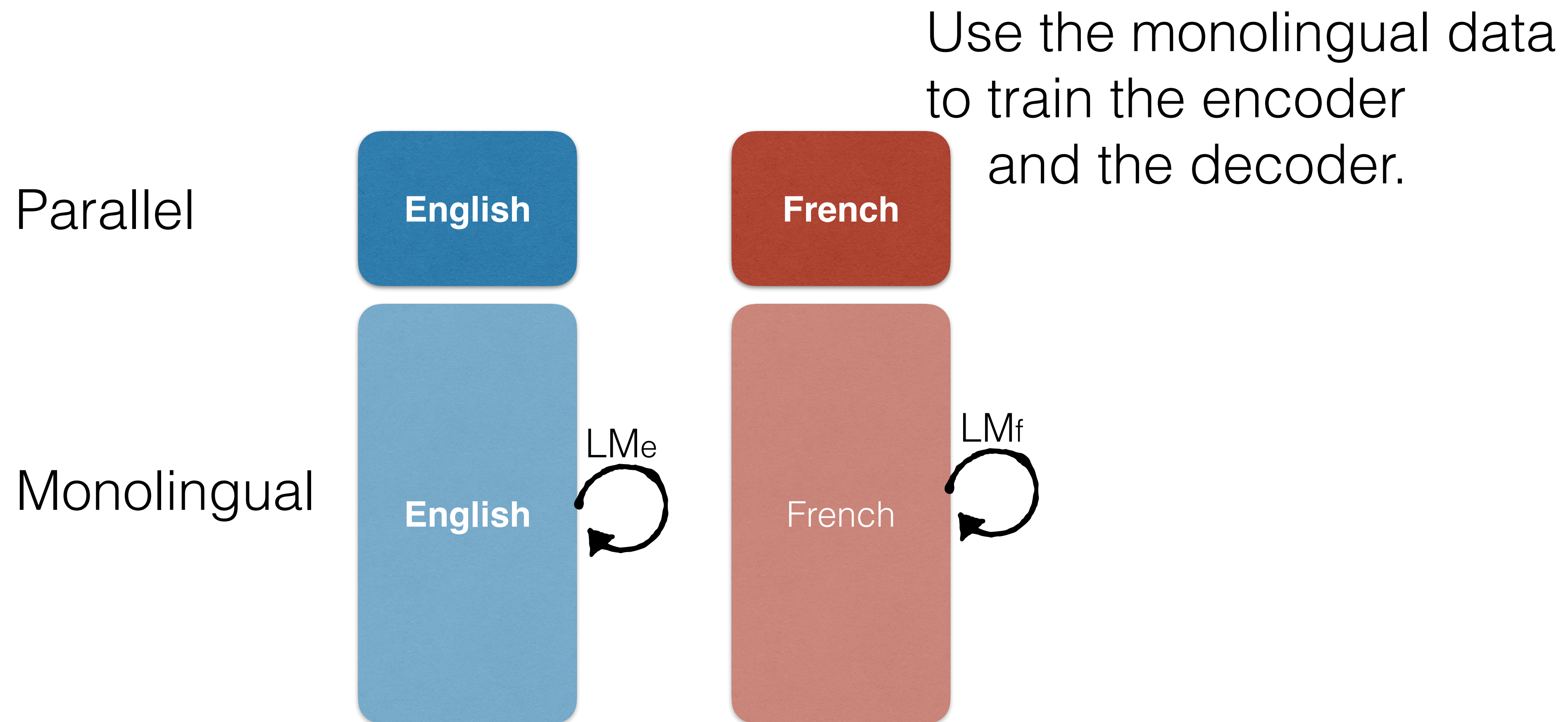
Semi-Supervised Learning for MT (Cheng et al. 2016)



Another idea: use monolingual
data to pretrain model components



Another idea: use monolingual data to pretrain model components



Unsupervised Translation

... at the core of it all:
decipherment

French

$$\arg \max_{\theta} \prod_f P_{\theta}(f)$$

From "Deciphering Foreign Language", Ravi and Knight 2011.

... at the core of it all:
decipherment

French

$$\arg \max_{\theta} \prod_f P_{\theta}(f)$$

Weaver (1955): *This is really English, encrypted in some strange symbols*

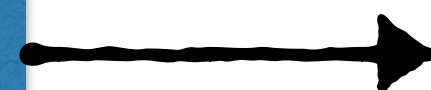
... at the core of it all:
decipherment

French

$$\arg \max_{\theta} \prod_f P_{\theta}(f)$$

Weaver (1955): *This is really English, encrypted in some strange symbols*

English



French

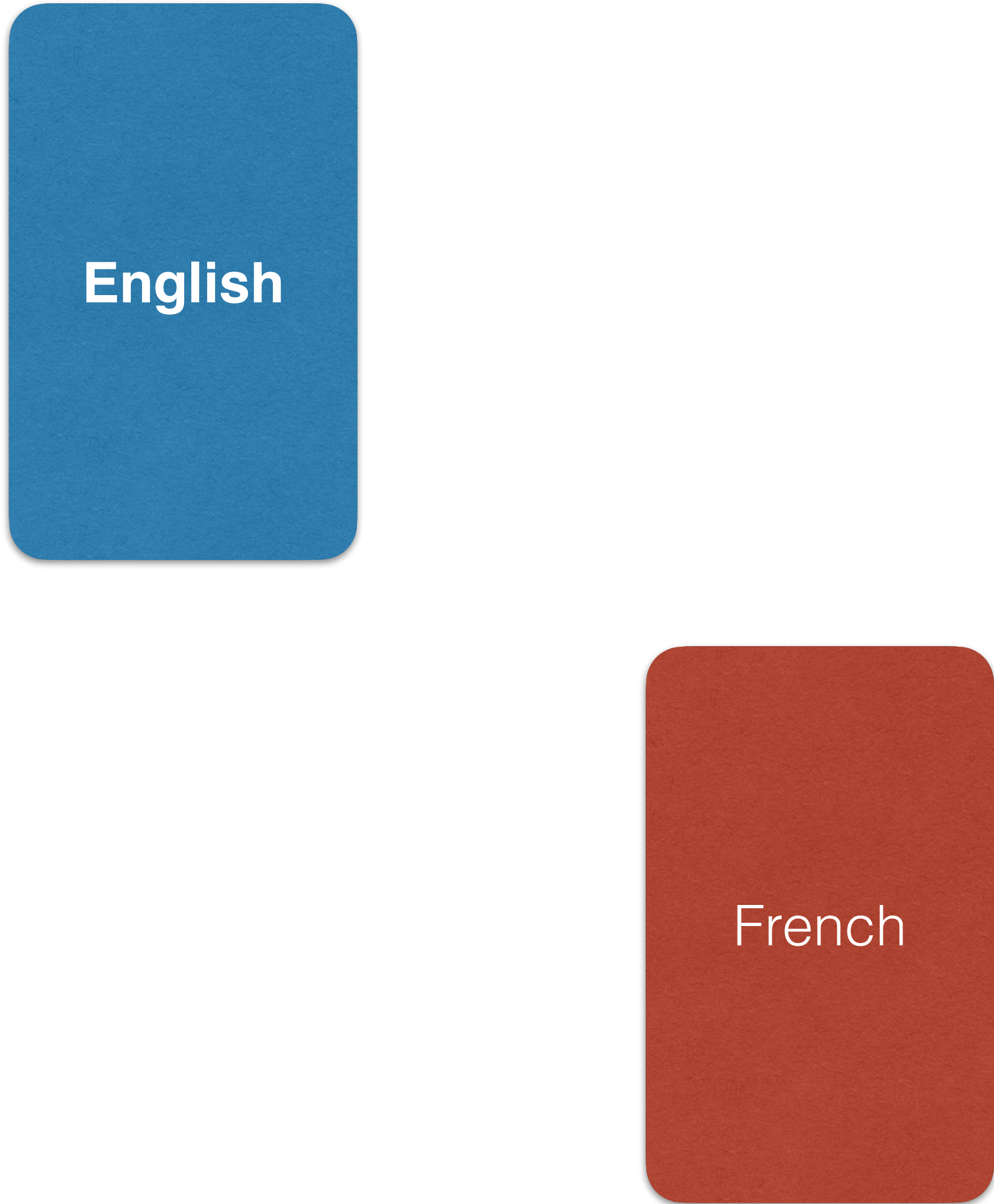
$$\arg \max_{\theta} \prod_f \sum_e P(e) \cdot P_{\theta}(f|e)$$

Unsupervised MT

(Lample et al. and Artetxe et al. 2018)

Unsupervised MT

(Lample et al. and Artetxe et al. 2018)



English

The diagram consists of two rounded rectangular boxes. The top box is blue and contains the word 'English' in white text. The bottom box is red and contains the word 'French' in white text. The boxes are positioned diagonally, with the English box at the top-left and the French box at the bottom-right.

French

Unsupervised MT

(Lample et al. and Artetxe et al. 2018)

1. Embeddings + Unsup. BLI



English



French

Unsupervised MT

(Lample et al. and Artetxe et al. 2018)



1. Embeddings + Unsup. BLI
2. BLI $\longrightarrow$ Word Translations

Unsupervised MT

(Lample et al. and Artetxe et al. 2018)

English

French

1. Embeddings + Unsup. BLI

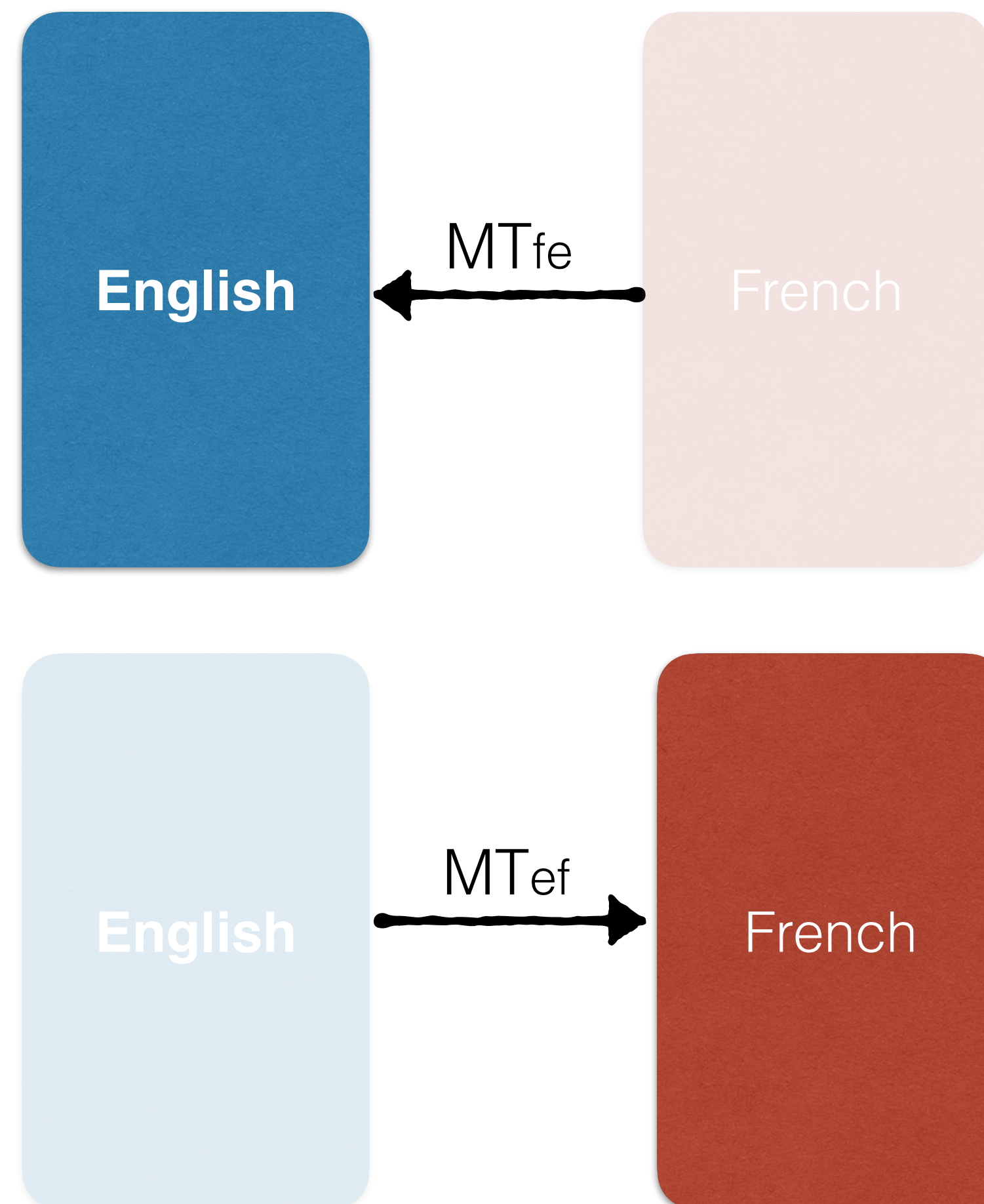
2. BLI $\rightarrow$ Word Translations

English

French

Unsupervised MT

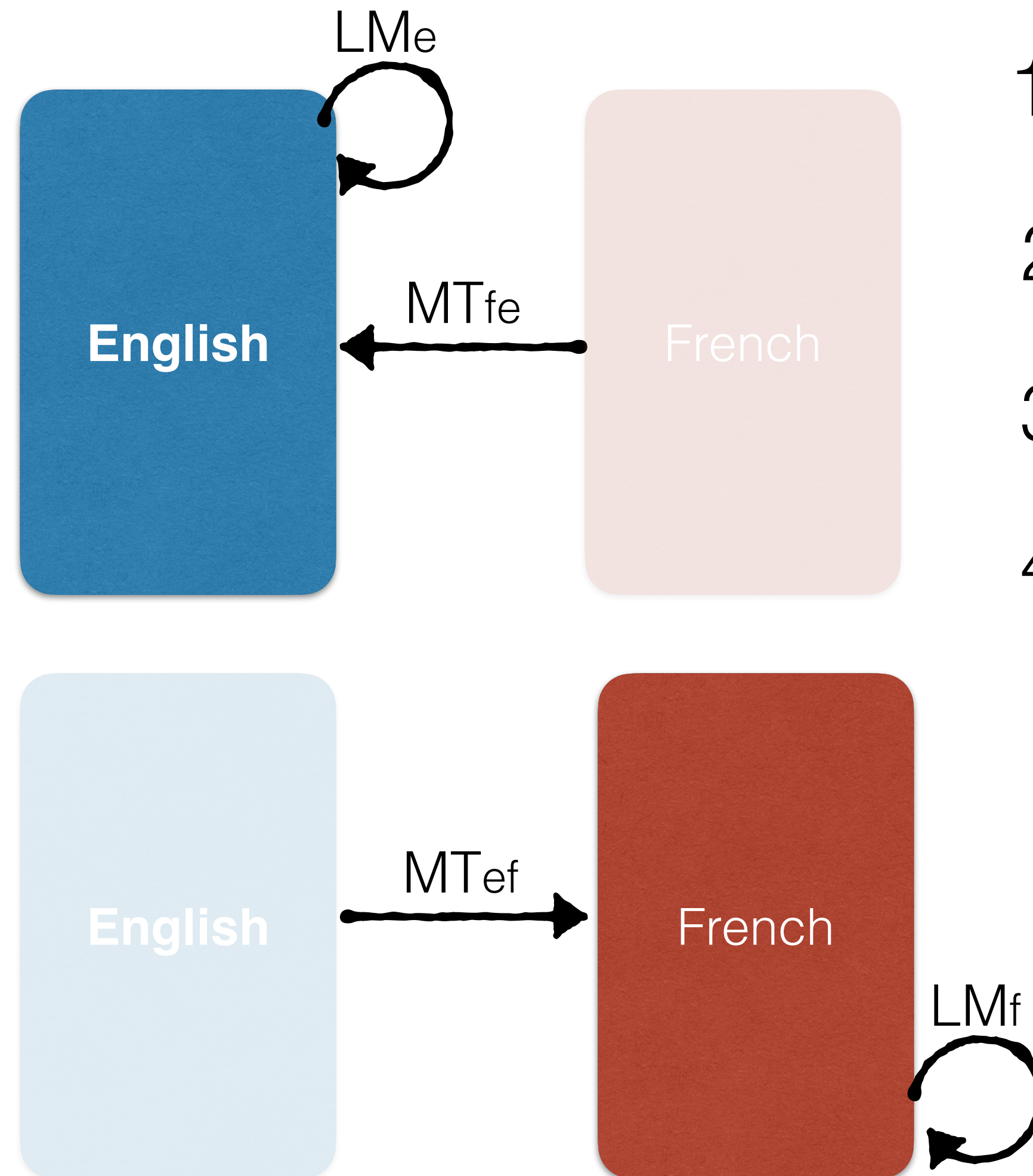
(Lample et al. and Artetxe et al. 2018)



1. Embeddings + Unsup. BLI
2. BLI $\rightarrow$ Word Translations
3. Train MT_{fe} and MT_{ef} systems

Unsupervised MT

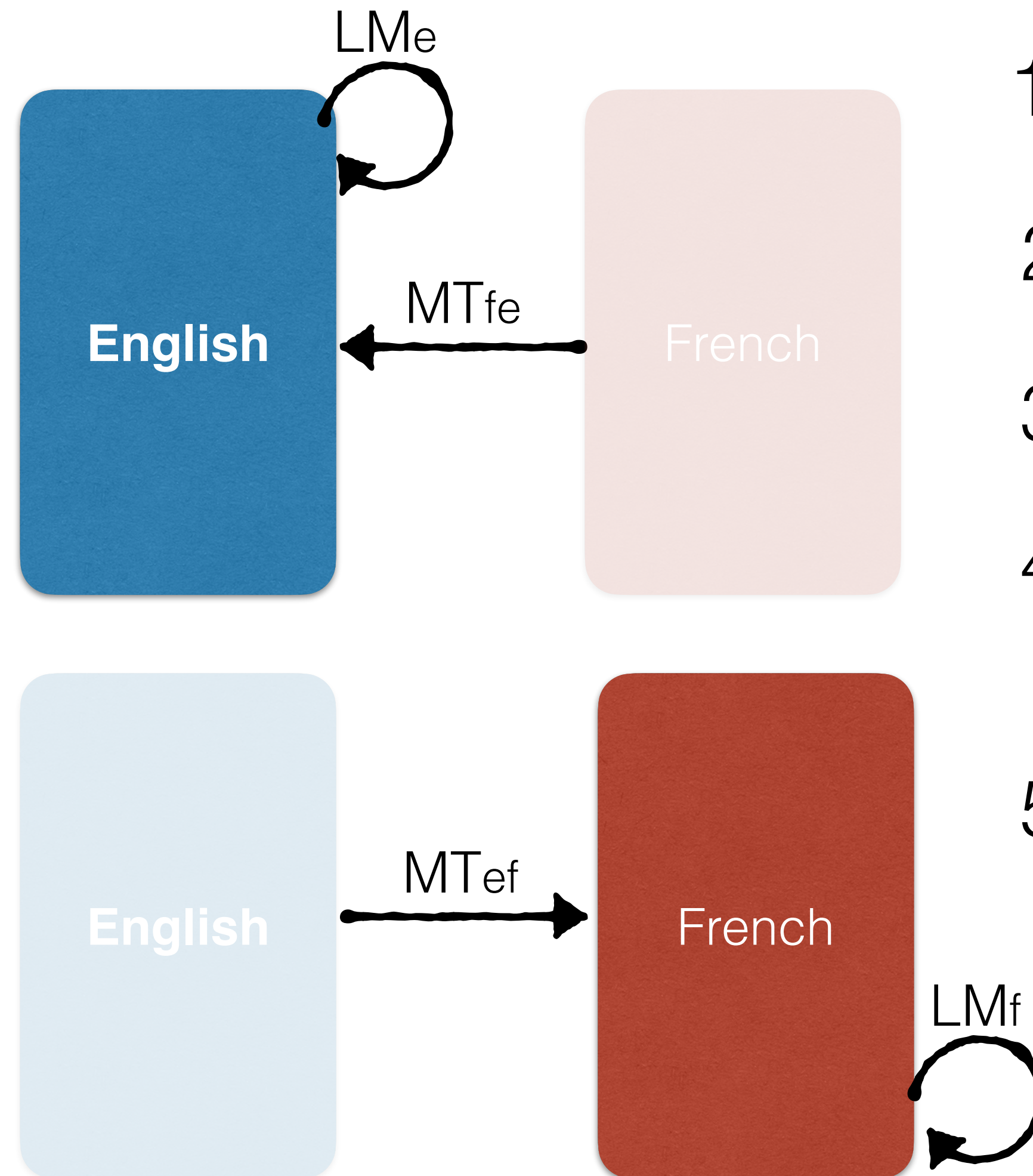
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1. Embeddings + Unsup. BLI
2. BLI $\rightarrow$ Word Translations
3. Train MT_{fe} and MT_{ef} systems
4. Meanwhile, use unsupervised objectives (denoising LM)

Unsupervised MT

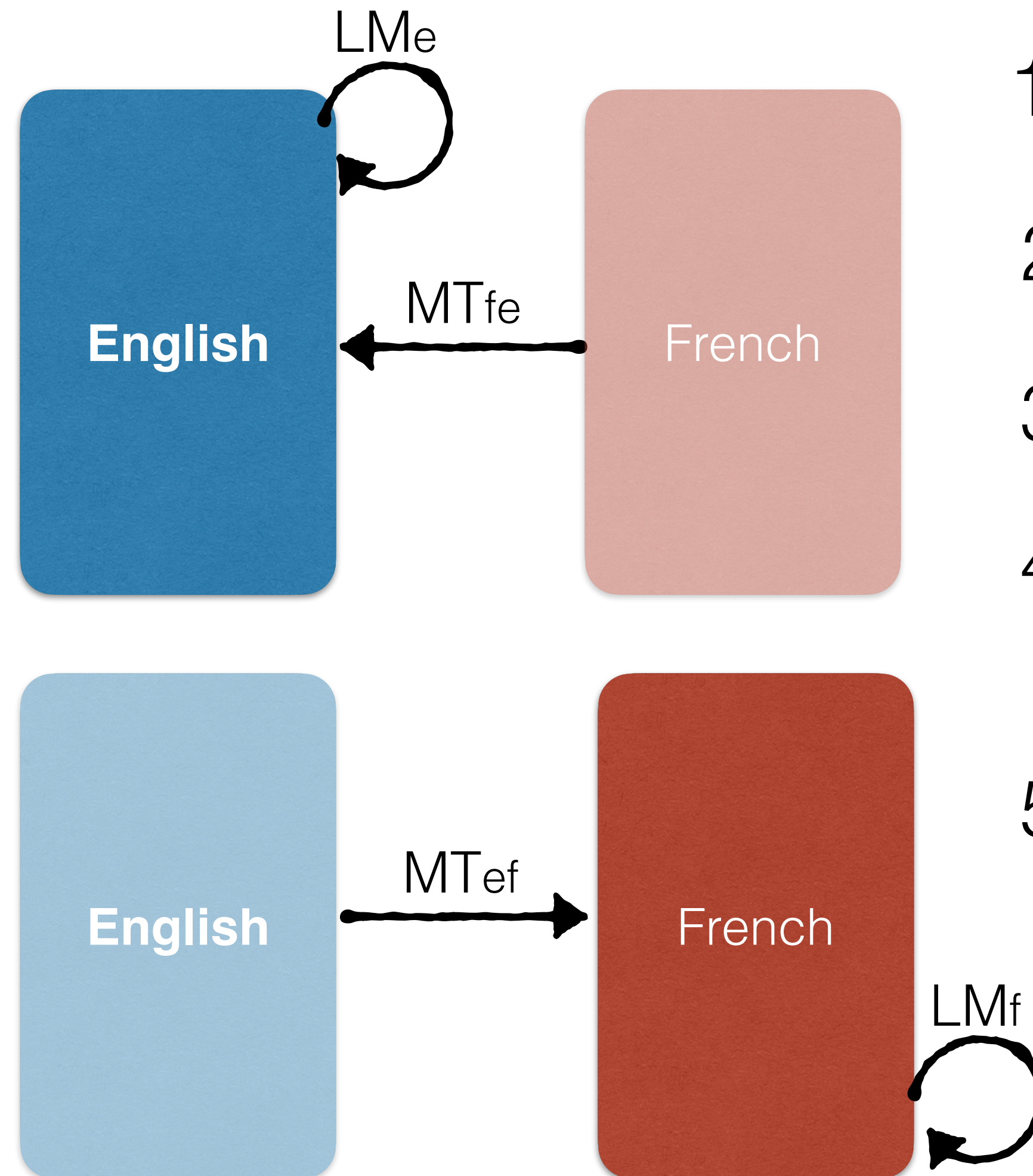
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1. Embeddings + Unsup. BLI
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5. Iterate

Unsupervised MT

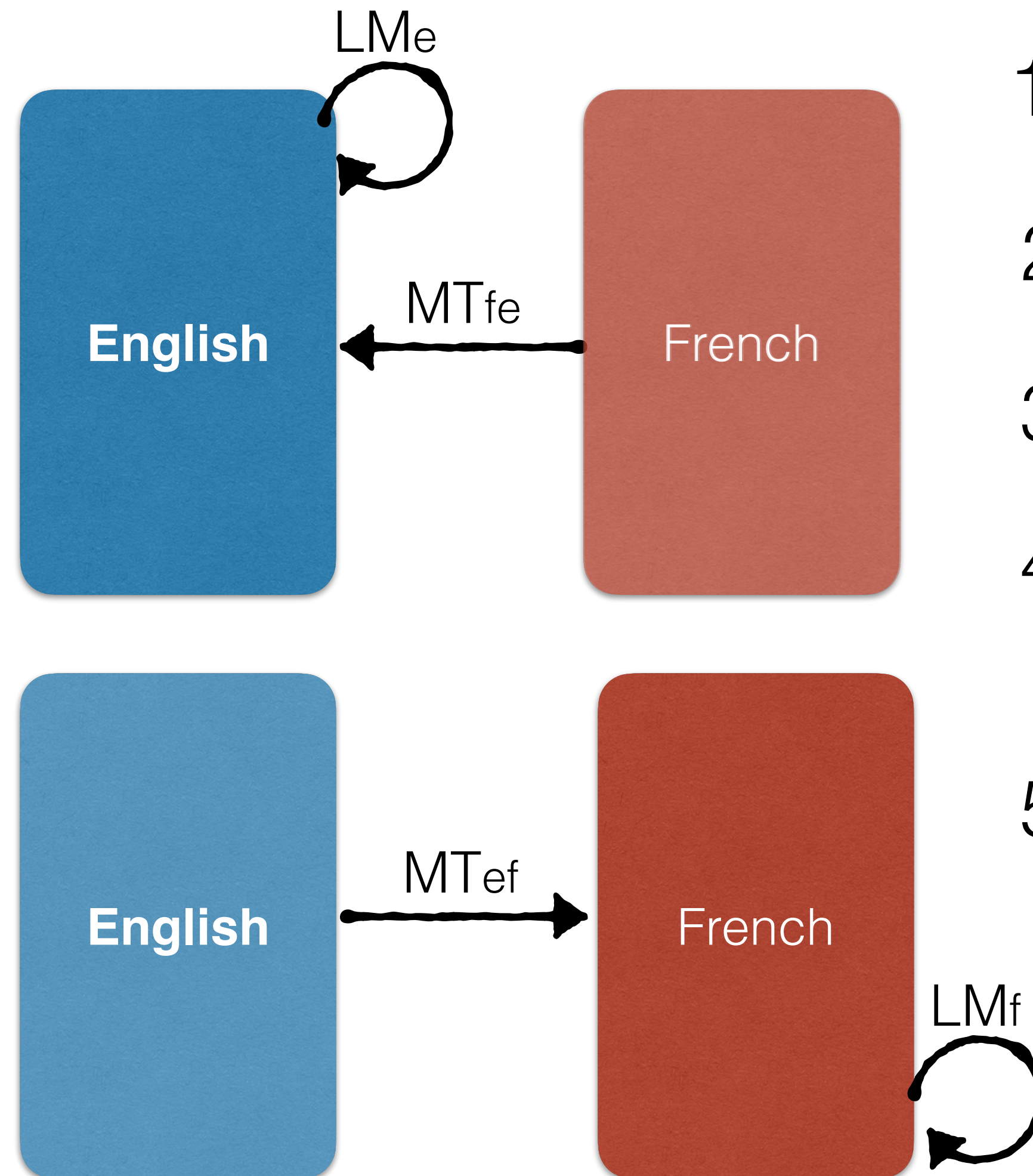
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NMT: the biggest success
story of NLP Deep Learning

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Neural Machine Translation went from a fringe research activity in 2014 to the leading standard method in 2016

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- 2014: First seq2seq paper published

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- **This is amazing!**

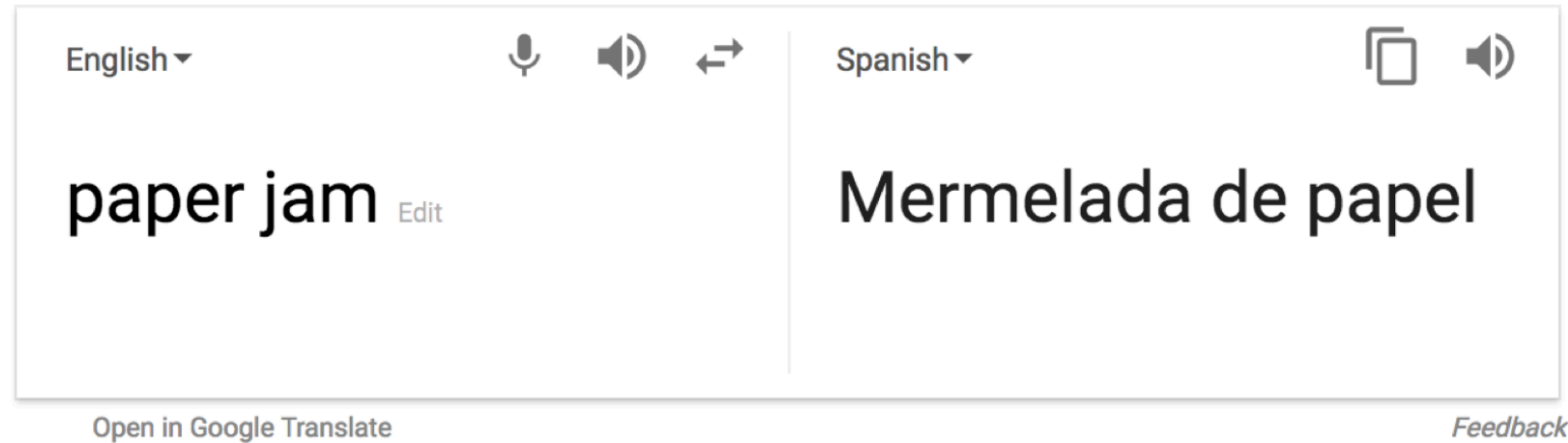
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- 2014: First seq2seq paper published
- 2016: Google Translate switches from SMT to NMT
- **This is amazing!**
- SMT systems, built by hundreds of engineers over many years, outperformed by NMT systems trained by a handful of engineers in a few months

So is Machine Translation solved?

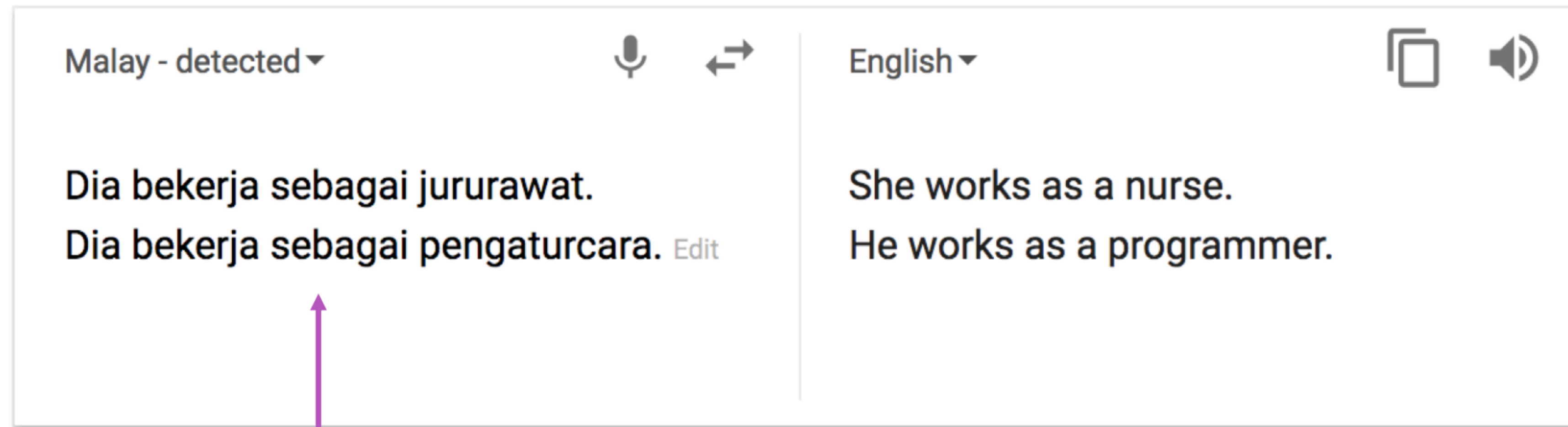
- Nope!
- Using common sense is still hard



?

So is Machine Translation solved?

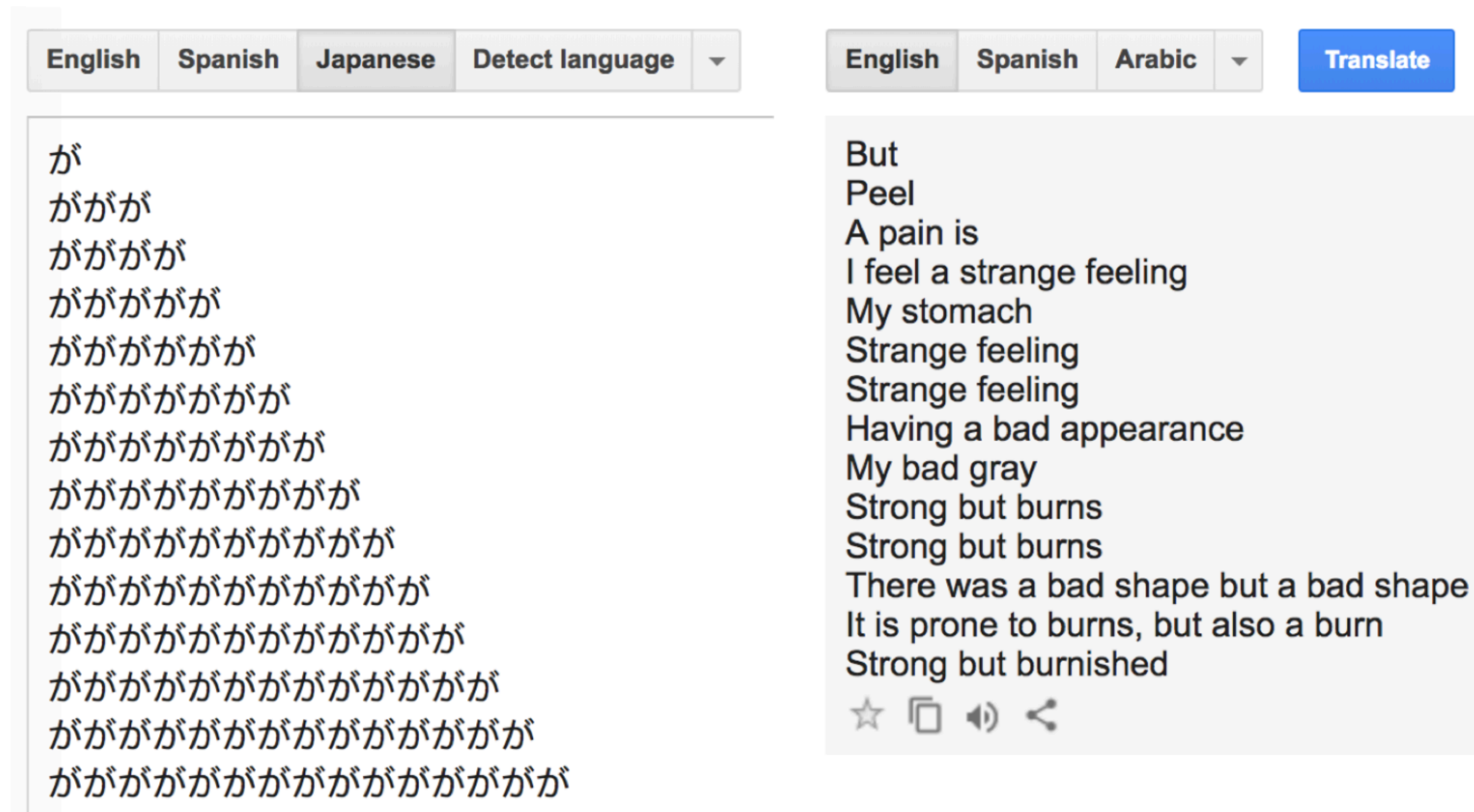
- Nope!
- NMT picks up biases in training data



Didn't specify gender

So is Machine Translation solved?

- Nope!
- Uninterpretable systems do strange things



The Last Translation Benchmark

- Colourful idioms, false friends, garden paths, cultural experience, signboards, menus, dialectal slang, archaic language, minority scripts, self-referential logic, social conventions, audio dialogue, daily life



Last Translation Benchmark

core authors	institutions: ETH, JHU, MBZUAI, KIT, UvA, QUB
Vilém Zouhar, Niyati Bafna, Mukund Choudhary, Maike Züfle, Sara Rajae, Pinzhen Chen	
other authors	
Jannis Vamvas, Sara Papi, Ona de Gibert, Bhavitvya Malik, Eliya Habba, Orfeas Menis Mastromichalakis, Patrícia Schmidtová, Michelle Wastl, Sheriff Issaka, Leshem Choshen, Stella Biderman, Antonis Anastasopoulos , Jan Niehues, Rico Sennrich, Mrinmaya Sachan, Ondřej Bojar, Kenton Murray, Jörg Tiedemann, Alham Fikri Aji, Philipp Koehn, Christof Monz, Alexandra Birch	
dataset authors in order of contributions	
Sowmya Vajjala, Chalamalasetti Kranti, Cristina España-Bonet, Nobin Sarwar, David Kaczér, Shunta Asano, Malik Marmonier, Daban Q. Jaff, Vaisakhi Mishra, Hend Al-Khalifa, Gabriele Sarti, Sourajit Saha, Nils Rehlinger, Juan Daniel Cuervo Villa, Jonathan Tonglet, Saugata Purkayastha, Dominik Macháček, Jagannathan Ramanujam, Heejin Do, Zuzana Nadova, Fred Philippy, Fabian Retkowski, Maria Lymperaiou, Silvia Casola, Hanna Yukhymenko, Shubhashis Roy Dipta, Sangwon Ryu, Andrés Jerez, Ron Keinan, Shuaib Shuaib Yusuf, Avantica Vempati, Maria Carmen Staiano, Sukannya Purkayastha, Adrian Cosma, Vitalii Babenko, Erivan Inan, Aviral Nigam, Wafa Aissa, Fatima Haouari, Venkata Prasanth Kumar Gummadi, Mehdi Jafarzadeh, Valentin Scourneau, Lukas Edman, Kaiser Sun, Shaomu Tan, Mohammad Sadegh Gholizadeh, Johannes-Rudolf David, Dipankar Srirag, Javier García Gilabert, Ruta Binkyte, Manar Ali, Ana-Maria Bucur, Sabry E. Farrag, Youssef Saber, Yihong Liu, Jean Maillard, Cojocaru Nicoleta, Xiaochuang Yuan, Sina Ahmadi, Philipp Mondorf, Kaustubh Dhole, Roman Wixinger, Shenbin Qian, Manuel Tuor, Sergey Troshin, Jonathan Yahav, Fida Mohammad Thoker, Amir Arsalan Rezapour, Lance Calvin Lim Gamboa, Manon Reusens, Kättriin Kukk, Koel Dutta Chowdhury, Giuseppe Gallipoli, Christian Hoang, Shaswati Saha, Seth Aycock, Jan Kocoń, Bo Chen, Linh Vu, Vatsal Venkatkrishna, Arafat Ahsan, Luan Thanh Nguyen, Hassan Soliman, Daryna Dementieva, Theresia Veronika Rampisela, Ngoc Quynh Tram Do, Marius Huber, Kazuki Egashira, Azmine Toushik Wasi, Vladislav Poritski, Mike Zhang, Deep Shah, Paul Gavrikov, Luis Frentzen Salim, David Africa, R. Damanhuri, Bello Umar Bello, Anumit Garg, Gengyu Rao, Pawan Sasanka Ammanamanchi, Kamile Dementaviciute, Andrianos Michail, L D M S Sai Teja, Dawei Zhu, Yi Fan, Wei Liu, Farhan Farsi, Elias Herranen, Sankalan Pal Chowdhury, Karen Sanchez, Farzad Shami, Ashok Urlana, Zimu Wang, Tomasz Limisiewicz, Priyaranjan Pattnayak, Marii Ojastu, Hongbin Na, Emilian Radoi, Chenyi Zhao, Carlos Hinojosa, Andrea Gregor de Varda, Zaid Alyafeai, Reem Alzahrani, Nehal Kathrotia, Alex Flückiger, Ulysses Sekai Tully Carr, Jimson Paulo Layacan, Guy Kaplan, Ritwik Tiwari, Rishit Dagli, Oksana Volchek, Isaac R Caswell, Bowen Yi, Blanka Kövér, Amir Hossein Yari, Aicha Chorana, Zhengxiang Wang, Selja Keränen, Samuel Simko, Joy Olusanya, Jenny Chim, Enzo Doyen, Vivek Harsha Lakkamaneni, Sophia Conrad, Pouya Sadeghi, Panayiotis Panayiotou, Luis Lara, Jannatul Nayem, Eran Yahav, Debanshu Das, Antonia Karamolegkou, Anmol Goel, Aishik Mandal, Tommaso Cerruti, Raoyuan Zhao, Mykola Haltiuk, Thura Aung, Naser Almousa, Amir Hossein Kargaran, Rachel Bawden, Qiaoyuan Zheng, Mateusz Lango, Beni Egressy, Fidel Rodríguez Velásquez, Natchapon Jongwiriyanurak, Minh Ngoc Do, Marco Gaido, Lena Libon, Dzmitry Kuzmin, Badal Nyalang, Antoine Taroni, Andrei Niculae, Abdulaziz Nura Kani, Rushikesh Zawar, Marek Šuppa, Beatrice Savoldi, Andreas Simons, Rayyan Merchant, Ilai Yaron Levy, Francesco Pinto, Ziyi Yang, Yolanda Xavier, Samuel Frontull, Muhammad Ravi Shulthan Habibi, Kenneth Enevoldsen, Harris Abdul Majid, Francesca Padovani, Tim Graf, Tatiana Bielakova, Sharifa Djurabaeva, Shaoxiong Ji, Raia Abu Ahmad, Pavel Stepachev, Jirui Qi, Ayush Sunil Munot, Alireza Pakniat, Ayla Rigouts Terryn, Yuxing Lu, Yurii Paniv, Xiyan Fu, Tosin Adewumi, Sunisth Kumar, Stéphane J.P.S. Thunus, Shree Harsha Bokkahalli Satish, Shayan Bali, Prakhar Gupta, Papa Abdou Karim Karou Diallo, Matija Akrap, Marko Culjak, Kristýna Onderková, Joseph Attieh, Esrael Teferi Tensay, Elisabeth Fittschen, Benoît Sagot, Jingwei Ni, Yu Fan	

<https://arxiv.org/pdf/2609.04173>

The Last Translation Benchmark

Translate from

Spanish (Colombia)

Translate into

English

Input

1Ese man se palistorneó. Quién sabe qué le habrá pasado al toche ese.

Add image/audio/video

Add instructions/context

Translate by all models

Translations

2/10 pass verification

2Human translation	That guy went crazy. Who knows what happened to that guy.	5✓✓
3Gemini 3.1 Pro	That guy lost it. Who knows what happened to that idiot.	6✓×
Gemma 4	That guy lost it. Who knows what happened to that idiot.	✓×
Llama 4 Maverick	That dude got really drunk. Who knows what happened to that weirdo.	××
GPT-5.4 Mini	That guy went off the rails. Who knows what happened to that dumbass.	✓×
GPT-5.6 Sol	That dude got all messed up. Who knows what happened to that idiot.	××
Deepseek V4 Pro	That guy got himself beat up. Who knows what could have happened to that dude.	×✓
Claude Sonnet 4.5	That guy lost it. Who knows what happened to that dude.	✓✓
Gemini 3.5 Flash Lite	That guy lost his mind. Who knows what happened to that idiot.	✓×
Claude Haiku 4.5	That guy got completely messed up. Who knows what happened to that guy.	×✓

Verification rules (all must pass)

4

- Remove

In Colombia, the word “palistorneo” is used to say that someone is crazy.

- Remove

In Colombia, the word “toche” is used to refer to someone; it is NOT used in a derogatory way.

+ Add Rule

✓ Verify translations

Submit Input, Translations & Rule

Figure 1: Screenshot of the contributor interface. The contributor writes input ① and reference translation ②, and obtains the automatic translations ③. The contributor then writes verification rules ④. This contribution can be submitted if it is: translatable (the reference translation passes the rules ⑤), and difficult (most automatic translations fail at least one rule ⑥).